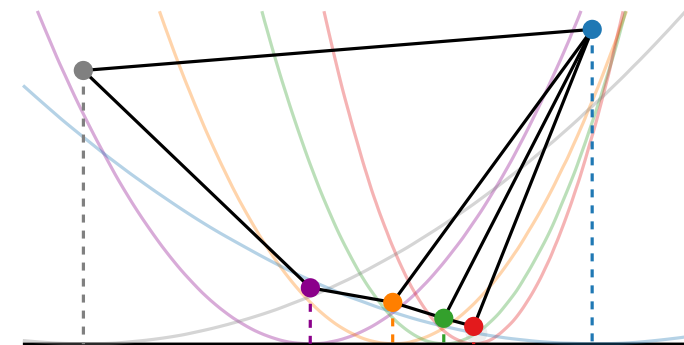
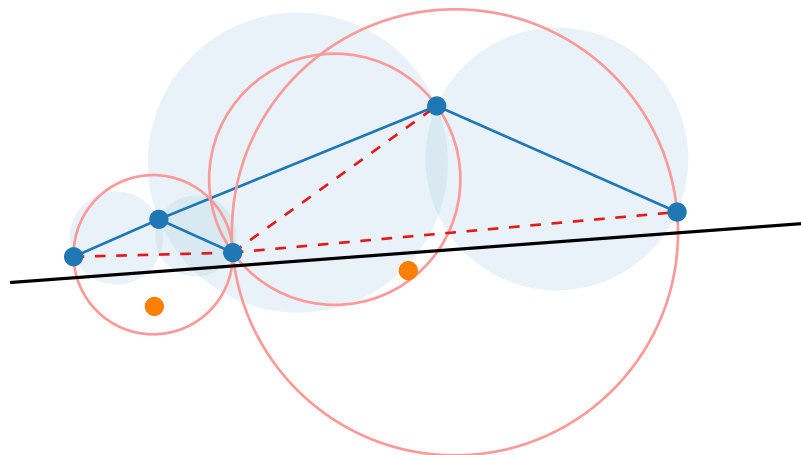




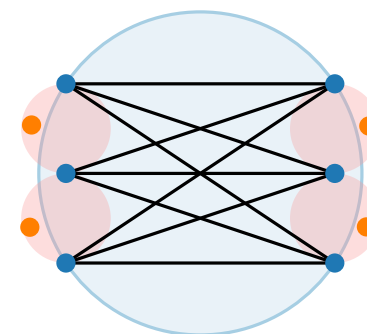
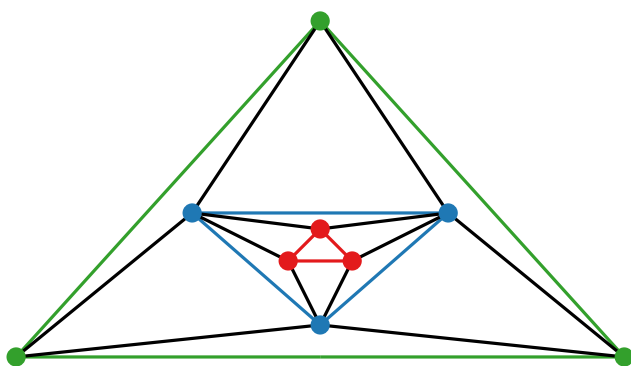
Separability of Witness Gabriel Drawings

Carolina Haase

William J. Lenhart

Philipp Kindermann

Giuseppe Liotta



Our Contribution

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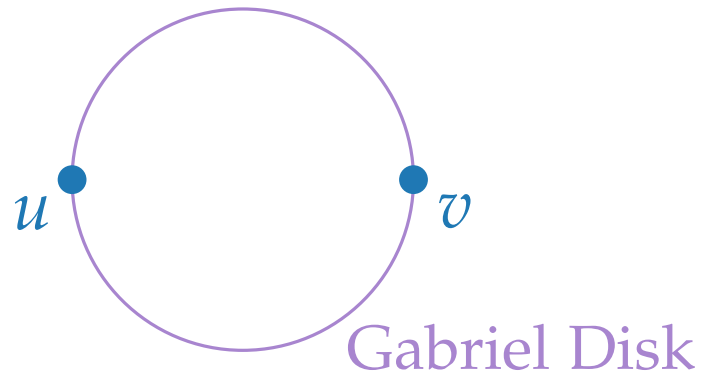
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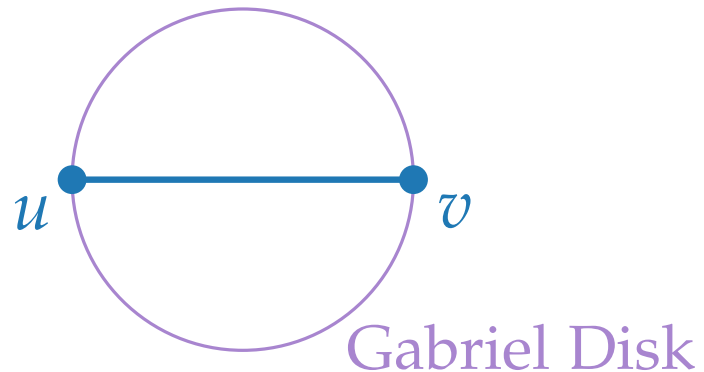
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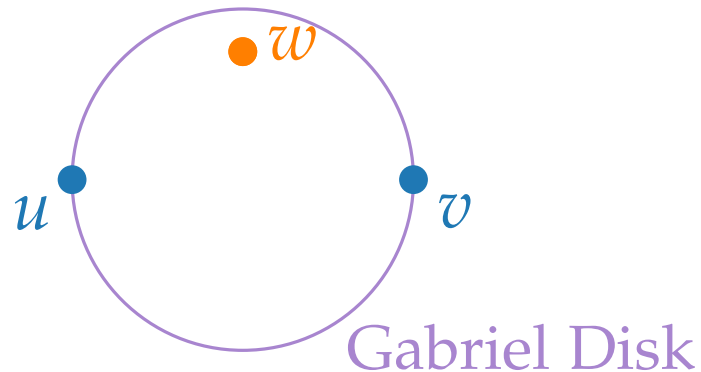
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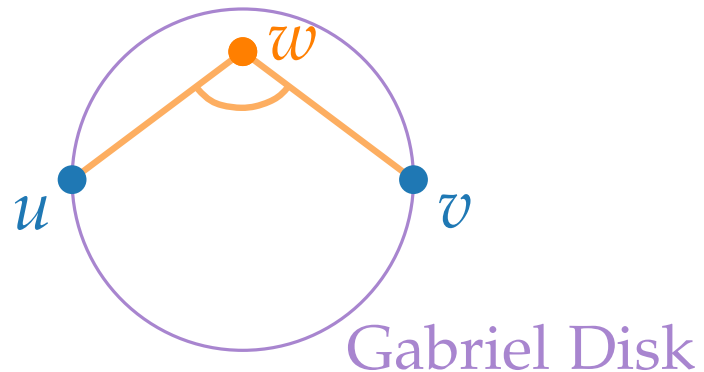
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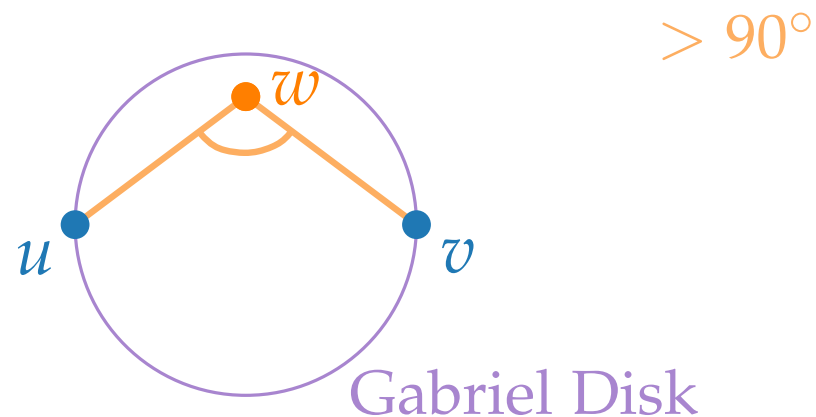
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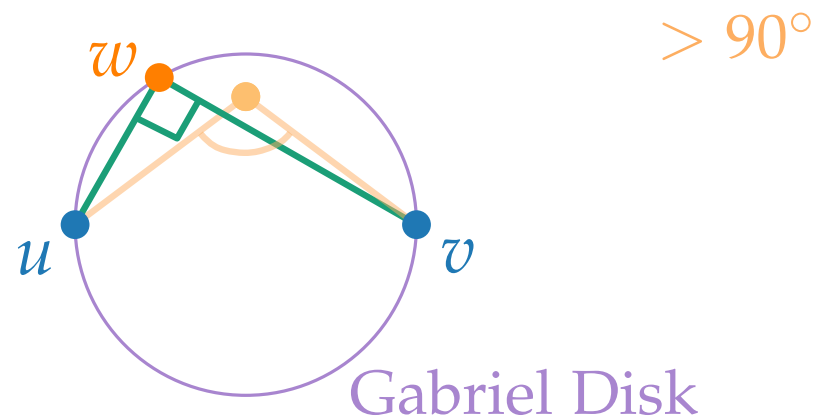
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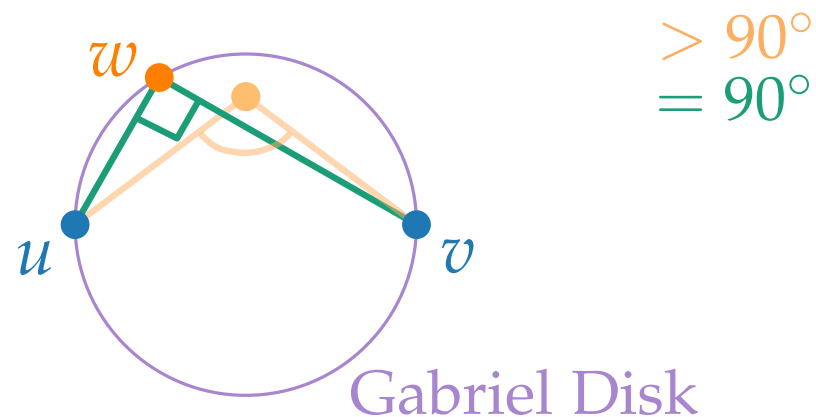
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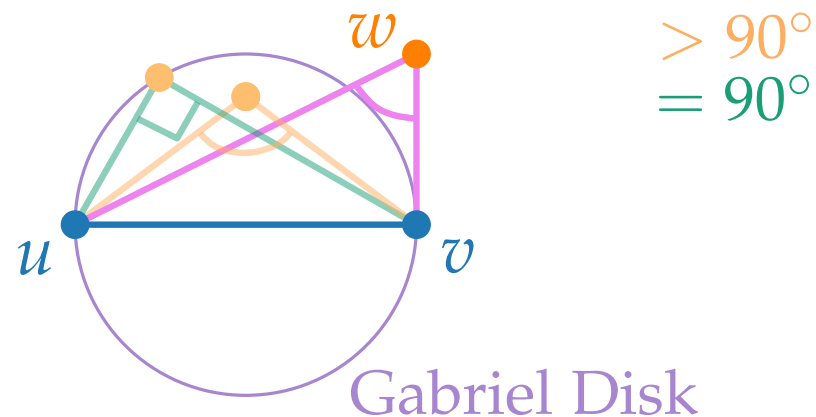
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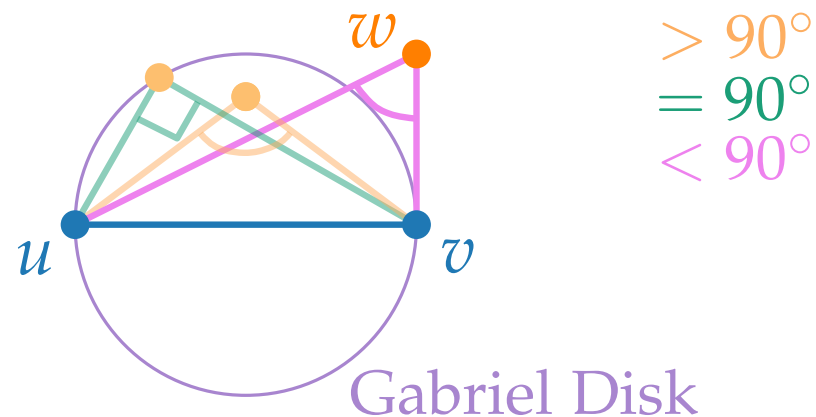
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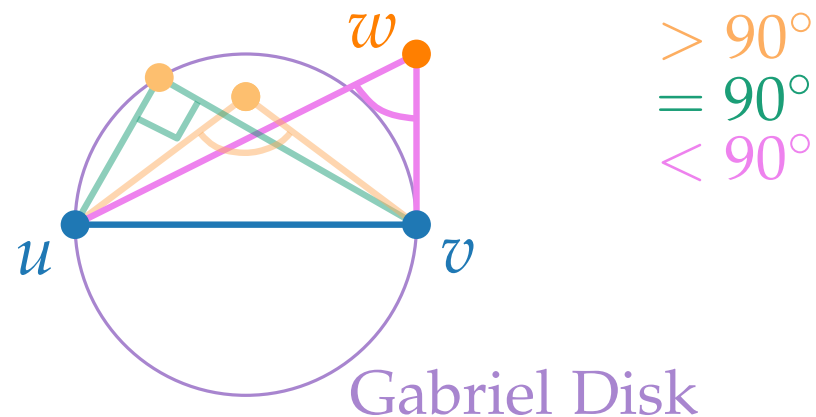
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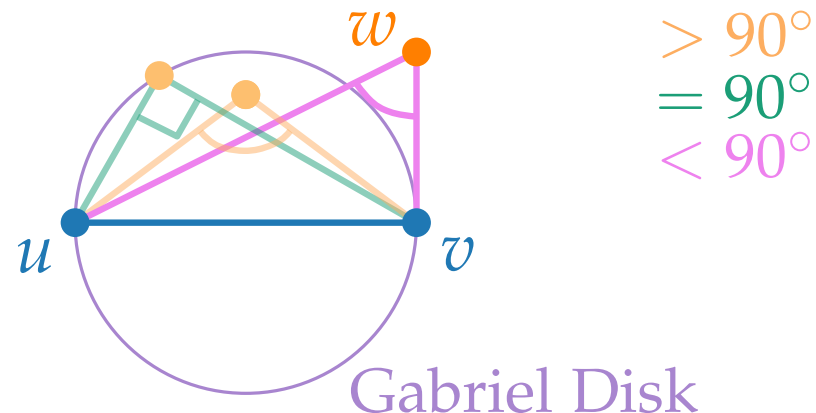
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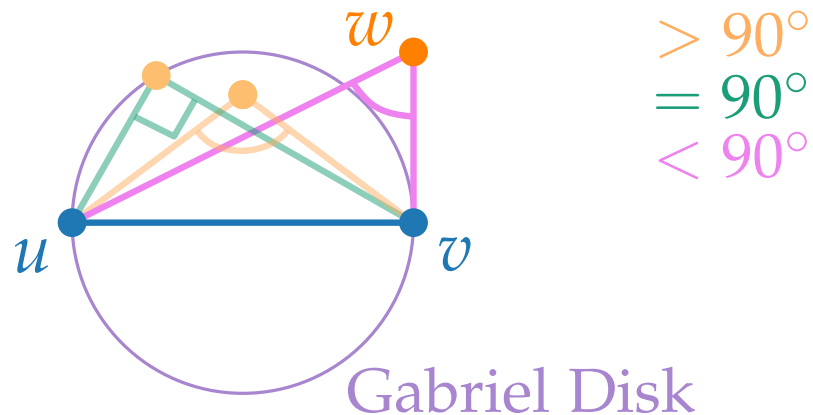
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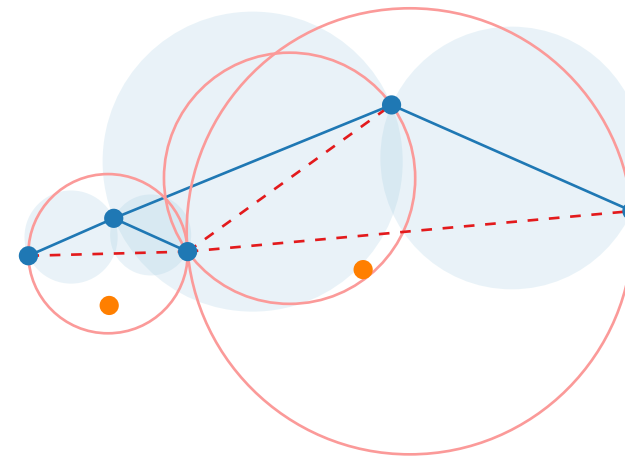
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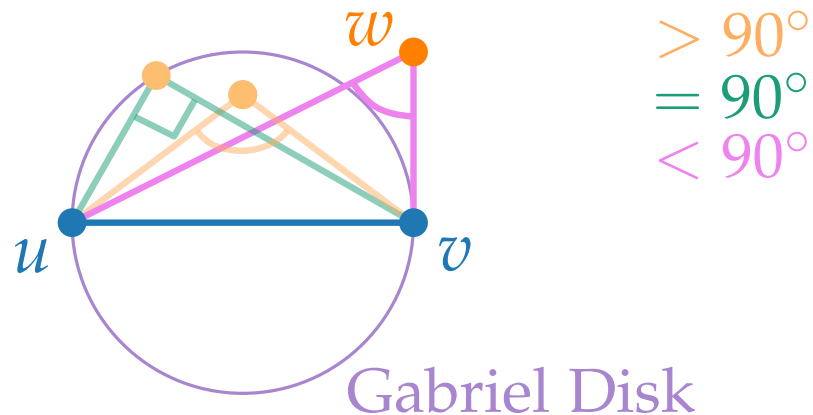
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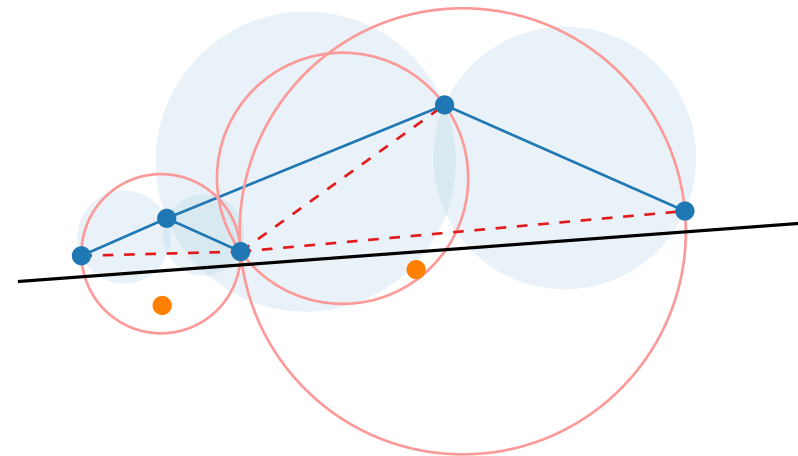
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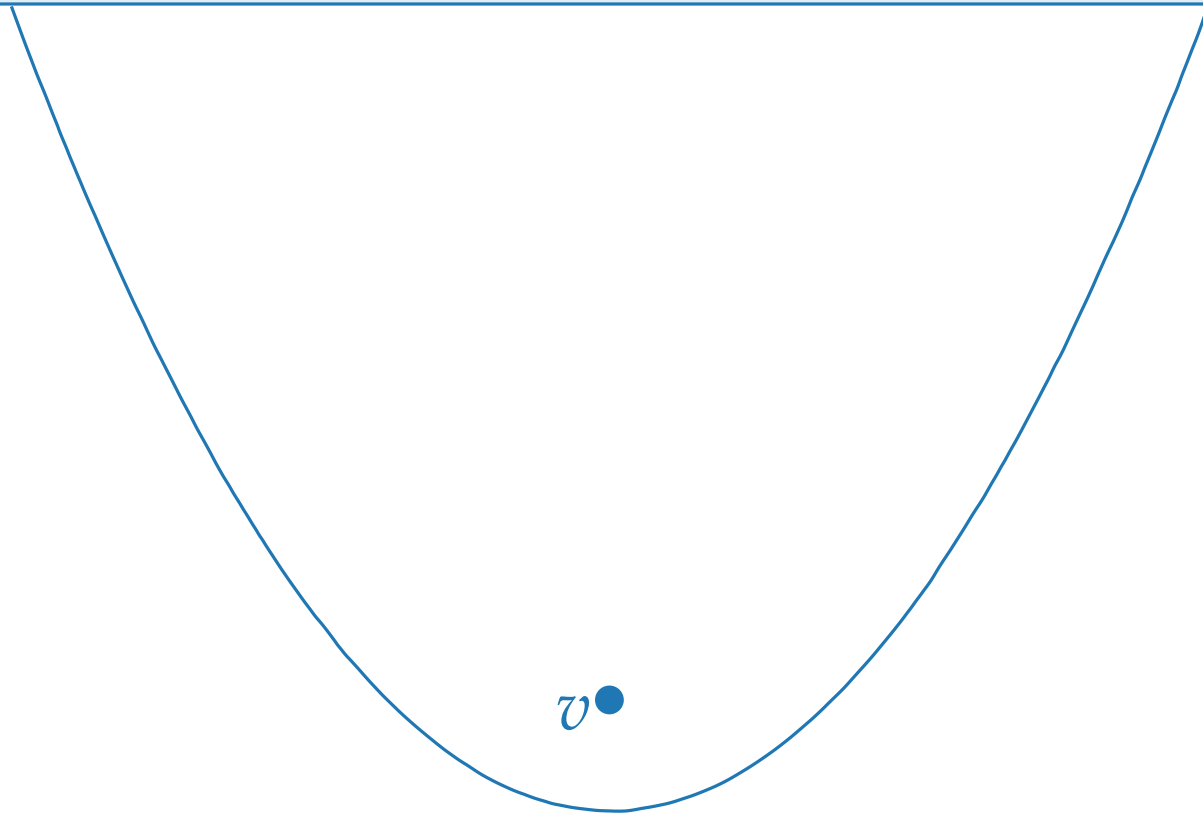
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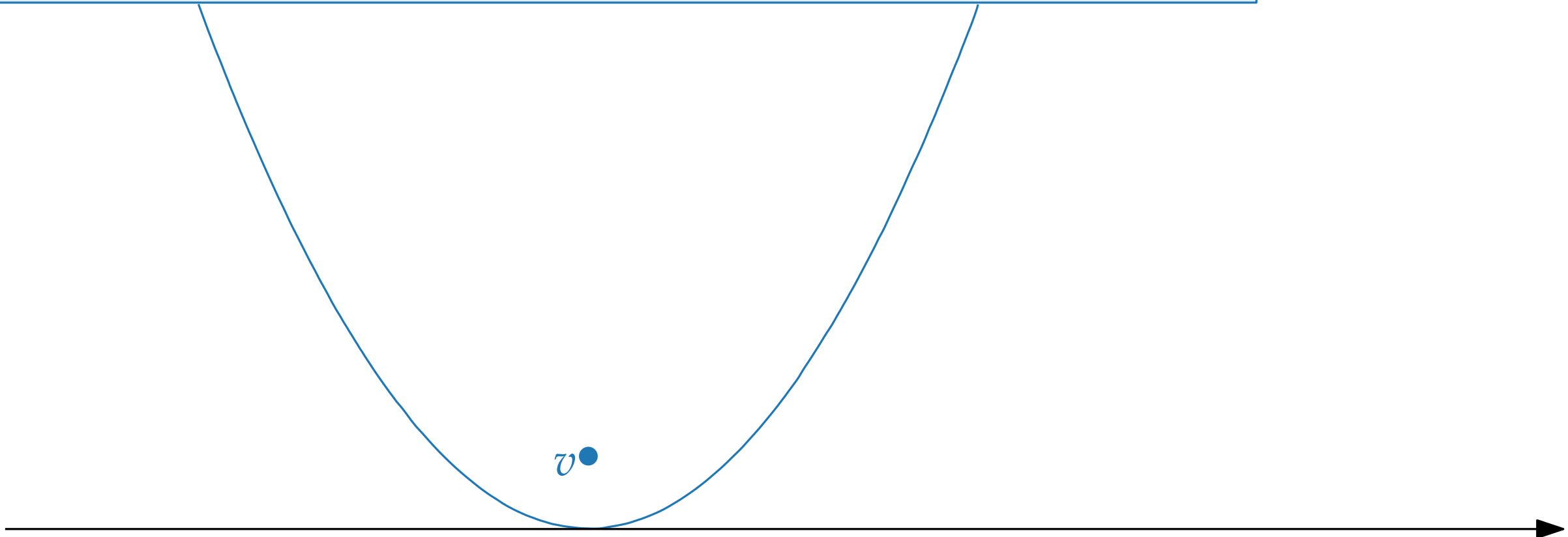


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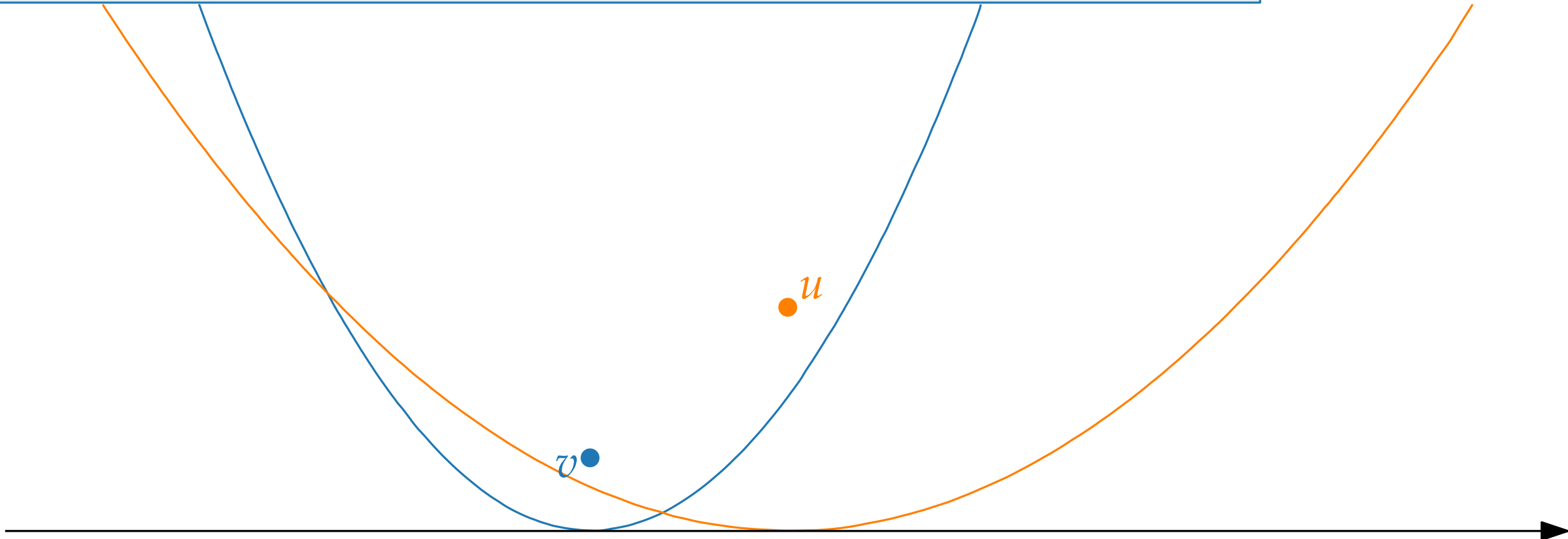


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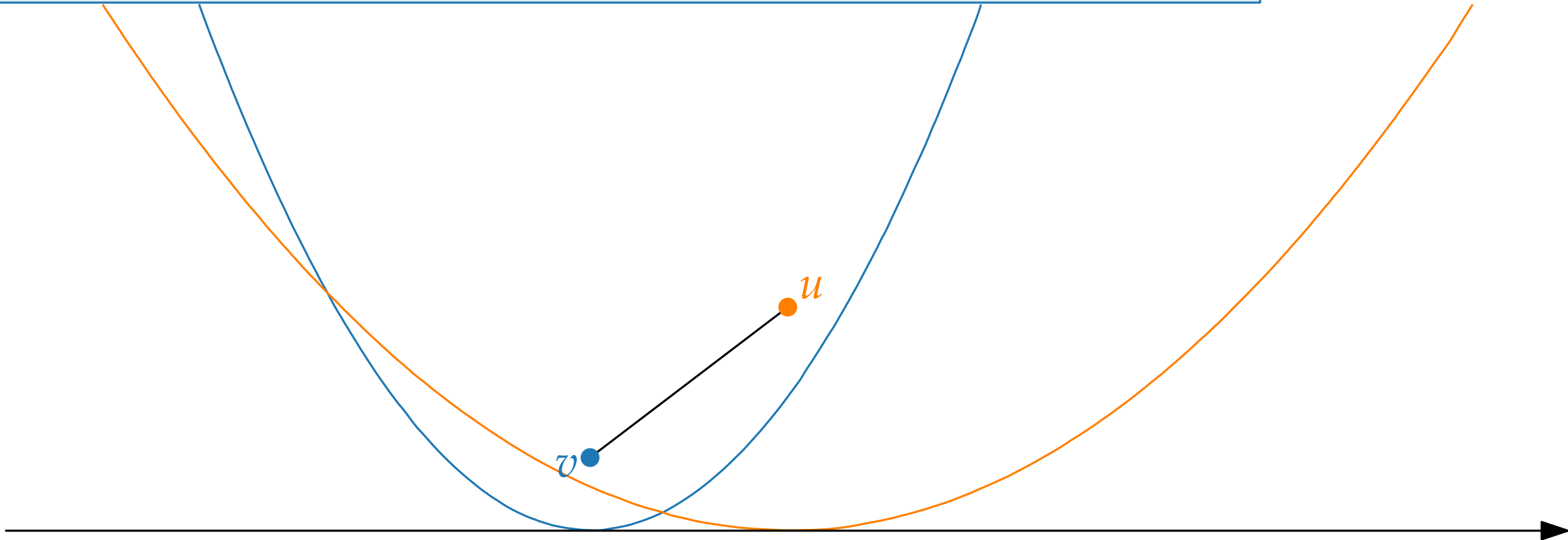


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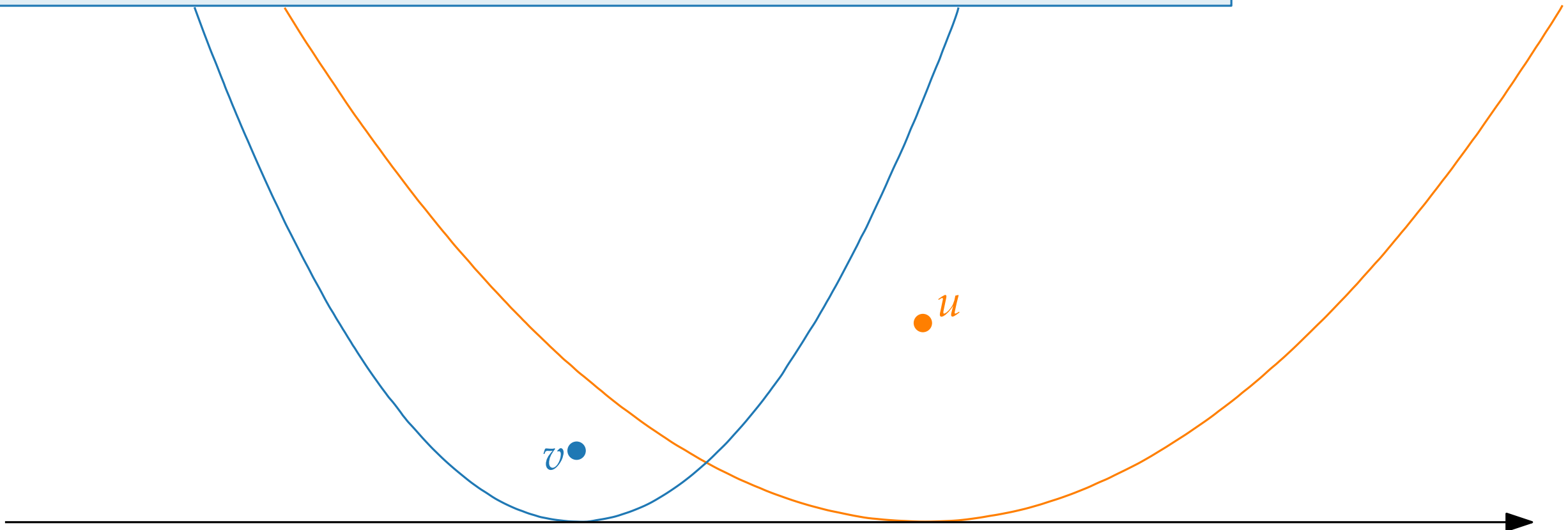


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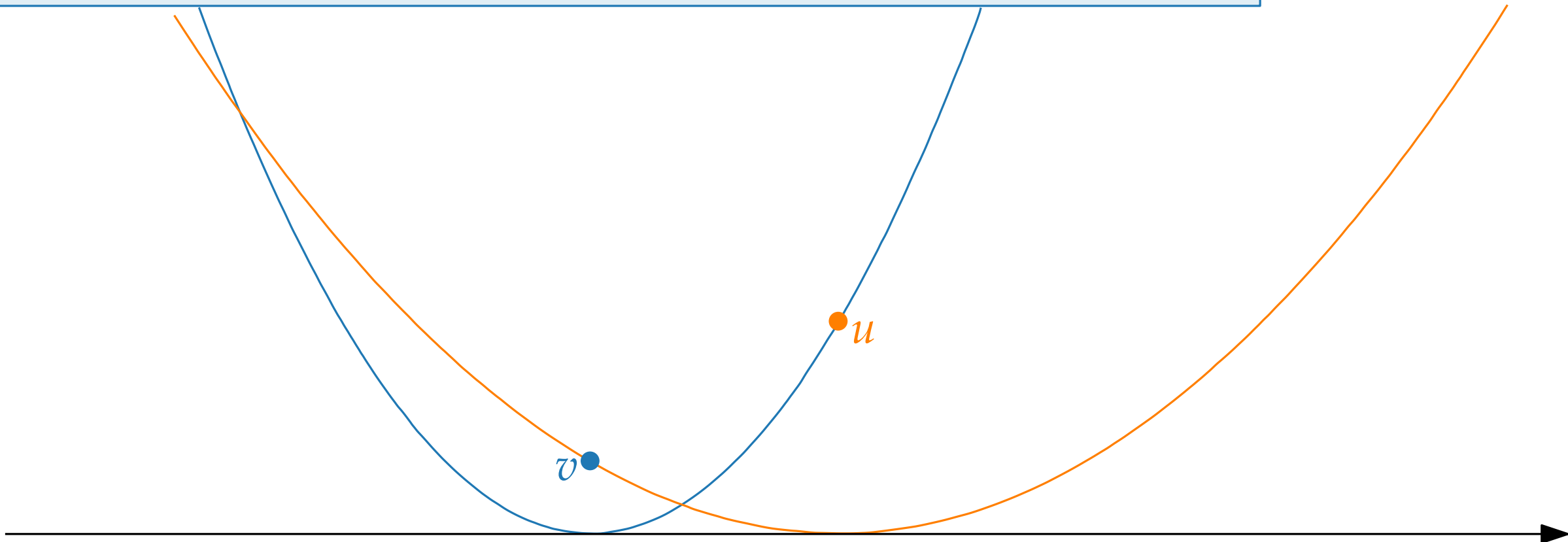


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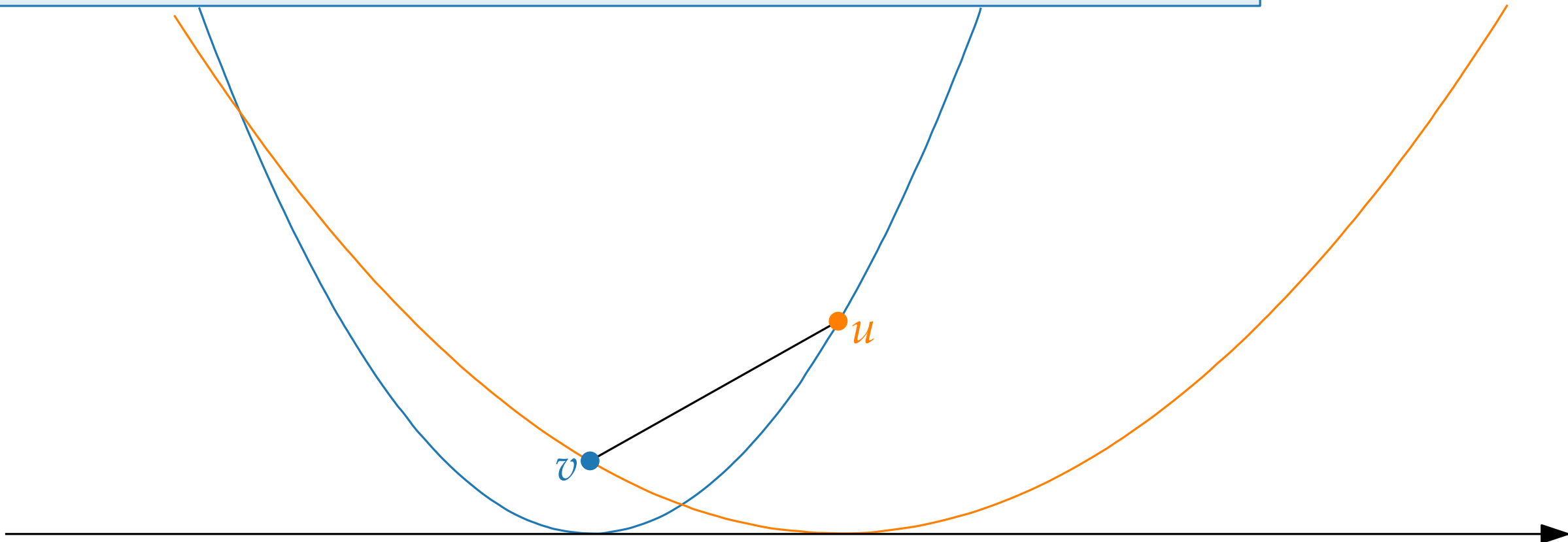


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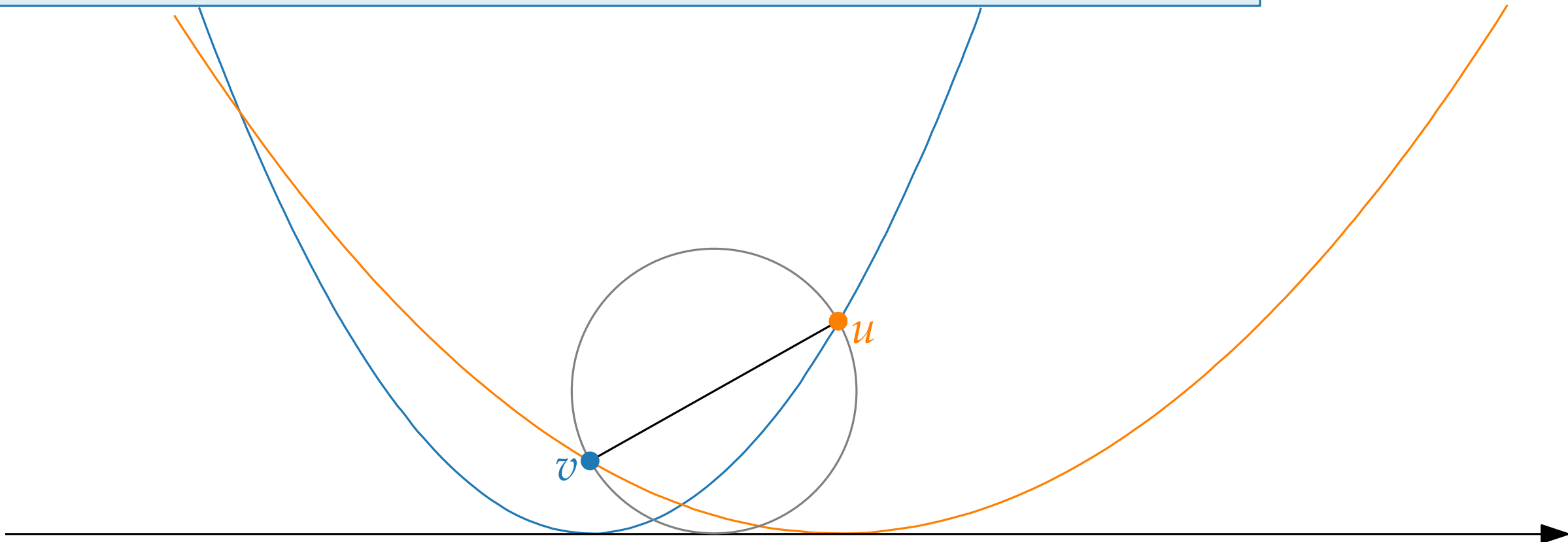


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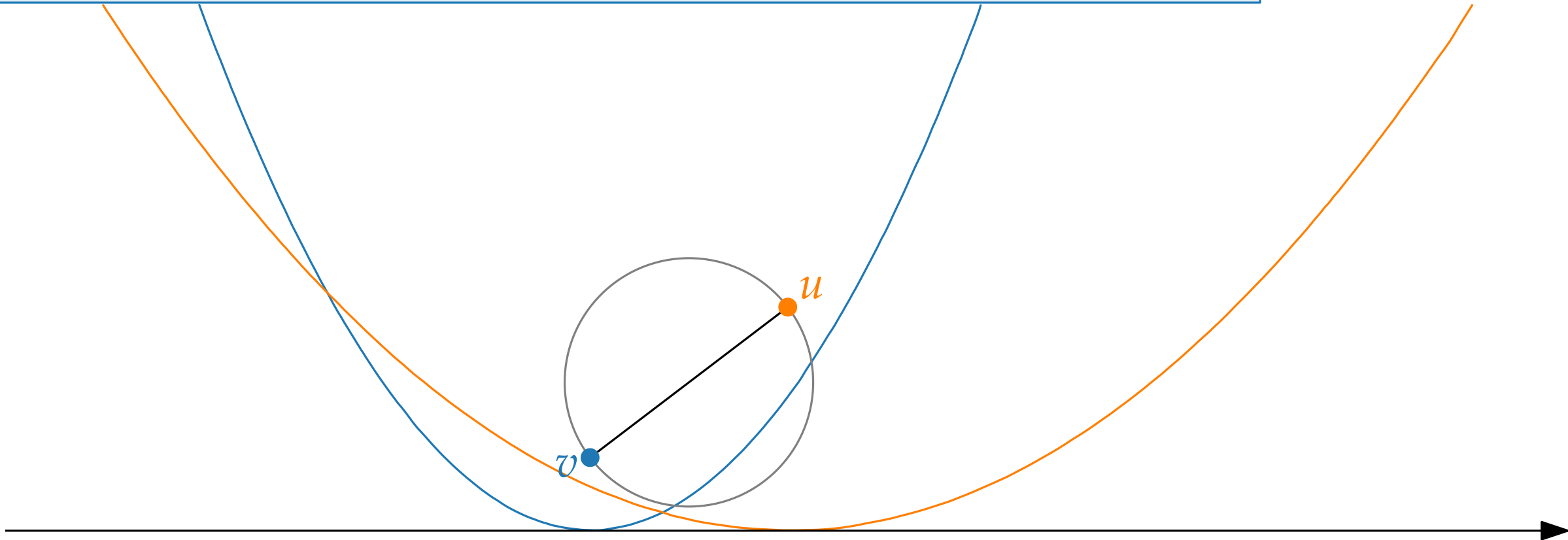


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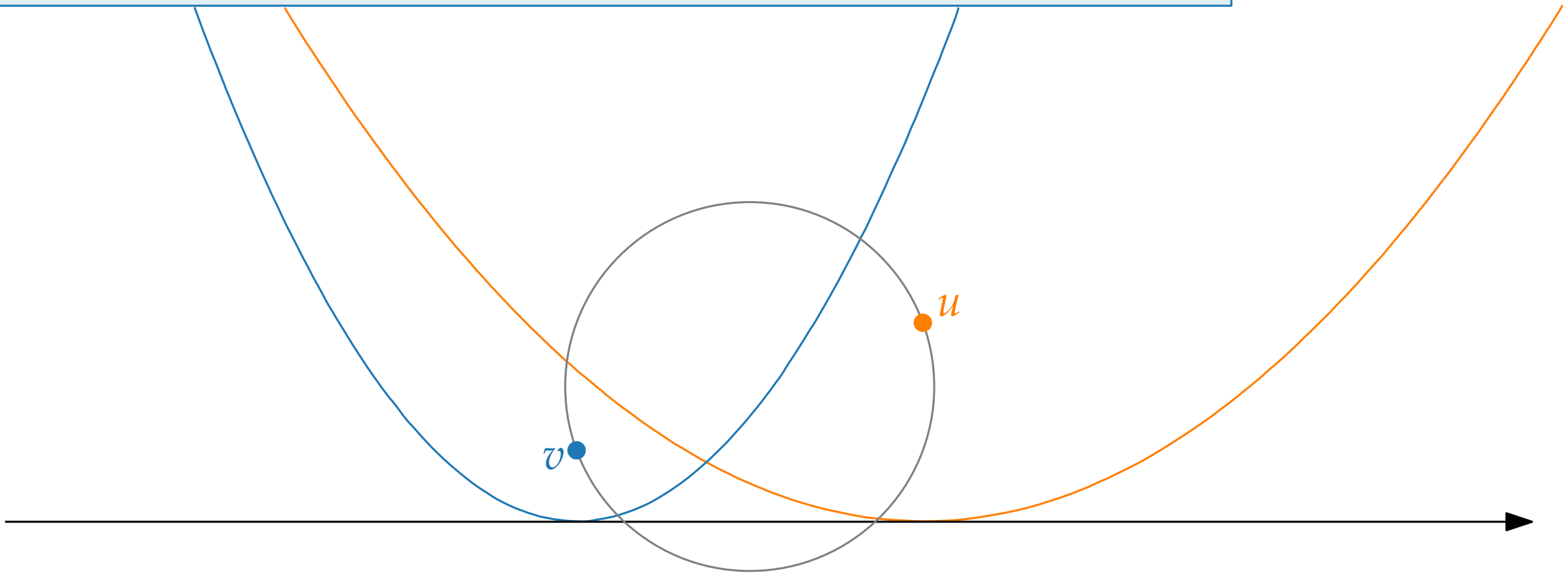


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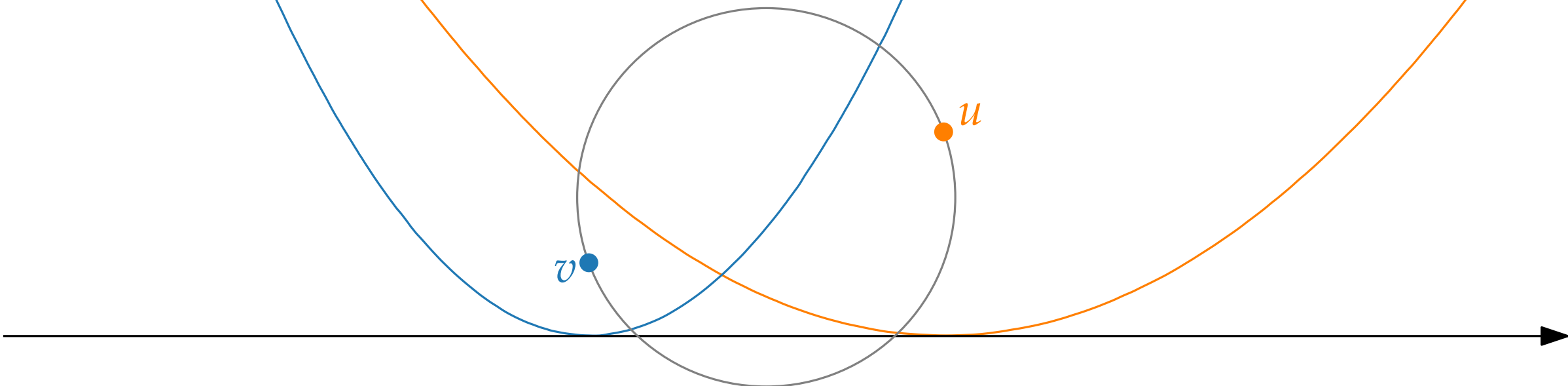
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only disks of *non-edges* cross the x-axis!



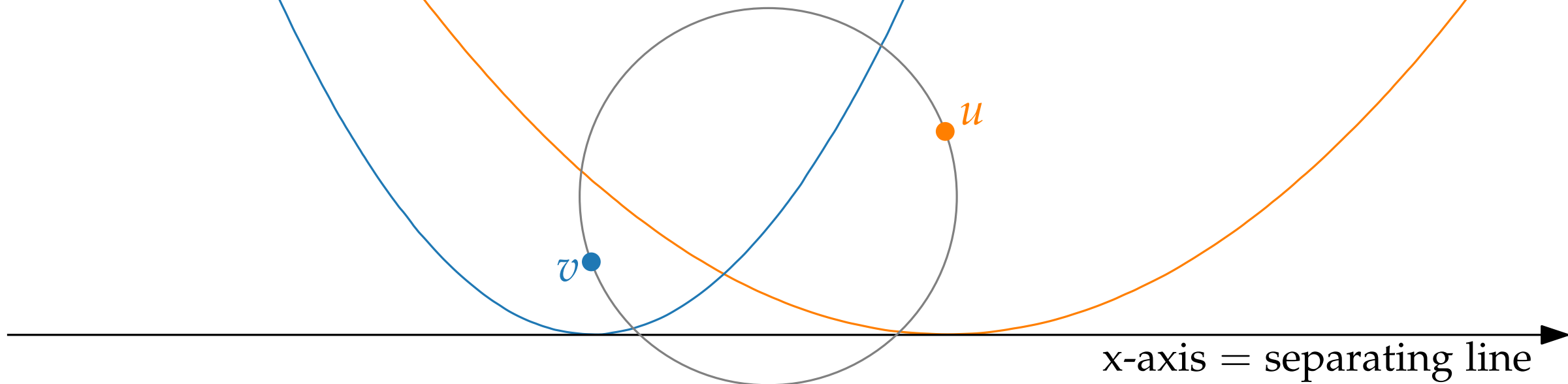
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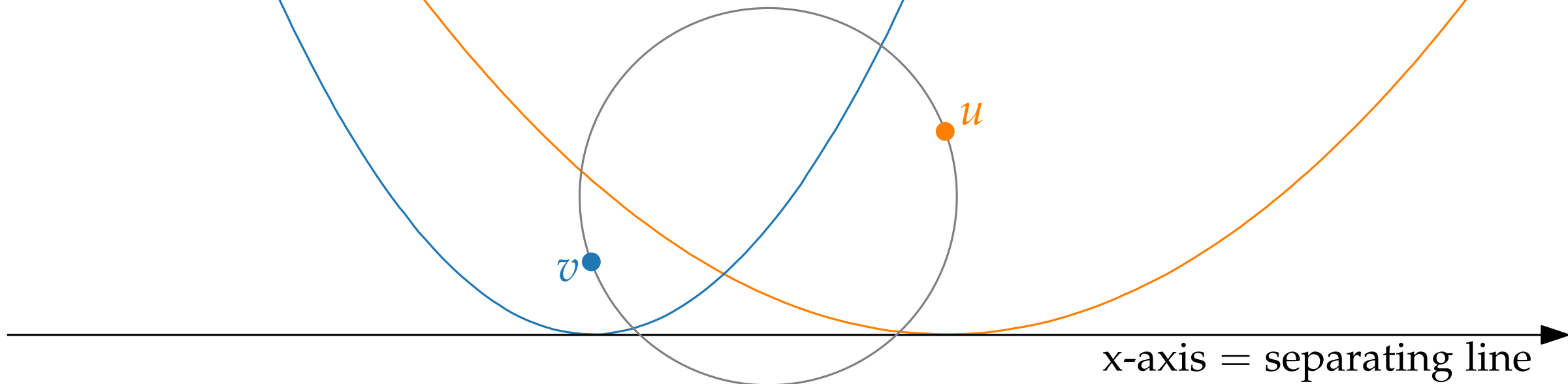
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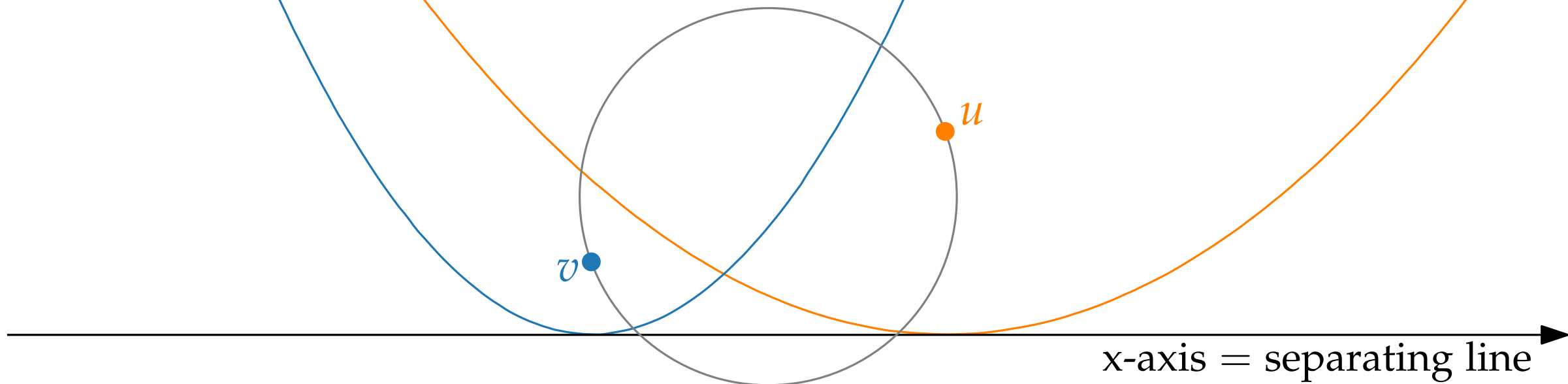
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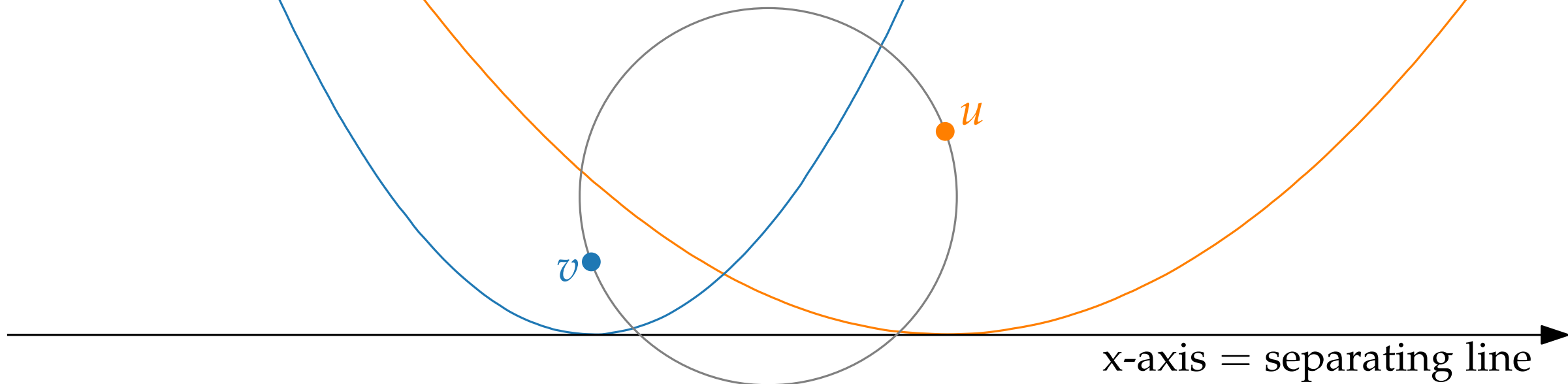
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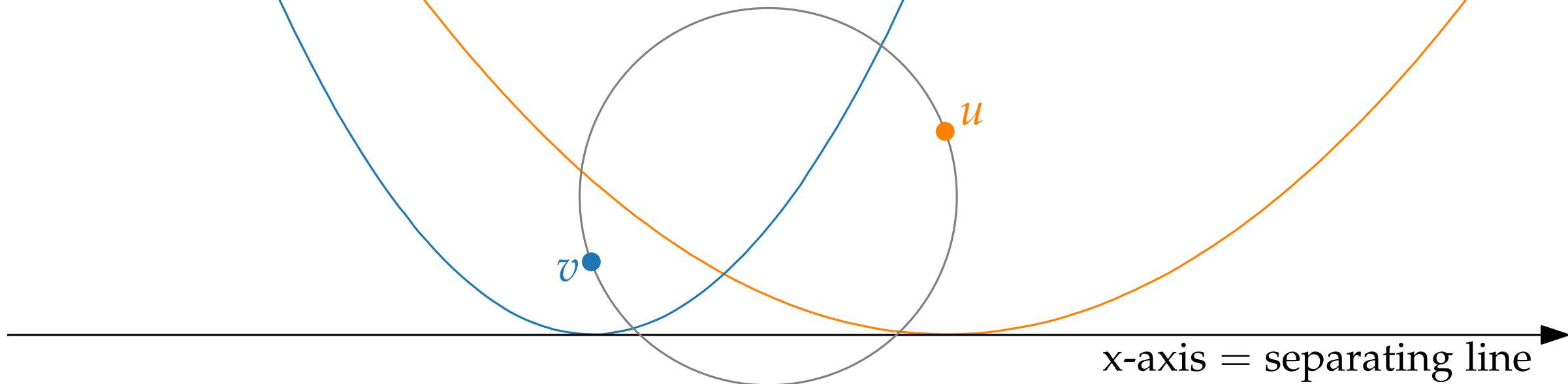
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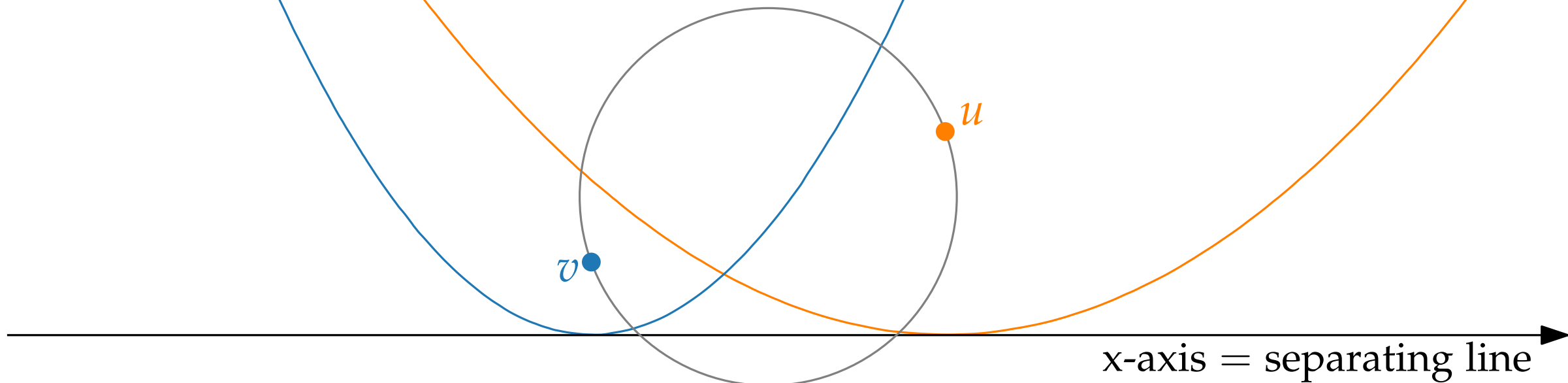
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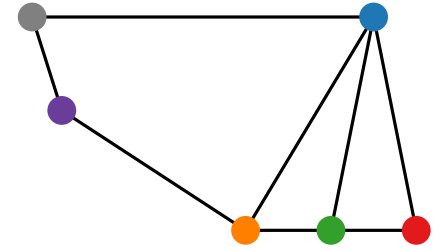


Theorem 1:

G is outerplanar $\Rightarrow G$ admits a border parabola drawing

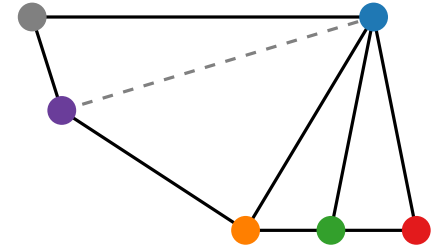
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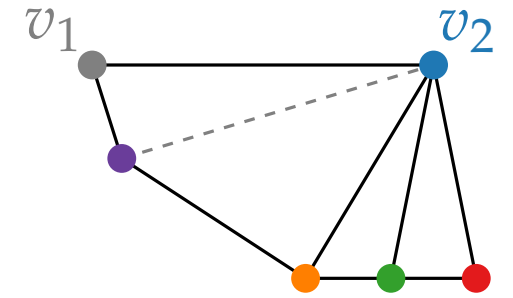
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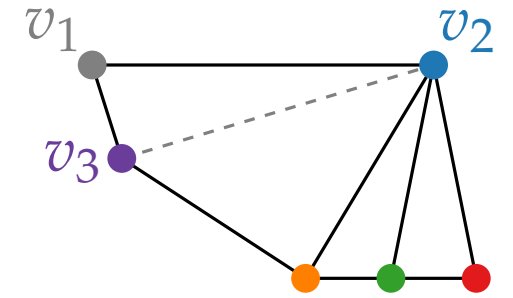
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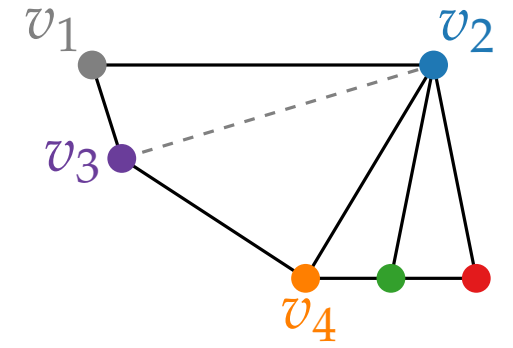
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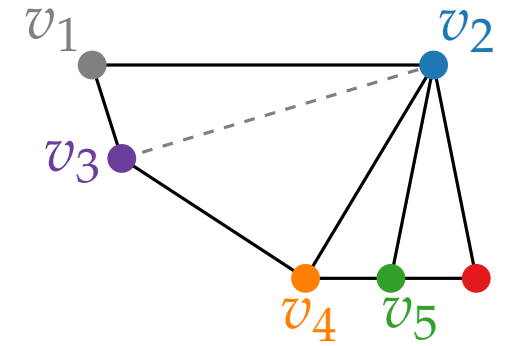
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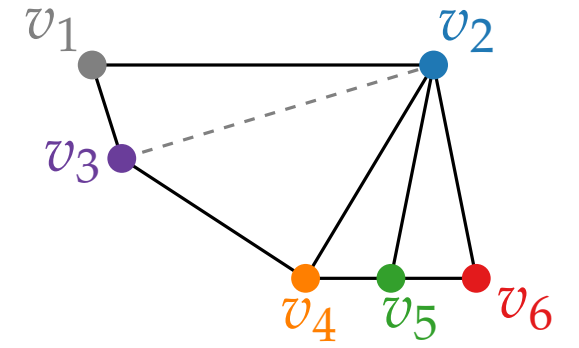
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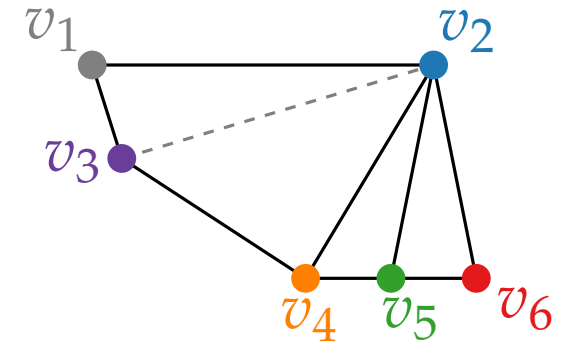
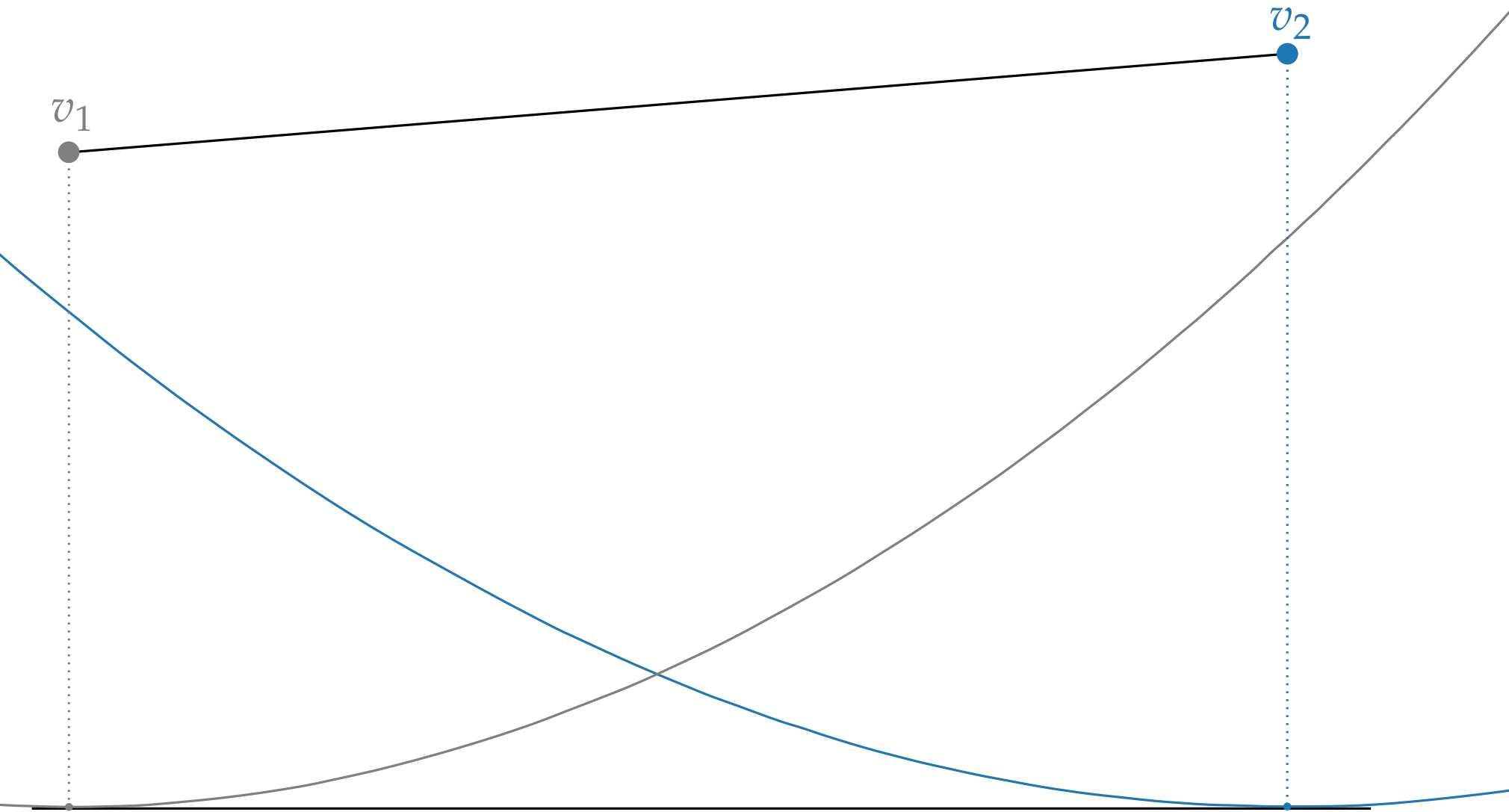
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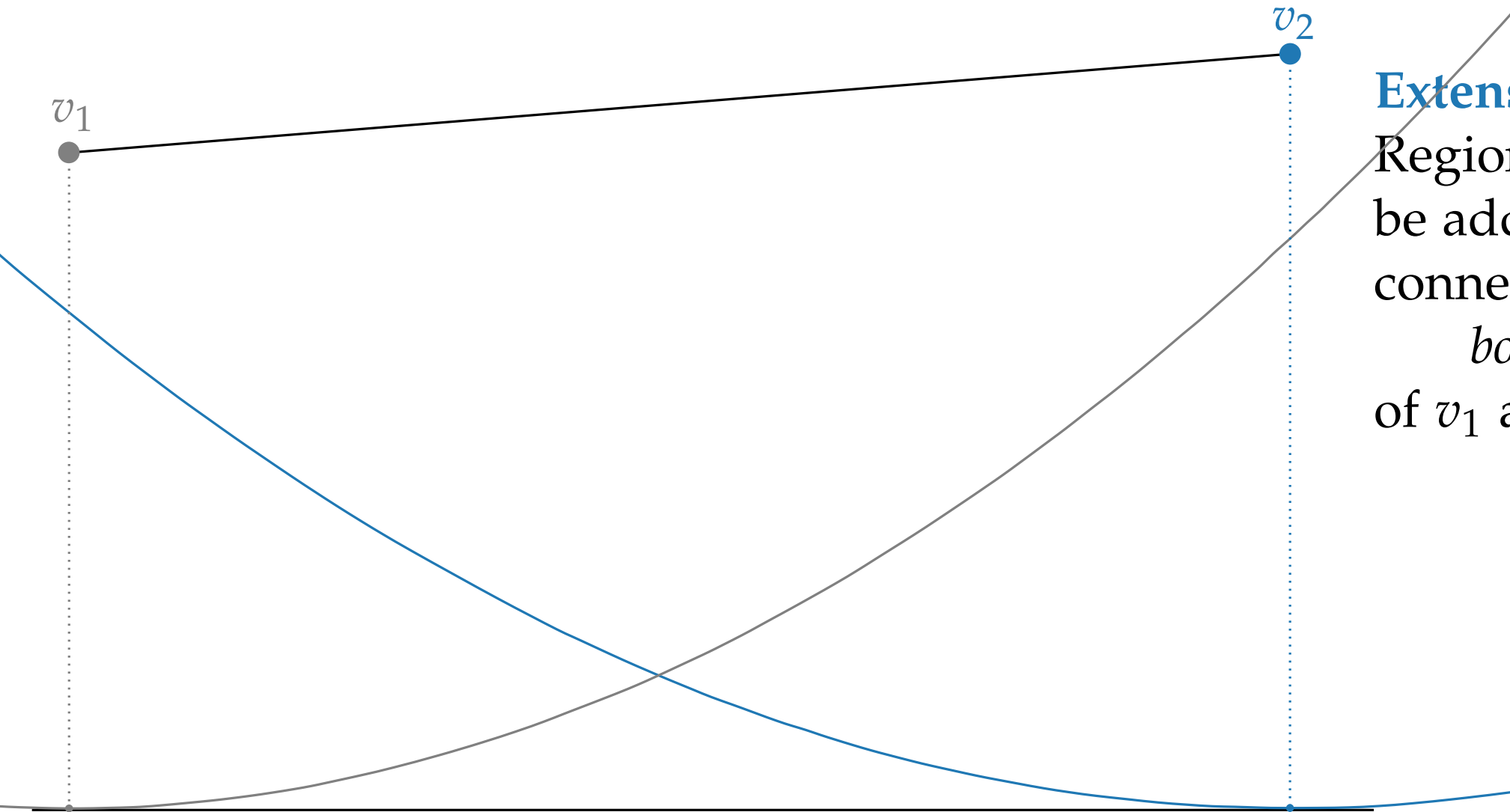
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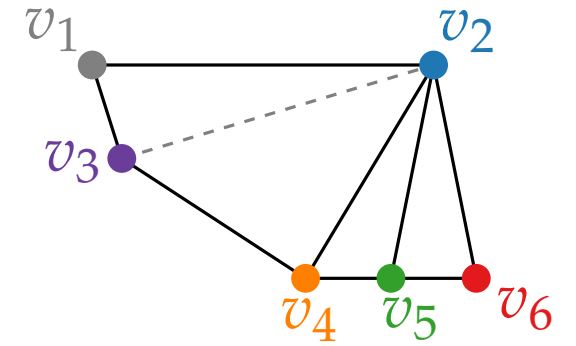
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Extension zone

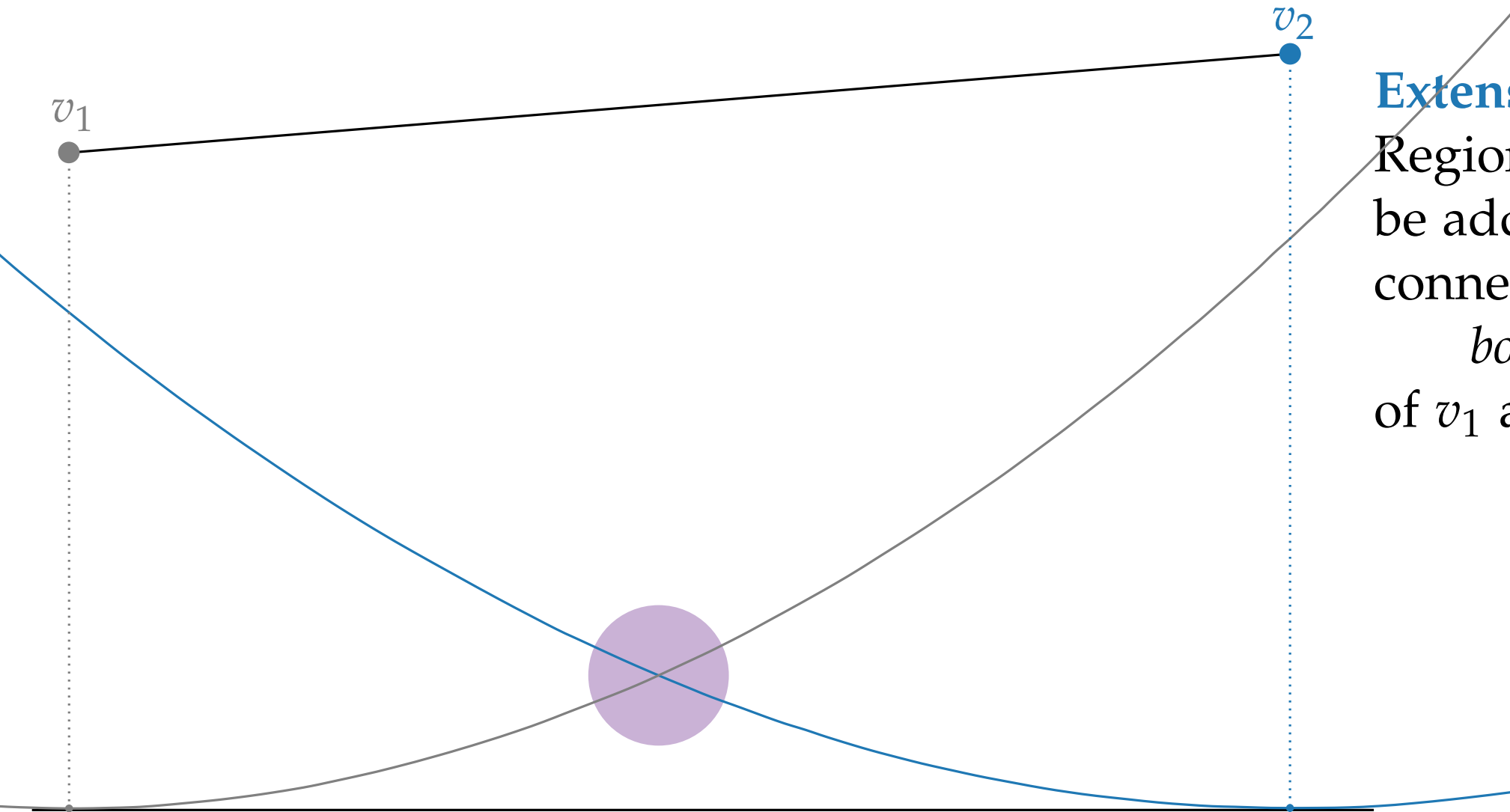
Region in which a vertex can be added, such that it is connected to

both, either or neither
of v_1 and v_2



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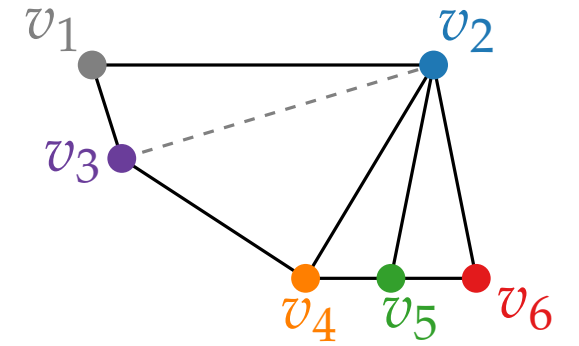
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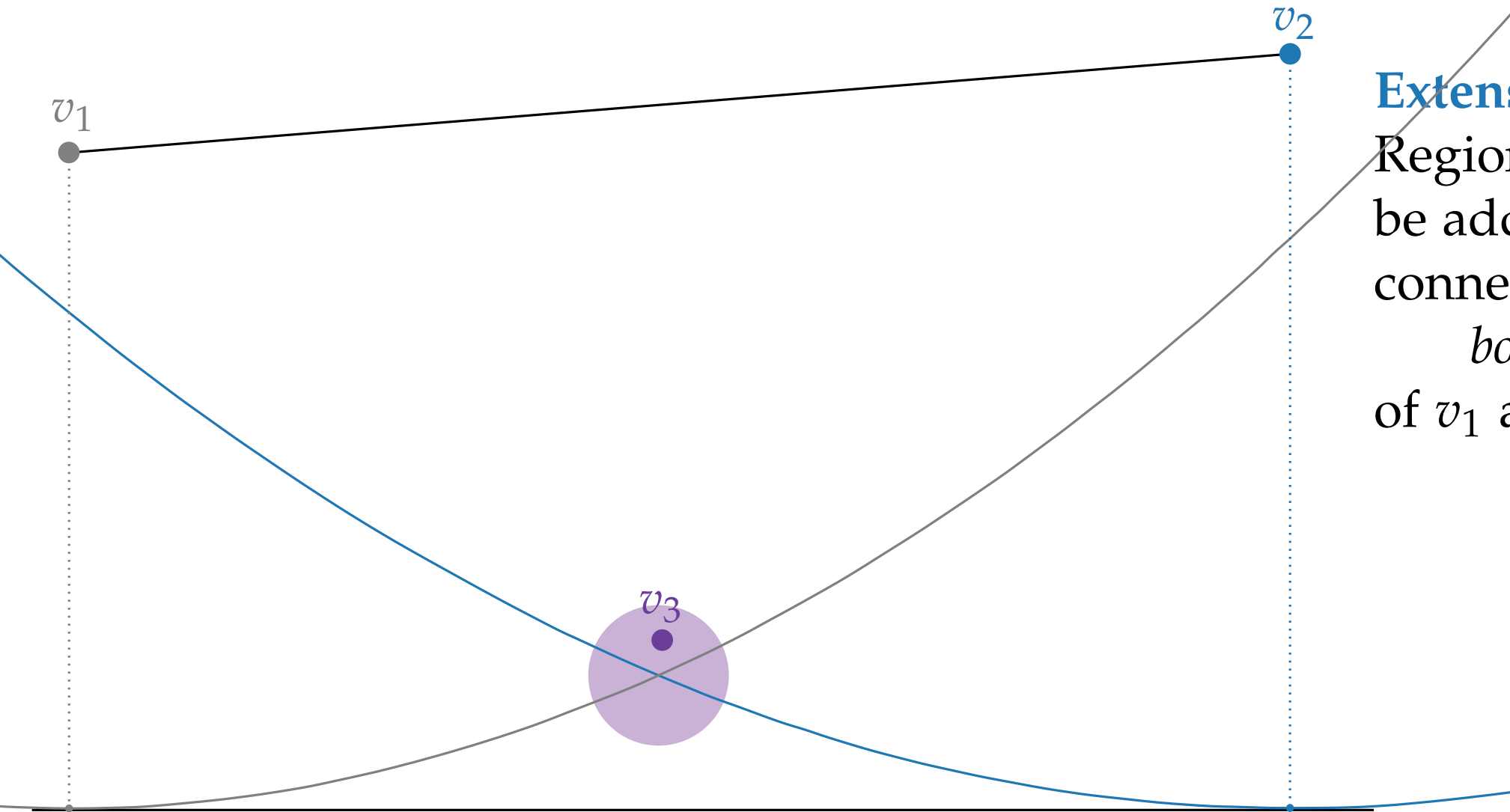
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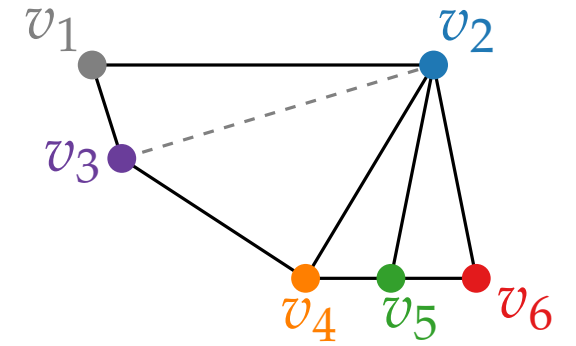
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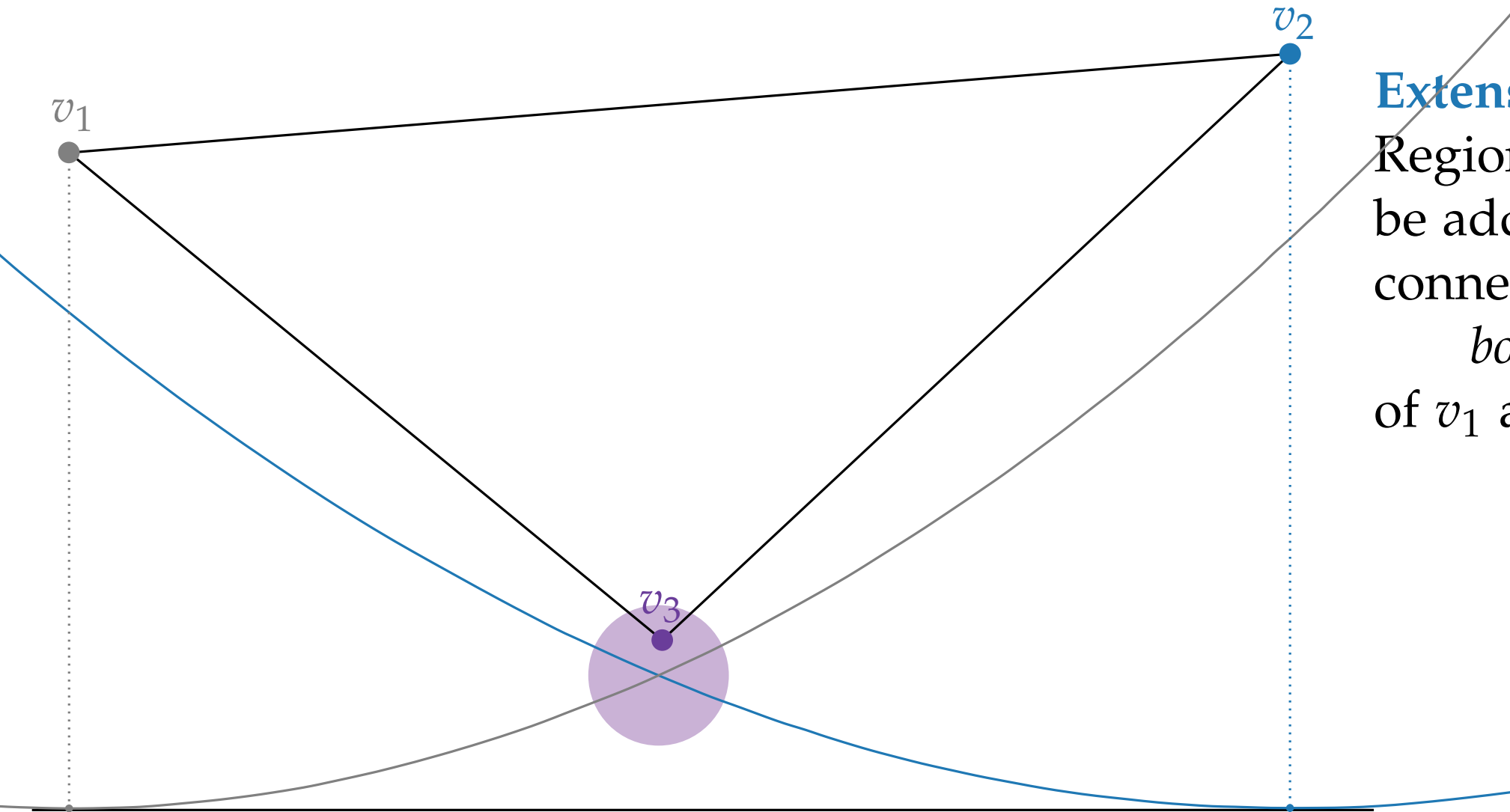
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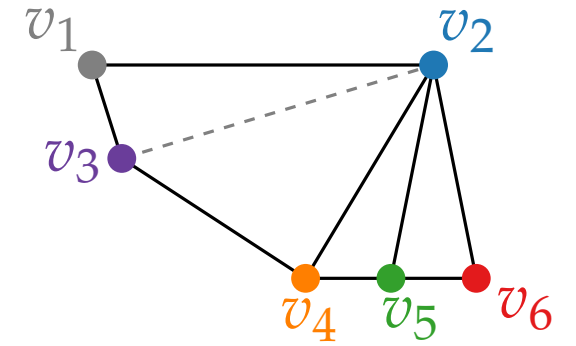
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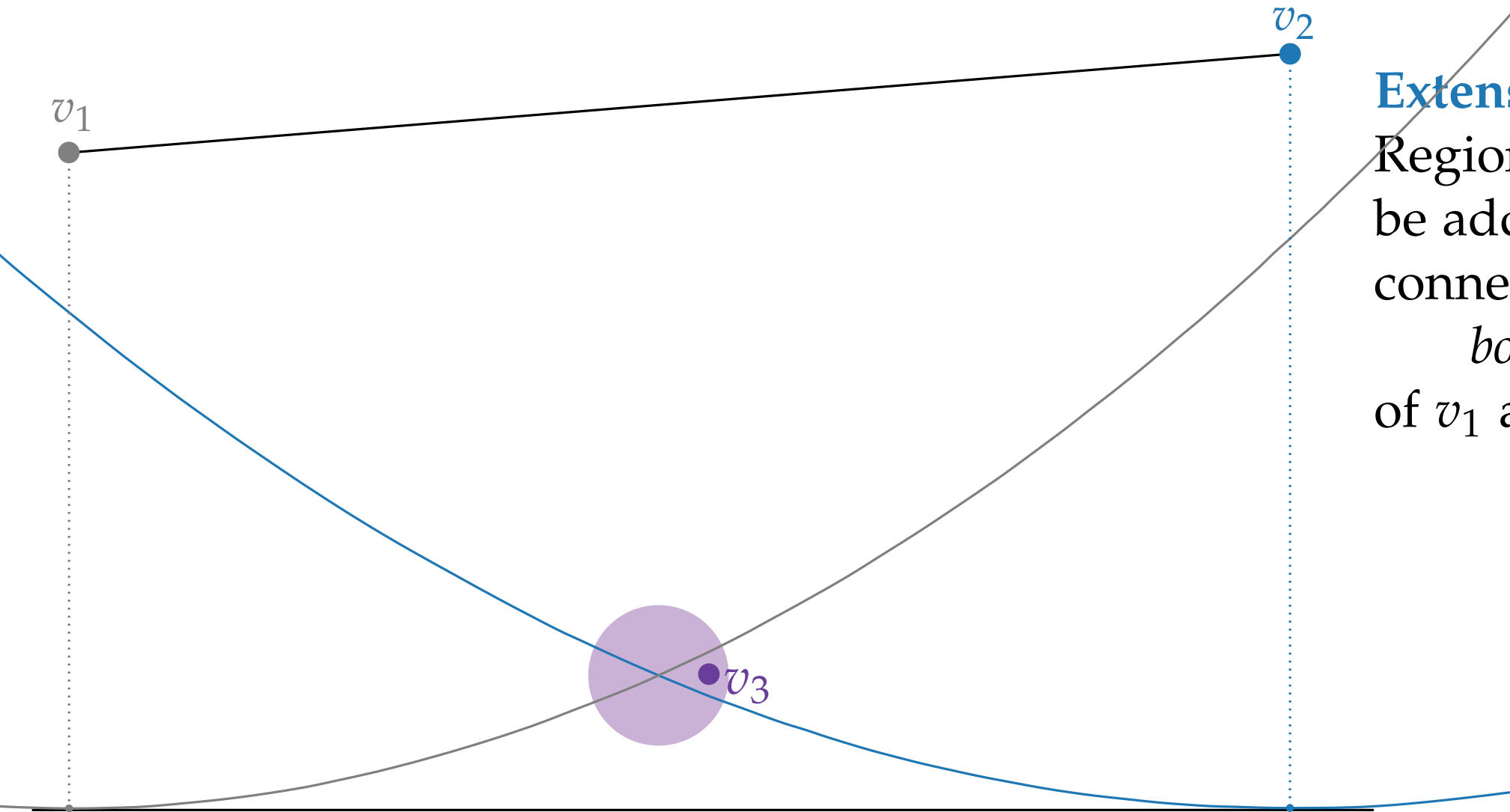
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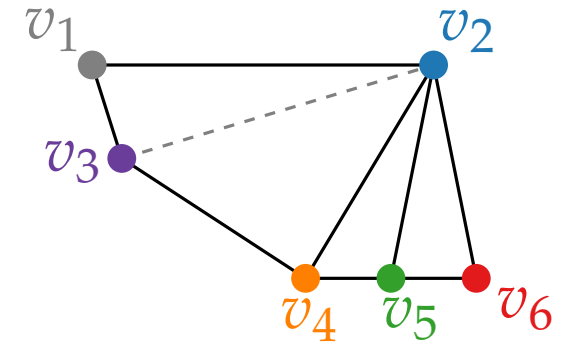
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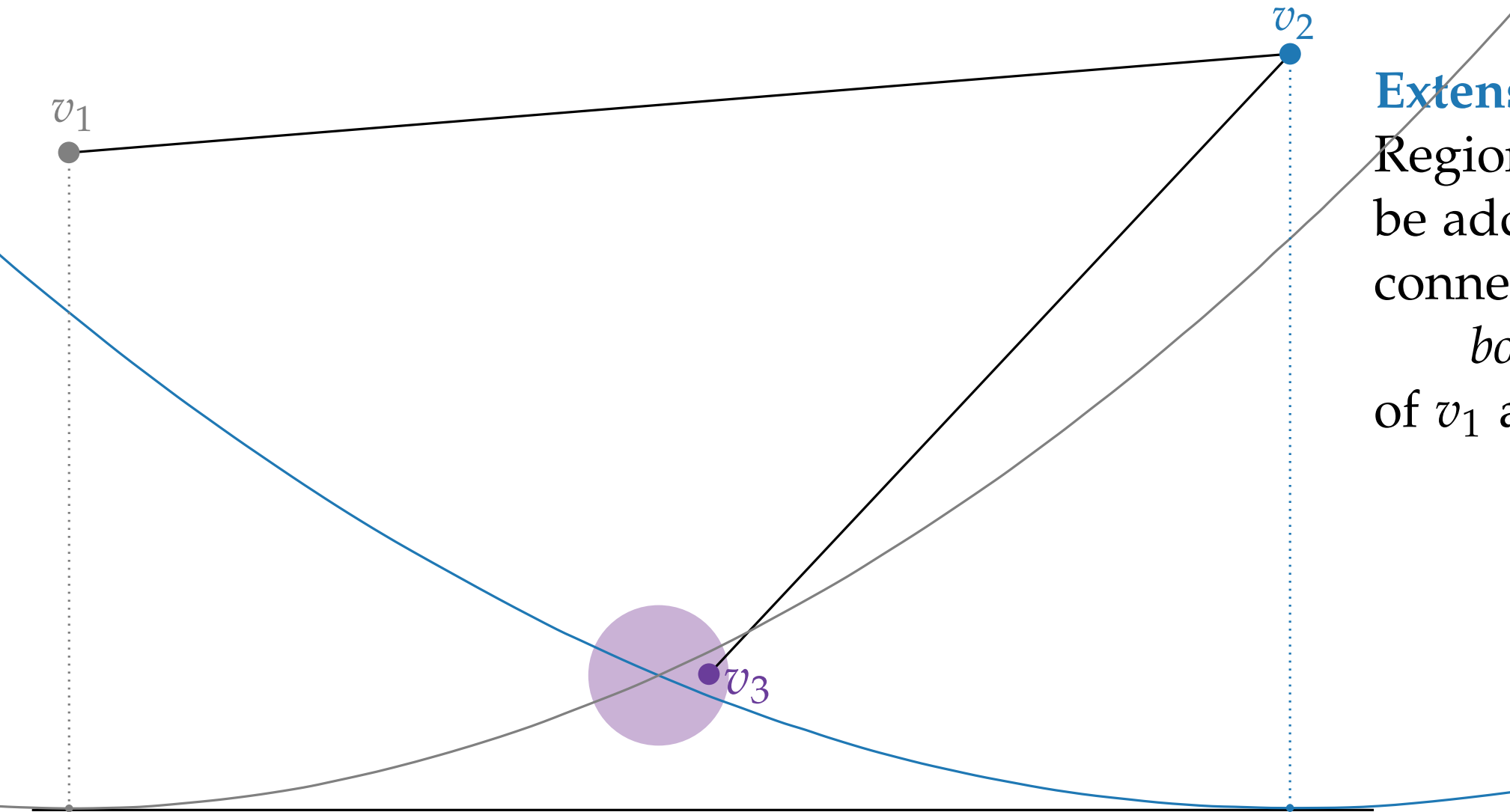
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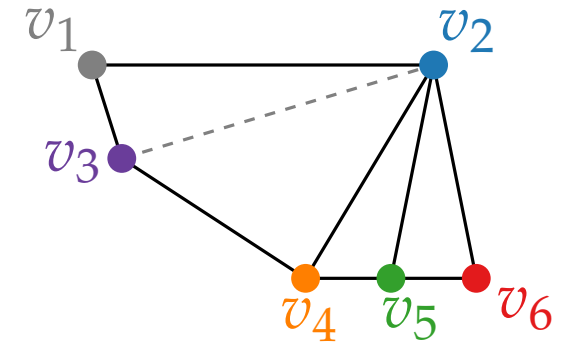
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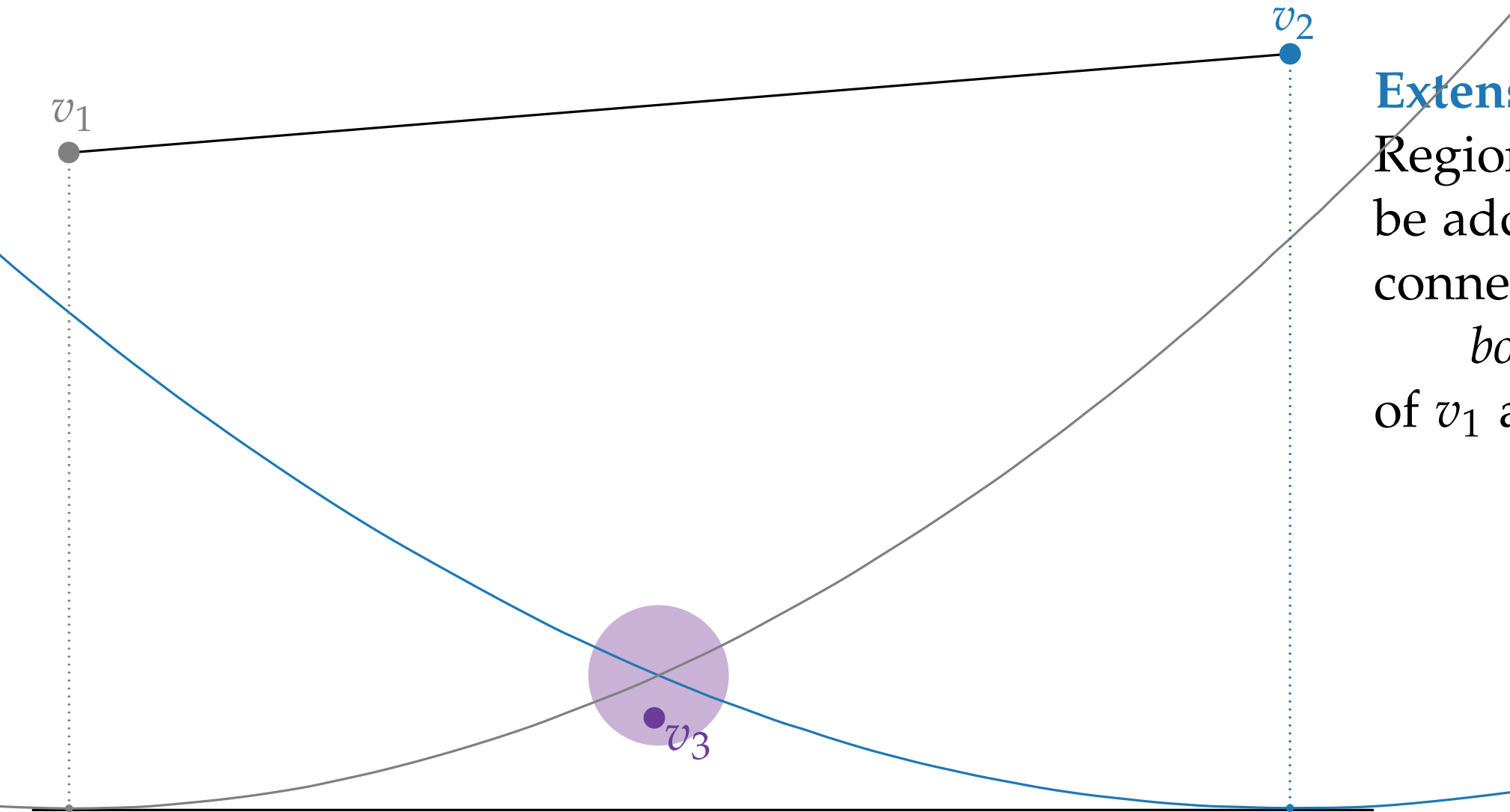
Region in which a vertex can be added, such that it is connected to

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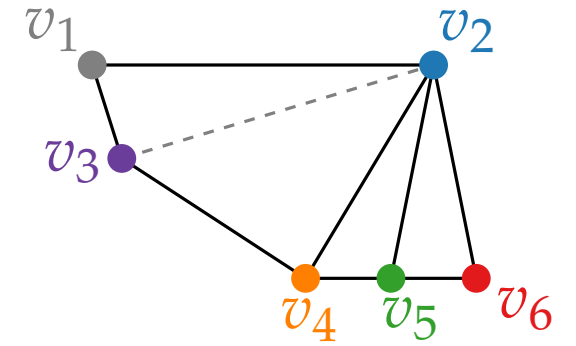
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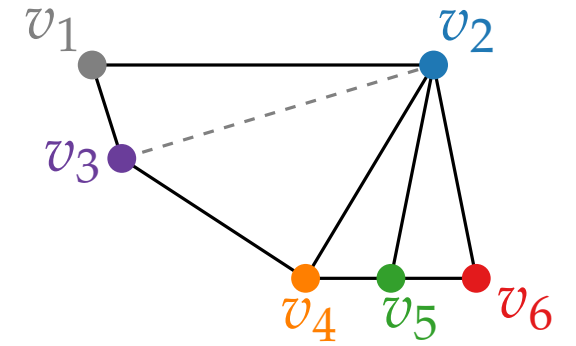
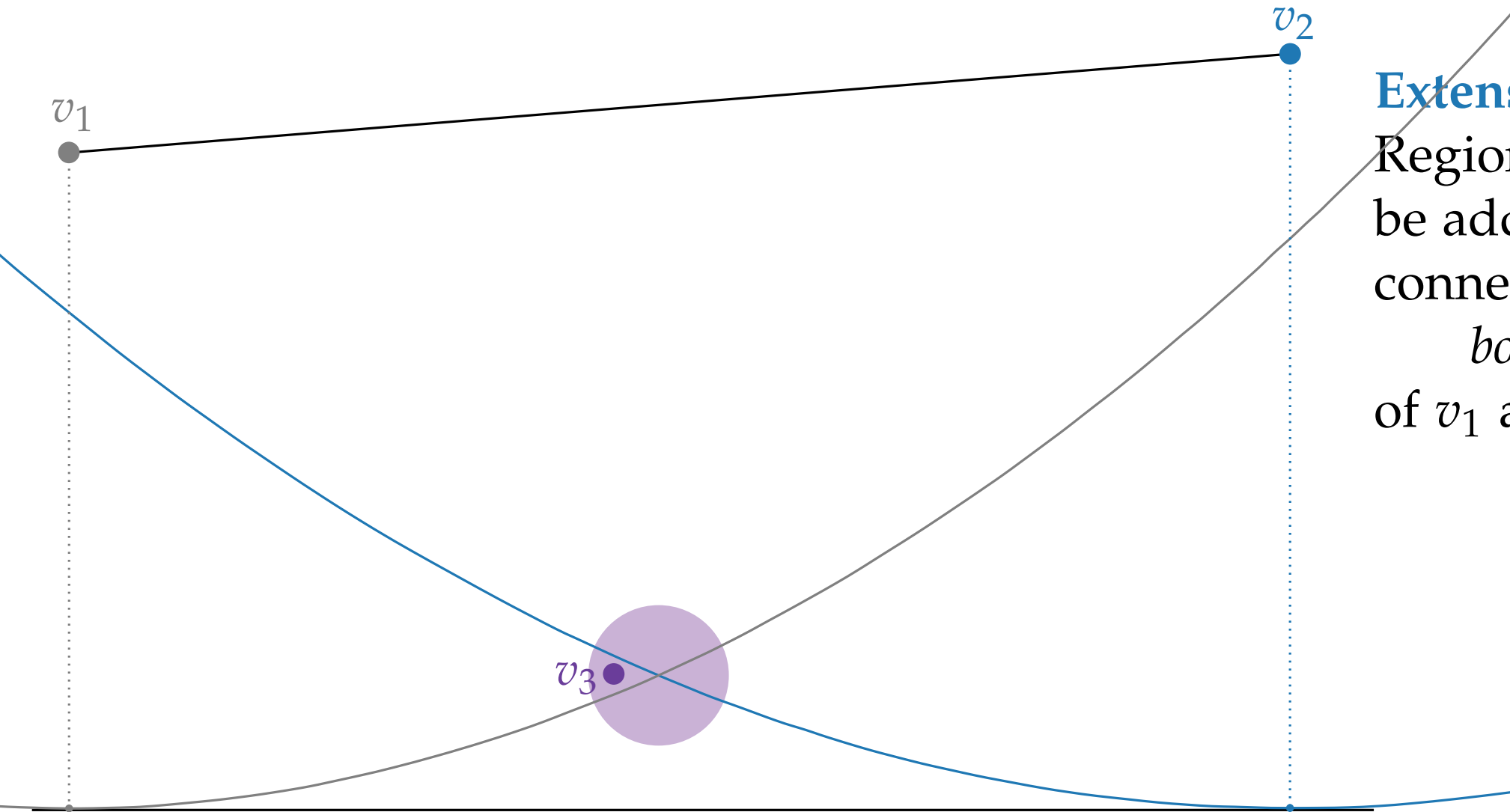
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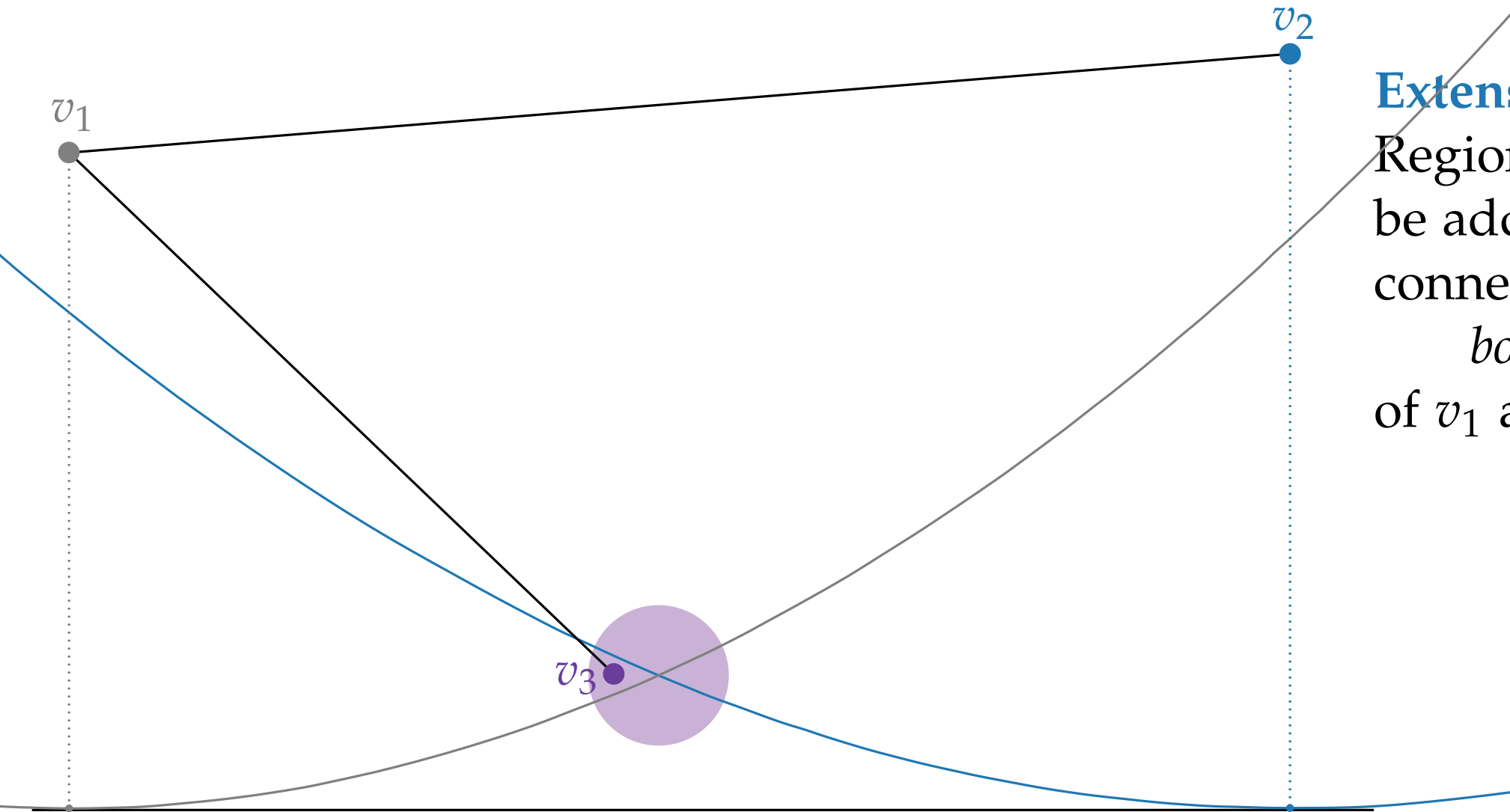
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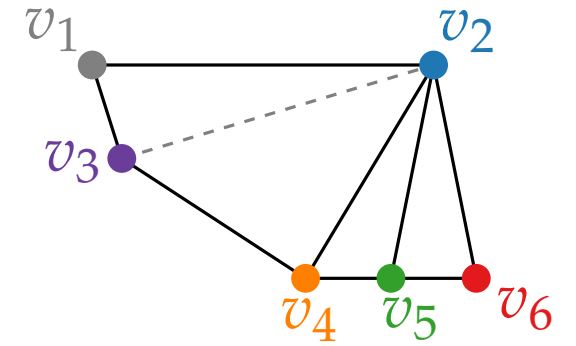
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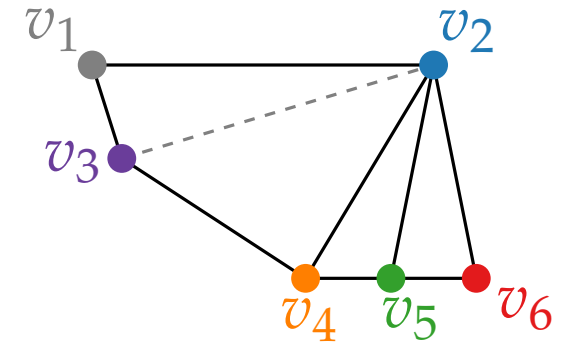
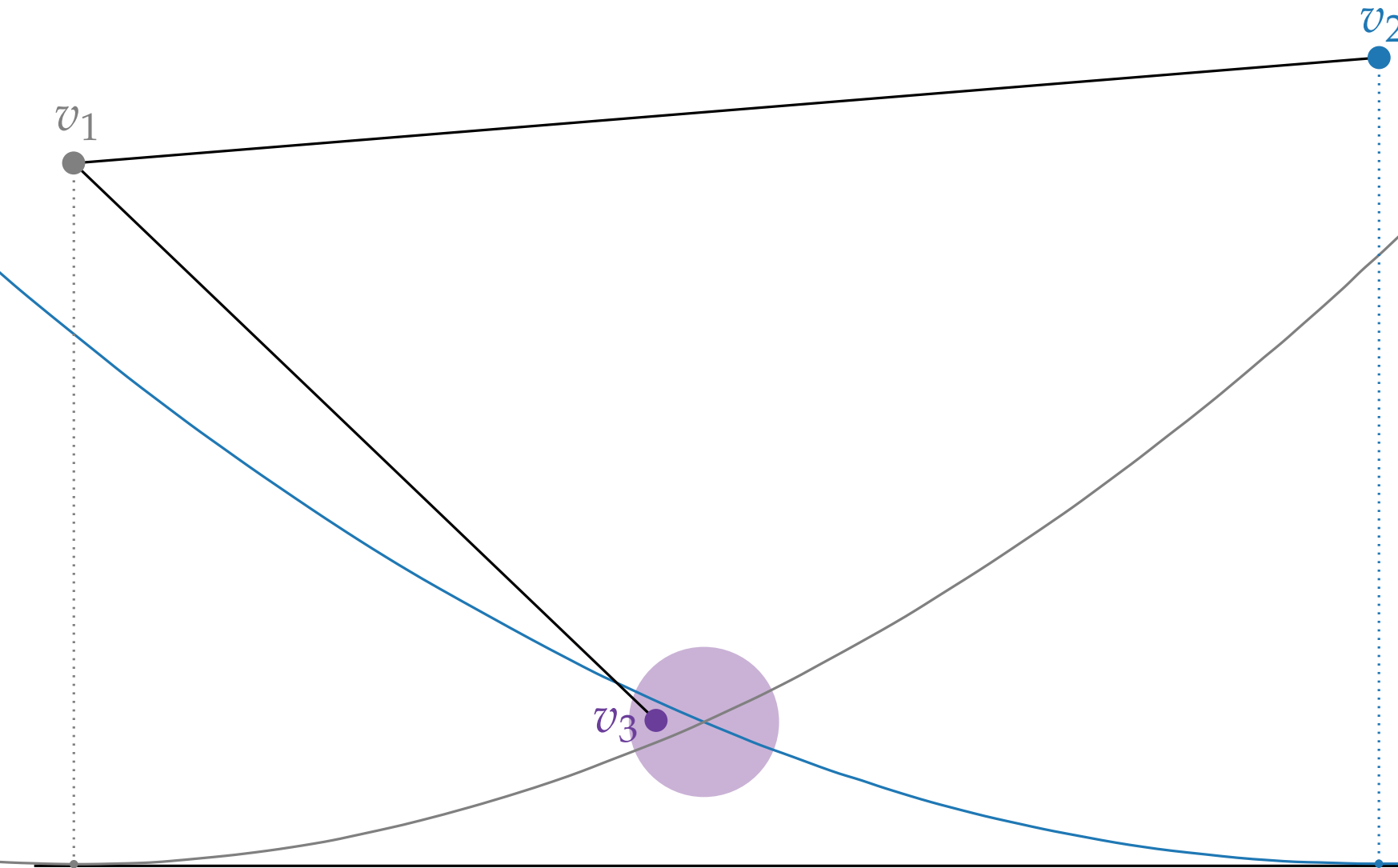
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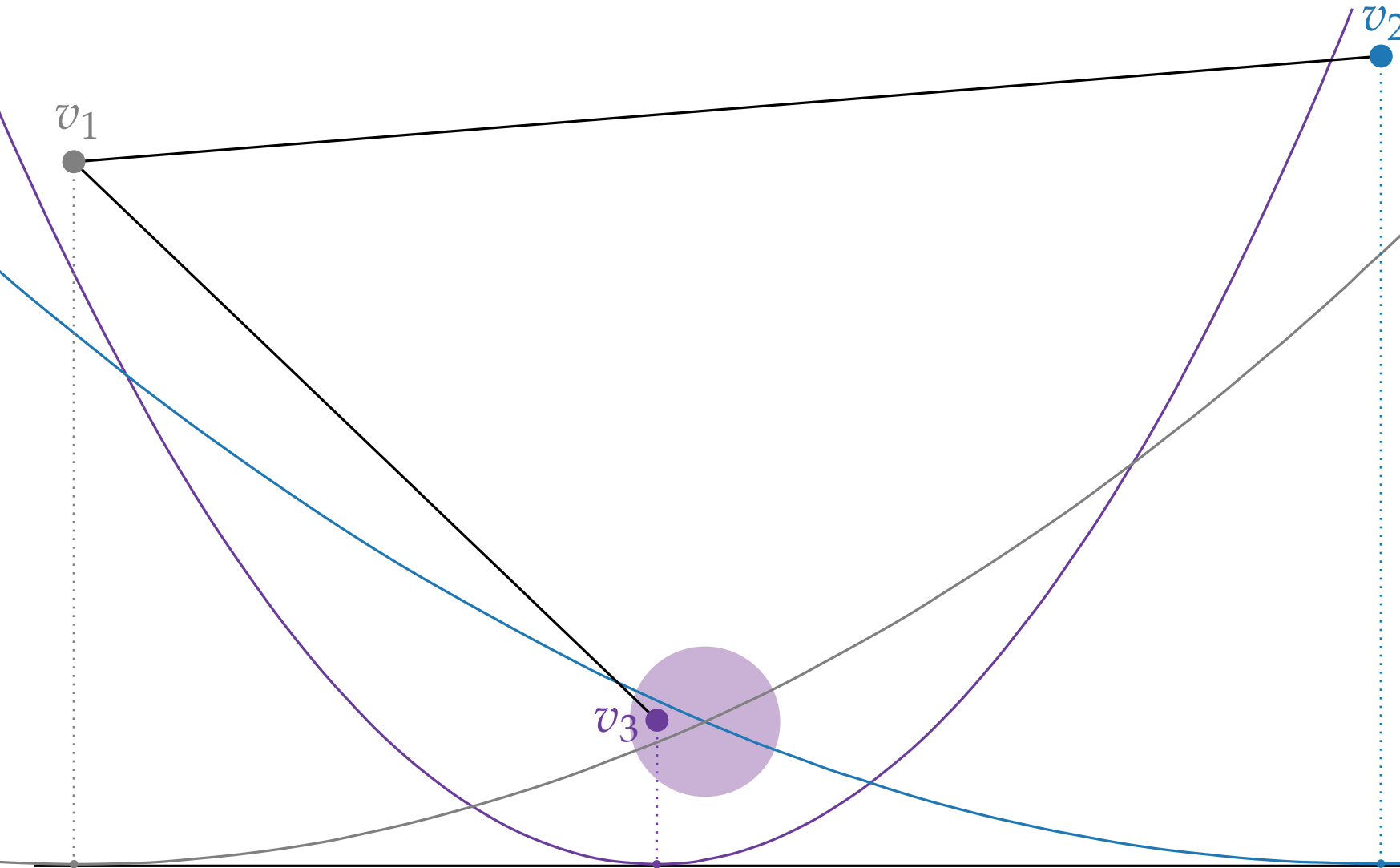
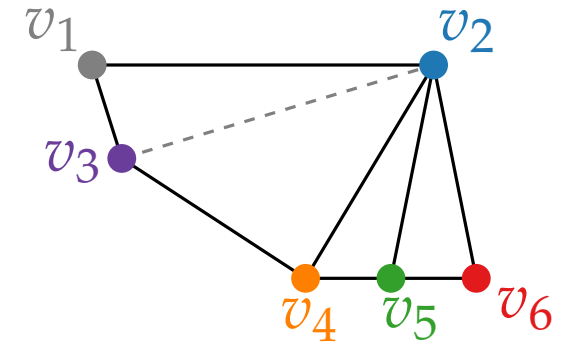
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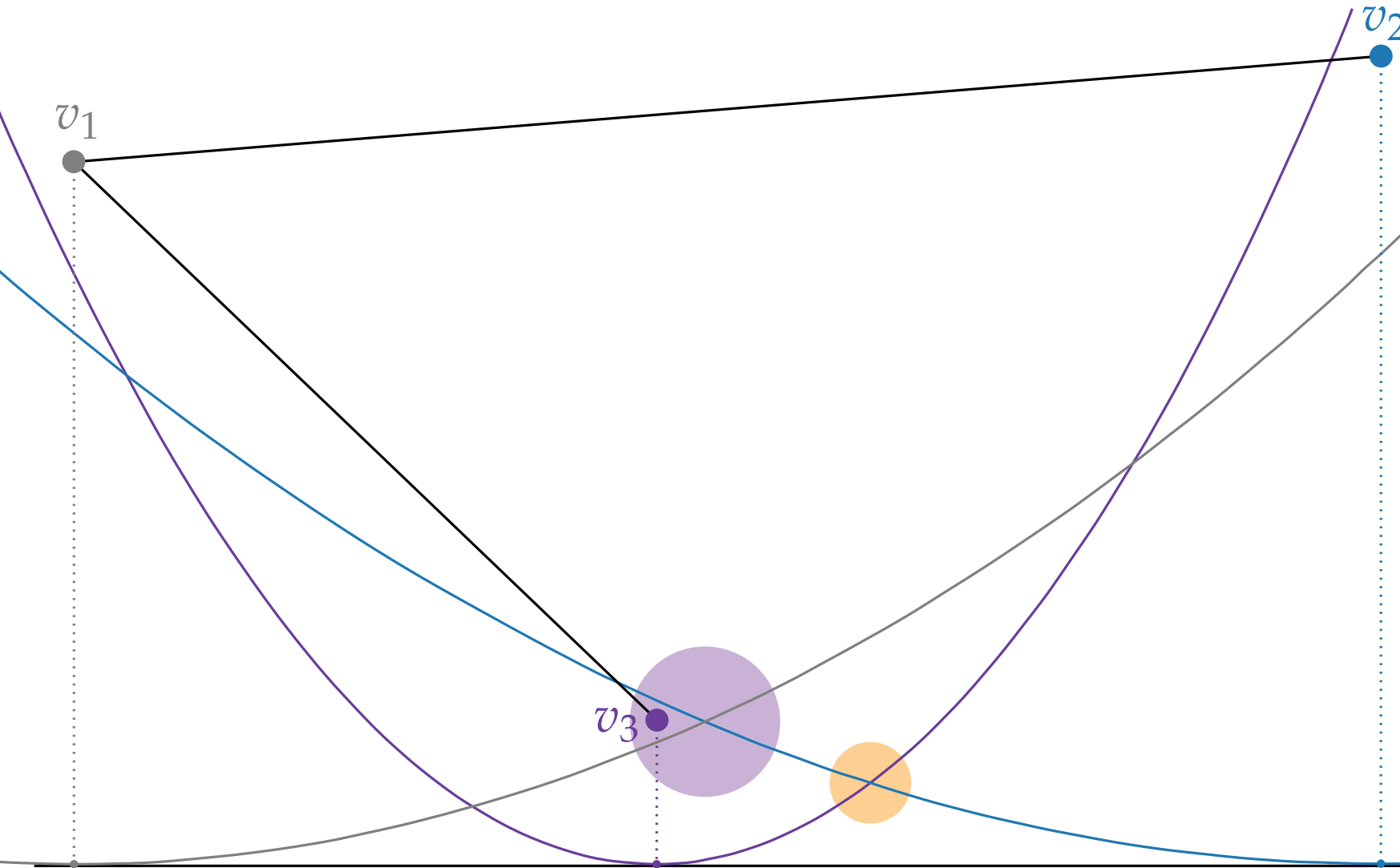
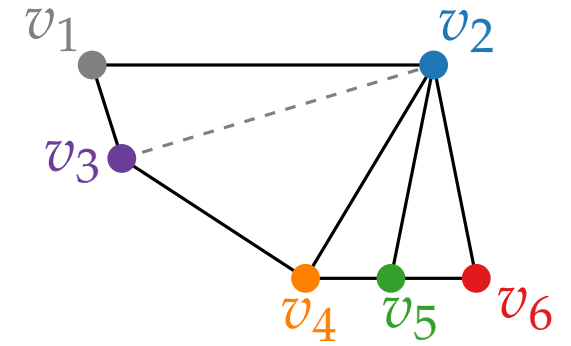
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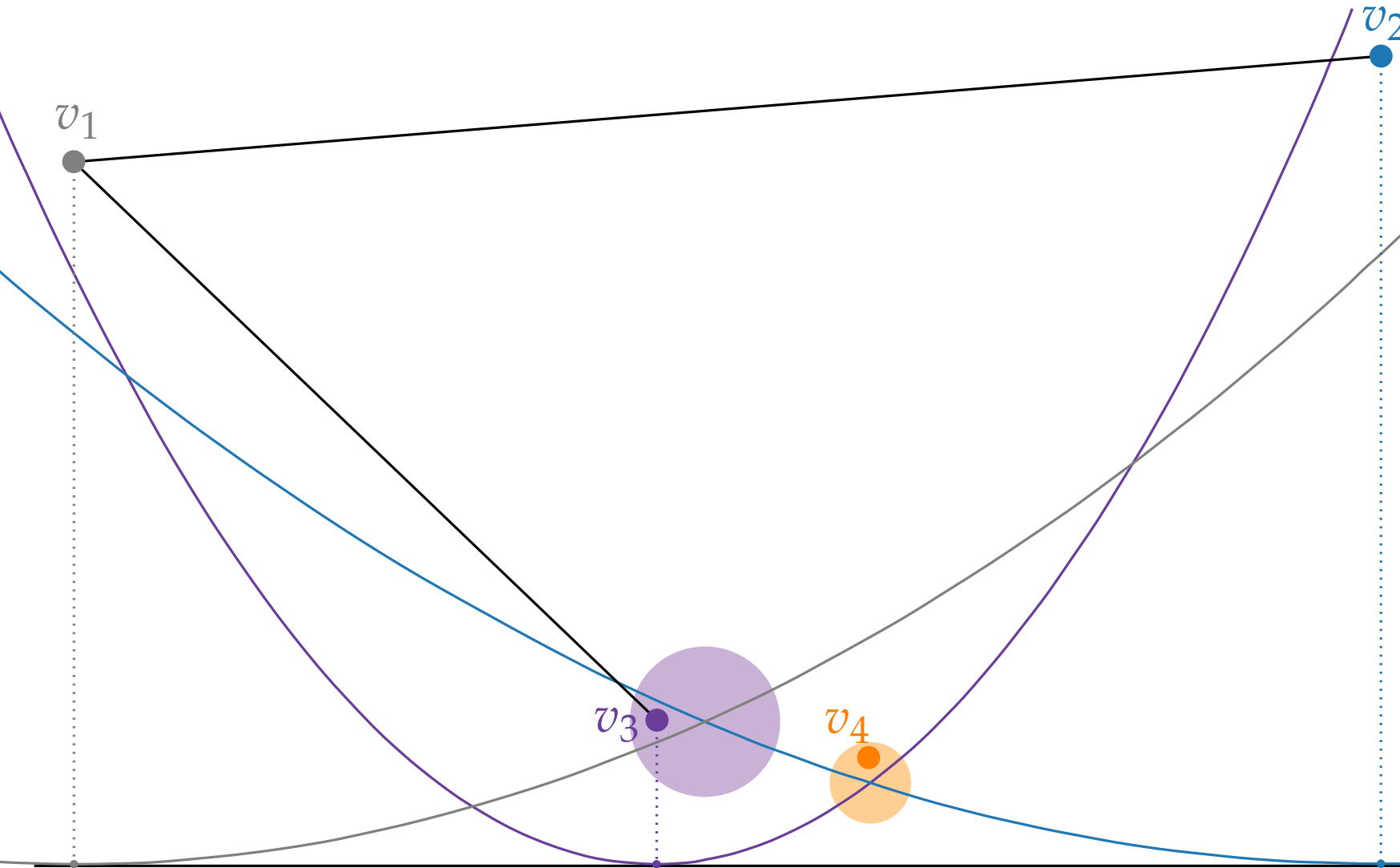
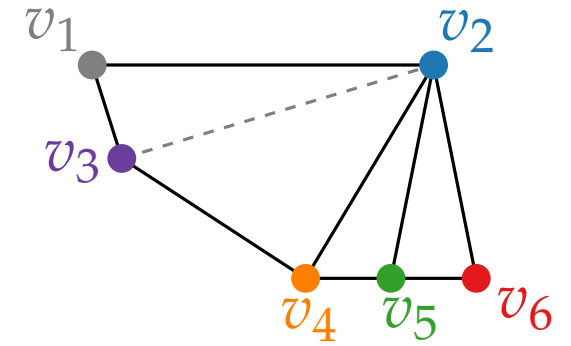
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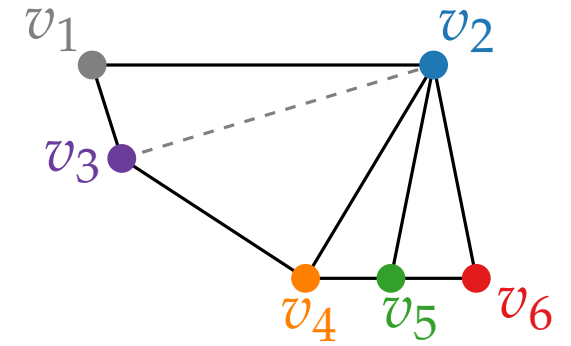
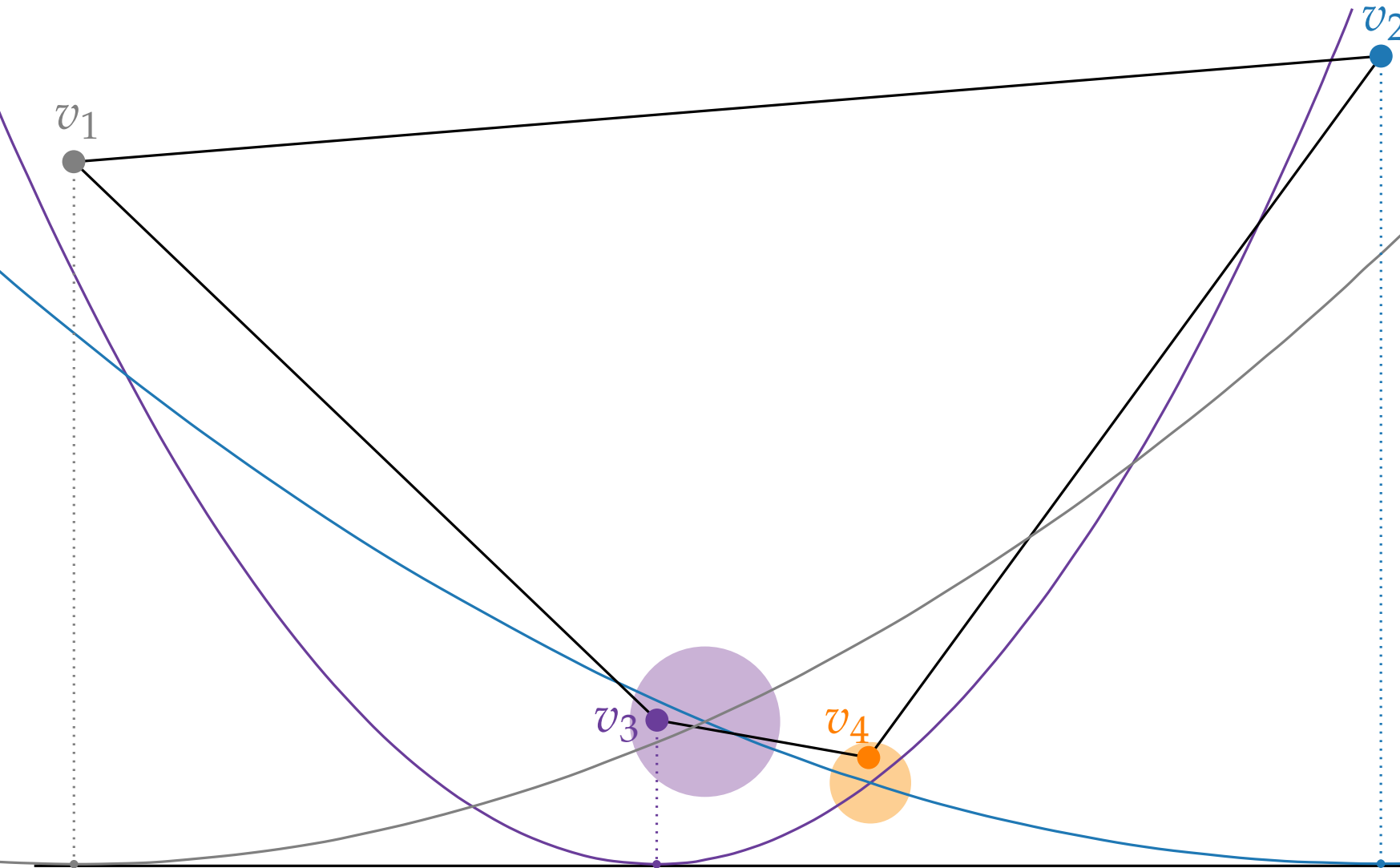
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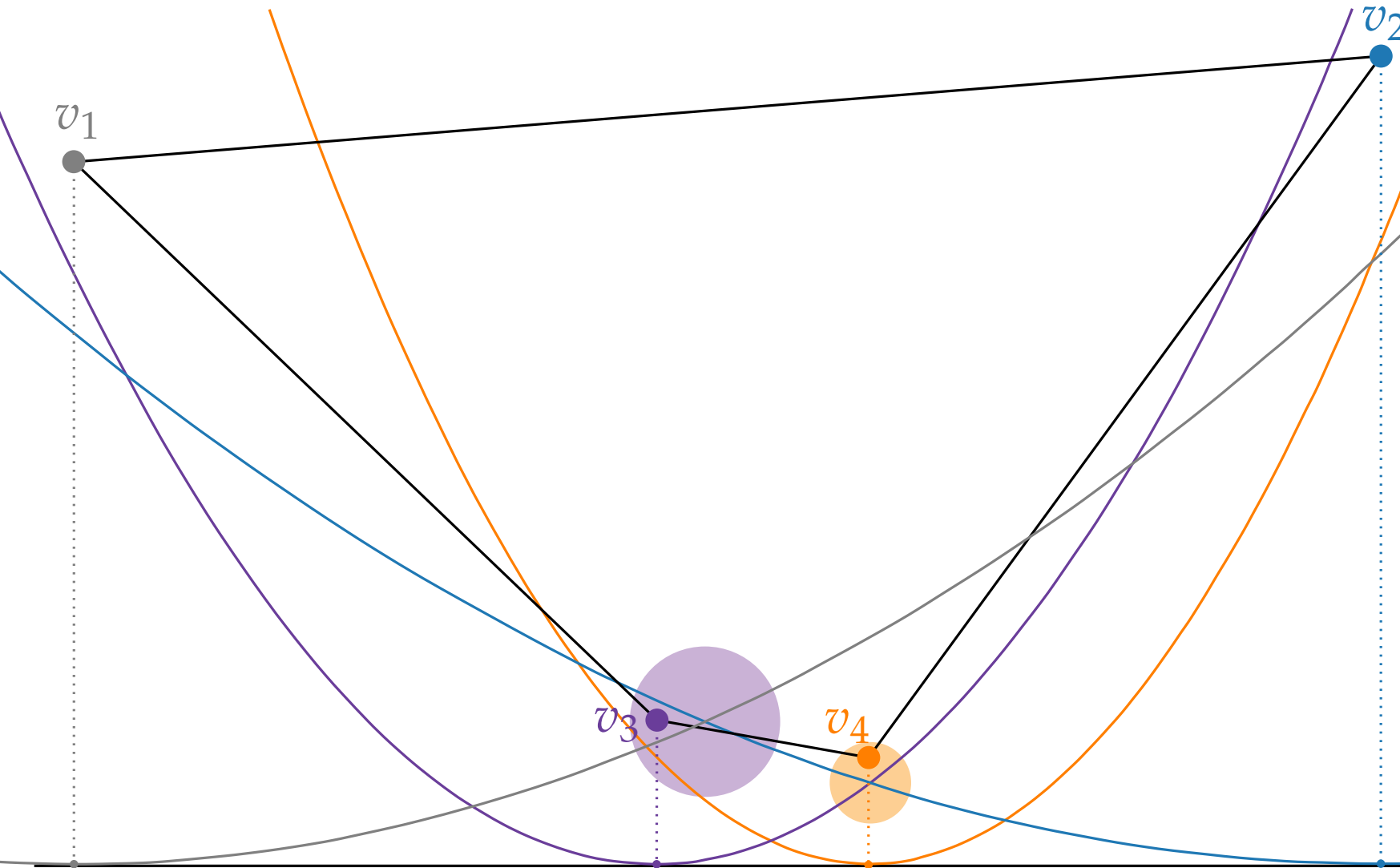
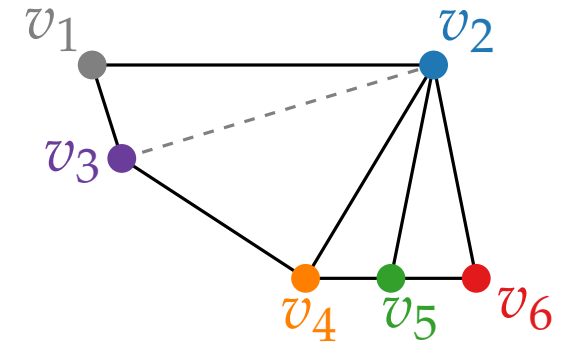
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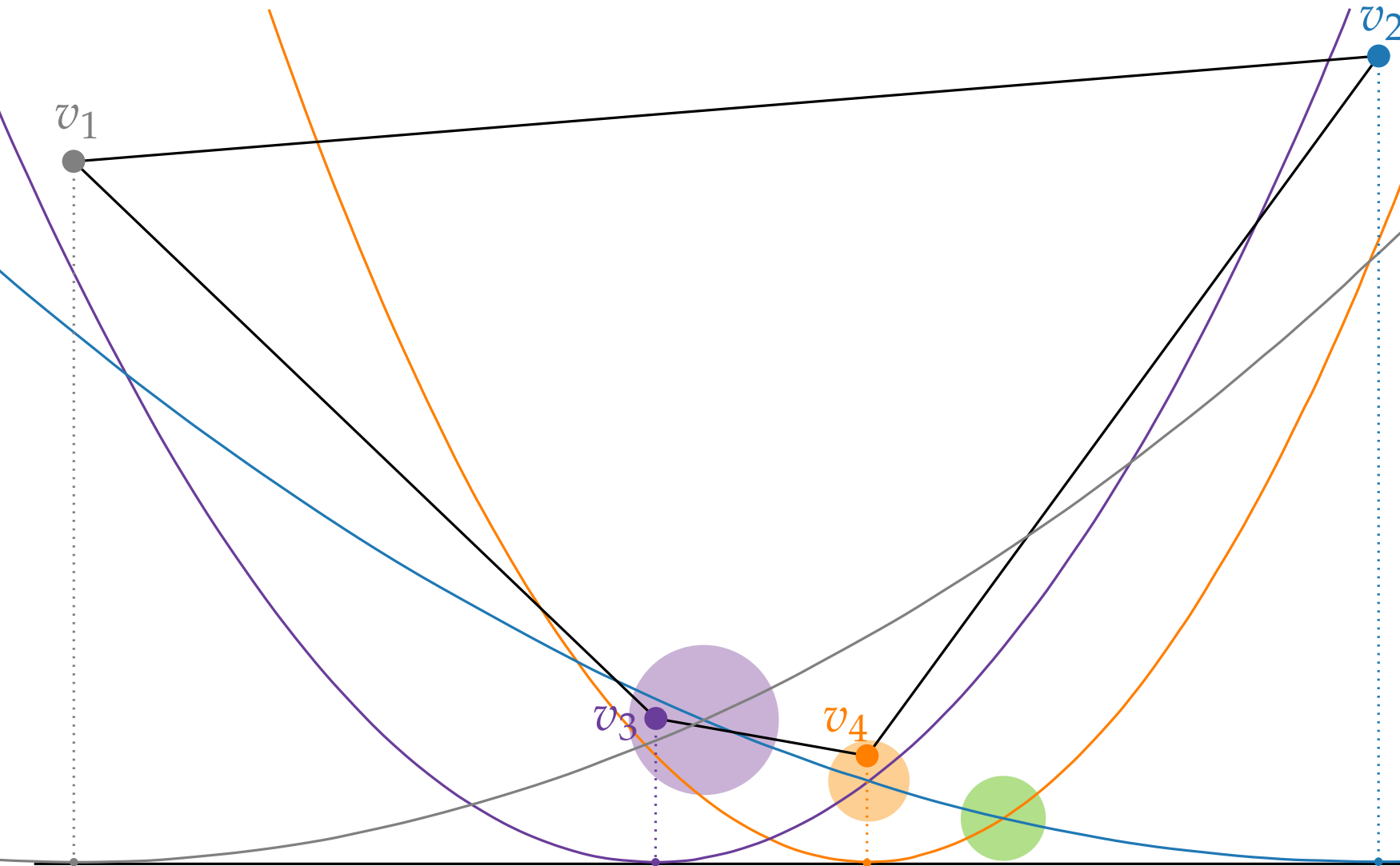
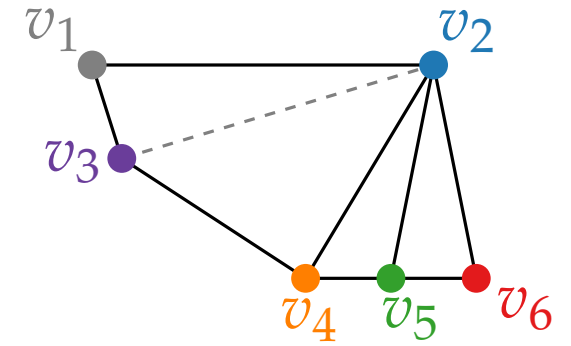
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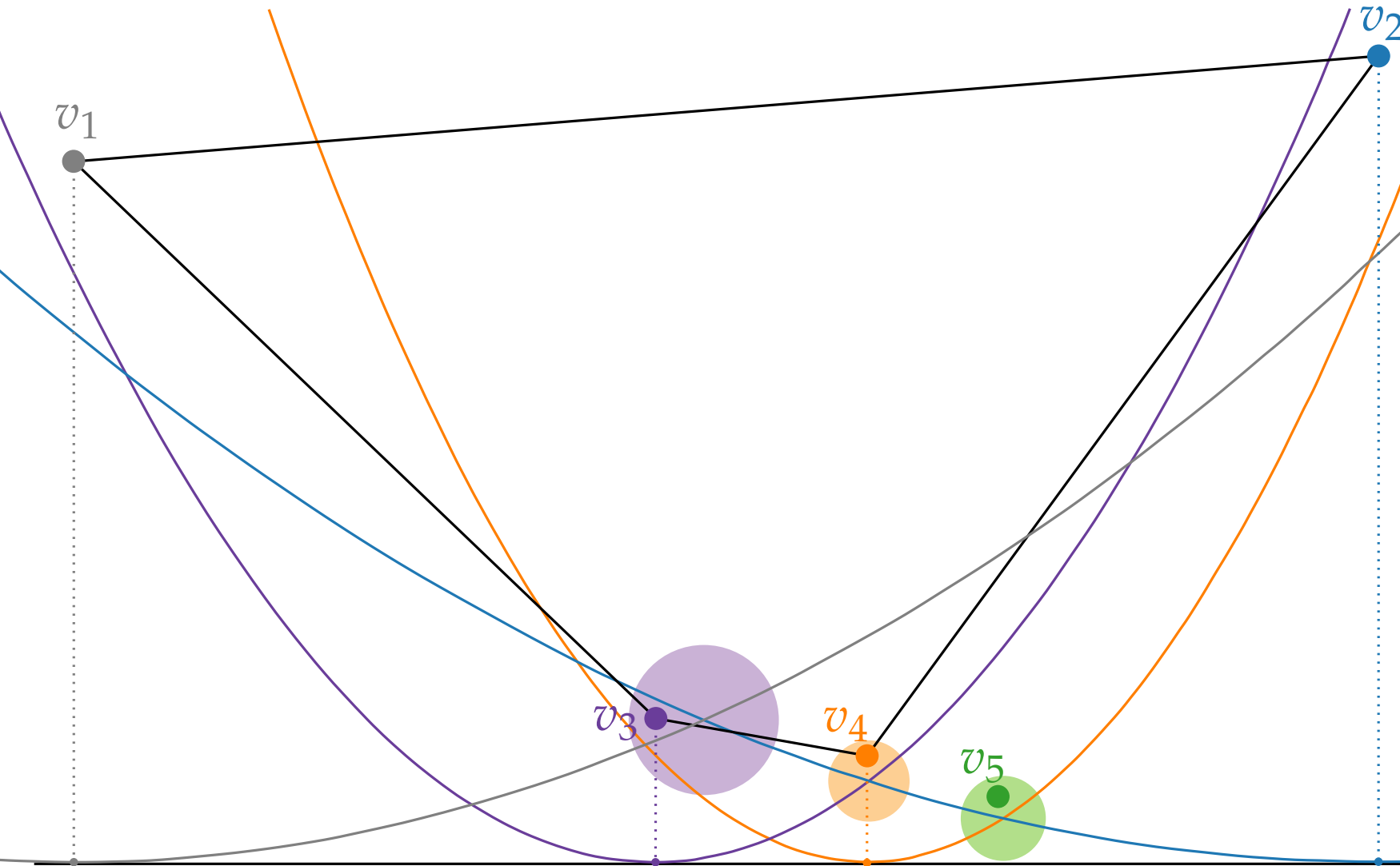
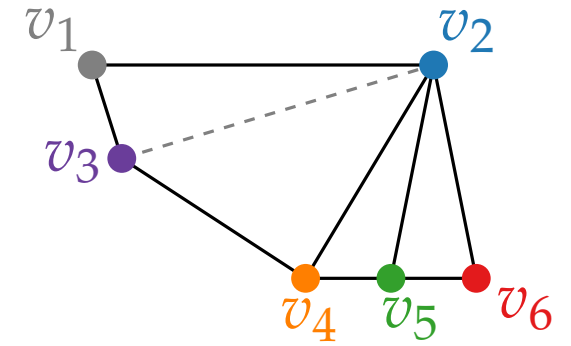
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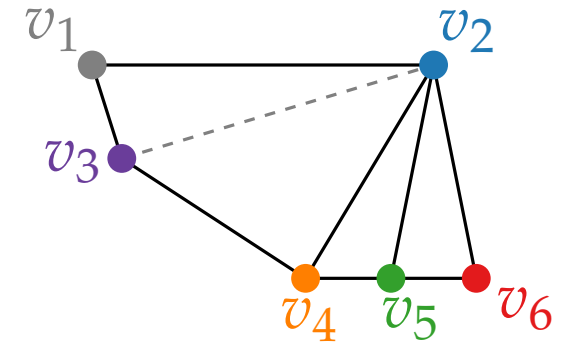
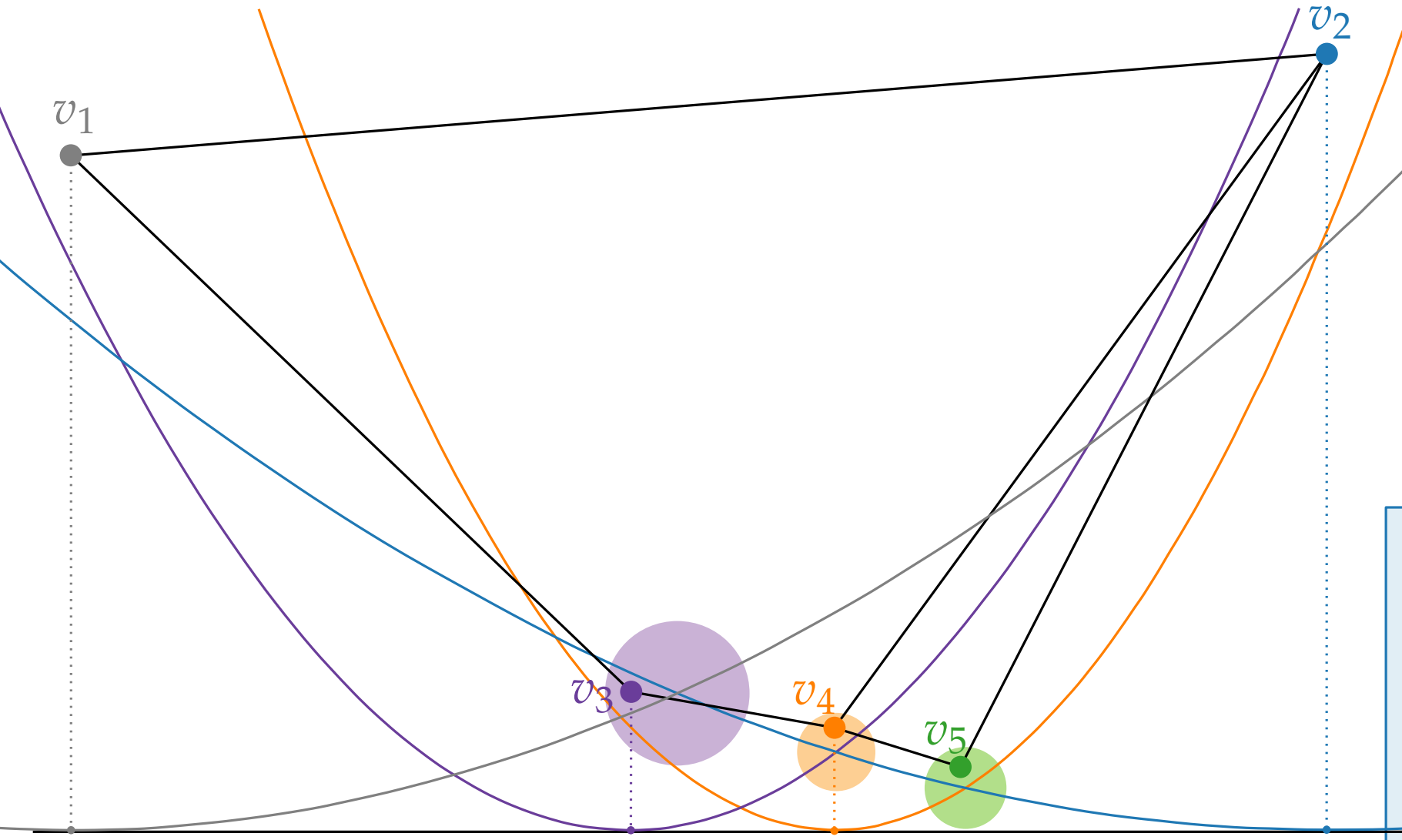
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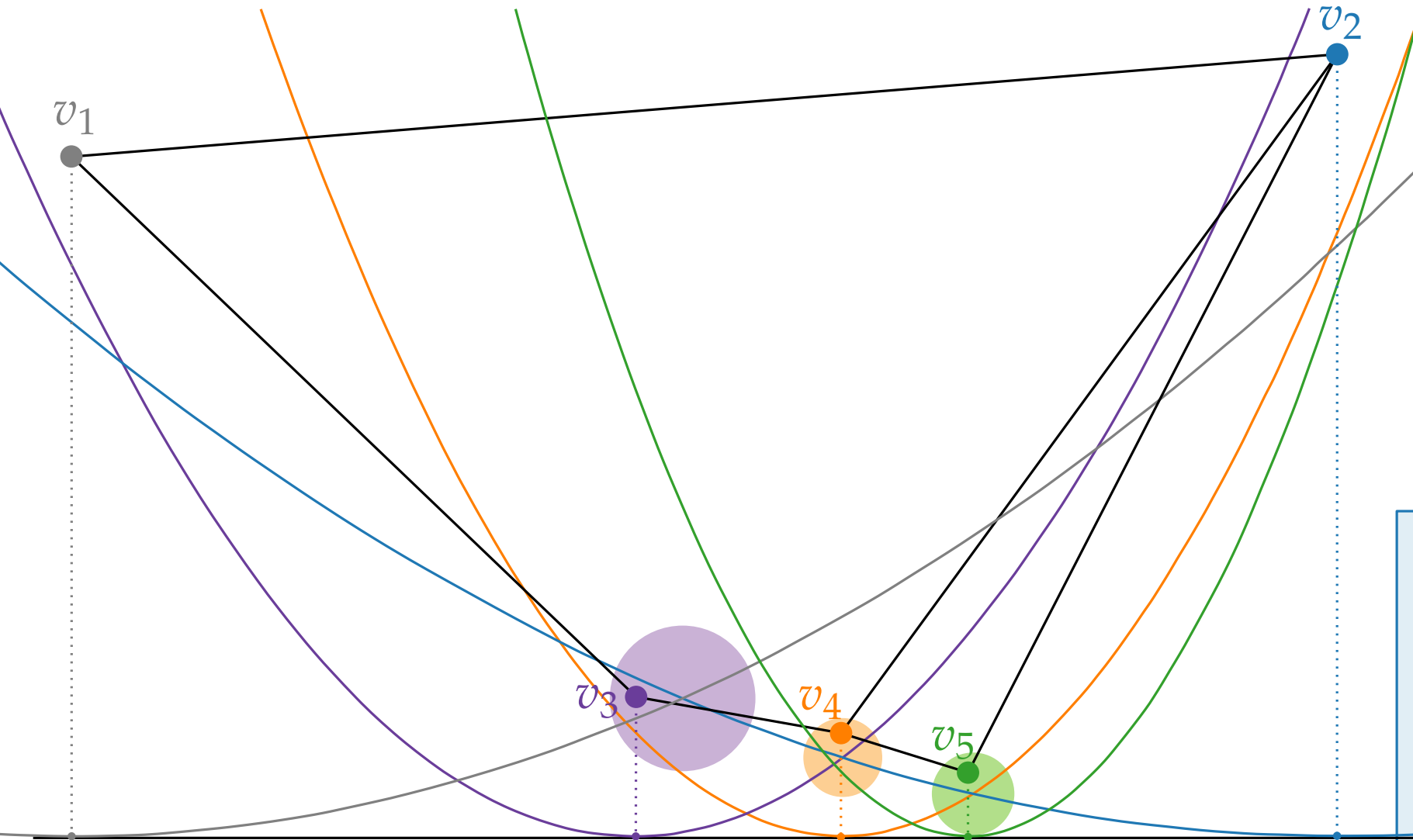
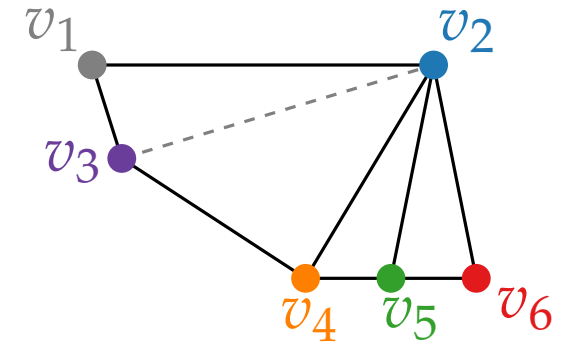
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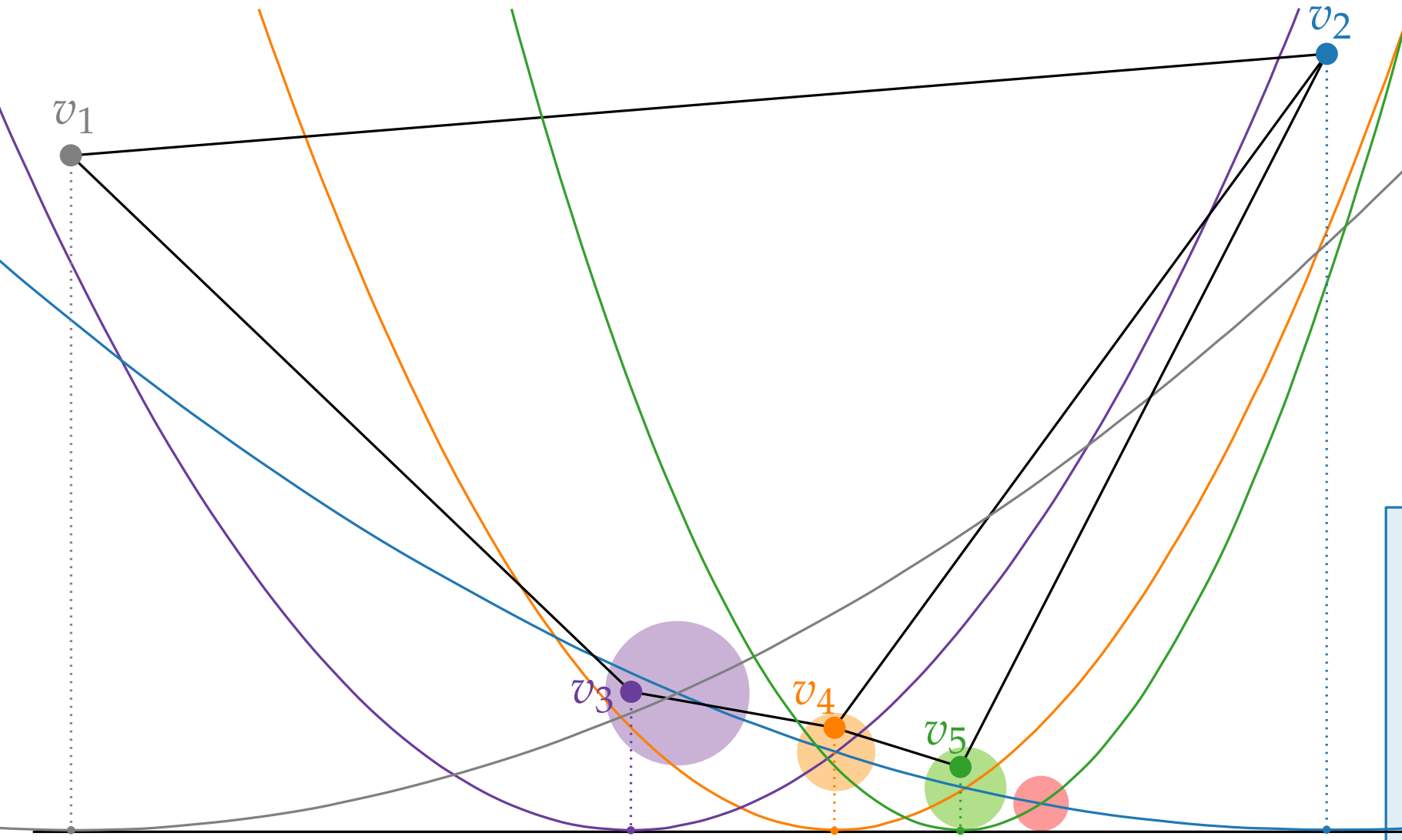
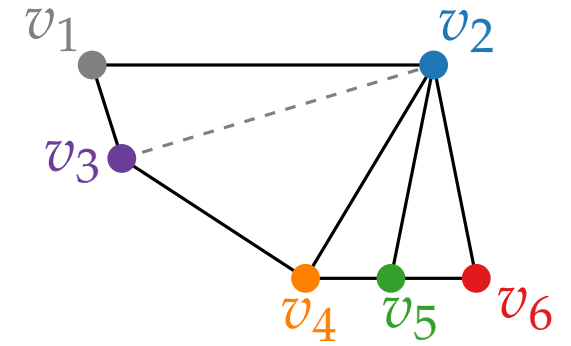
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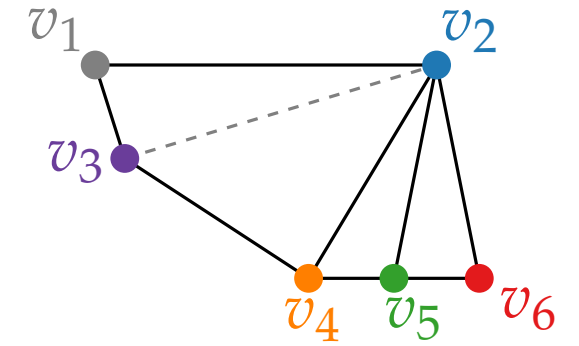
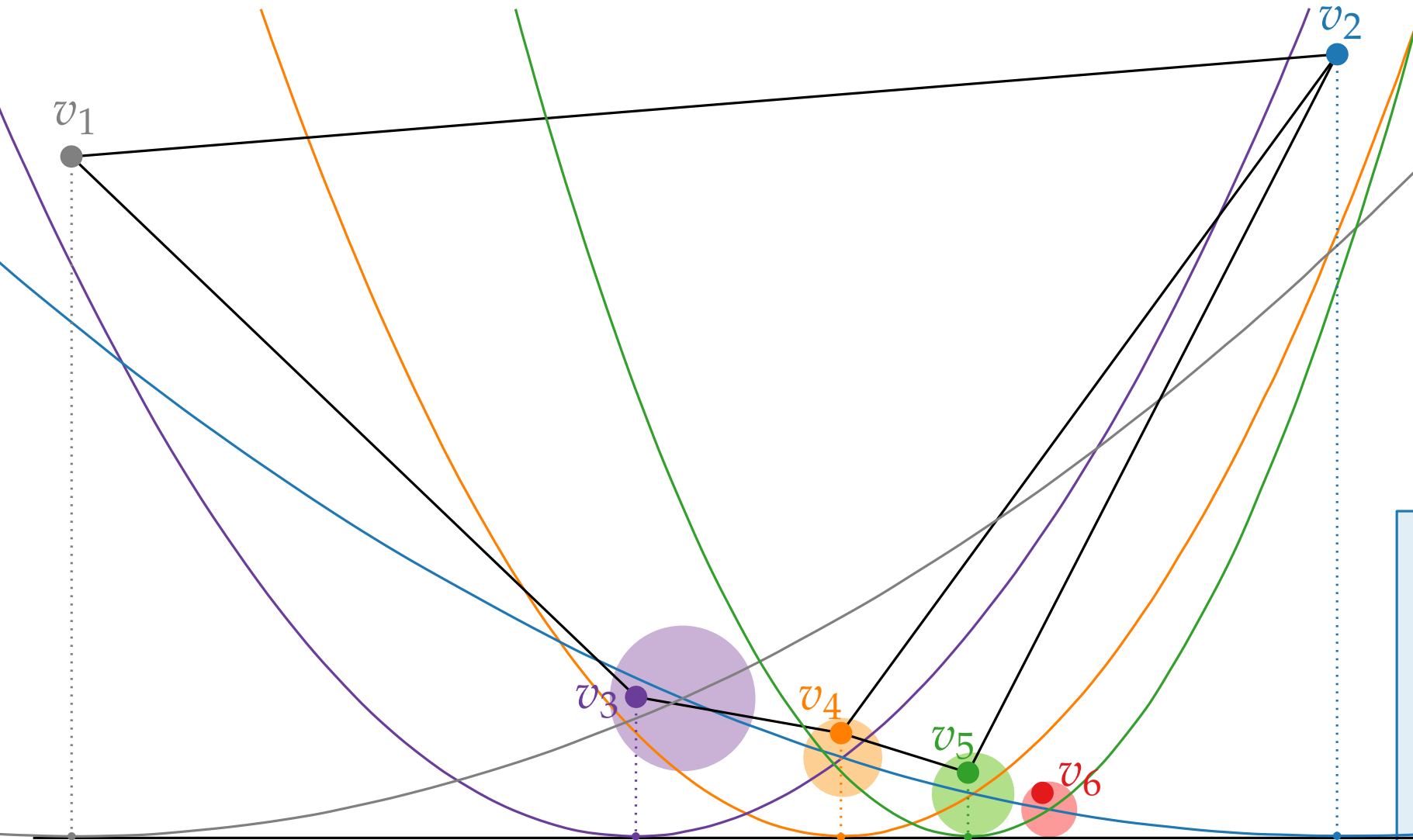
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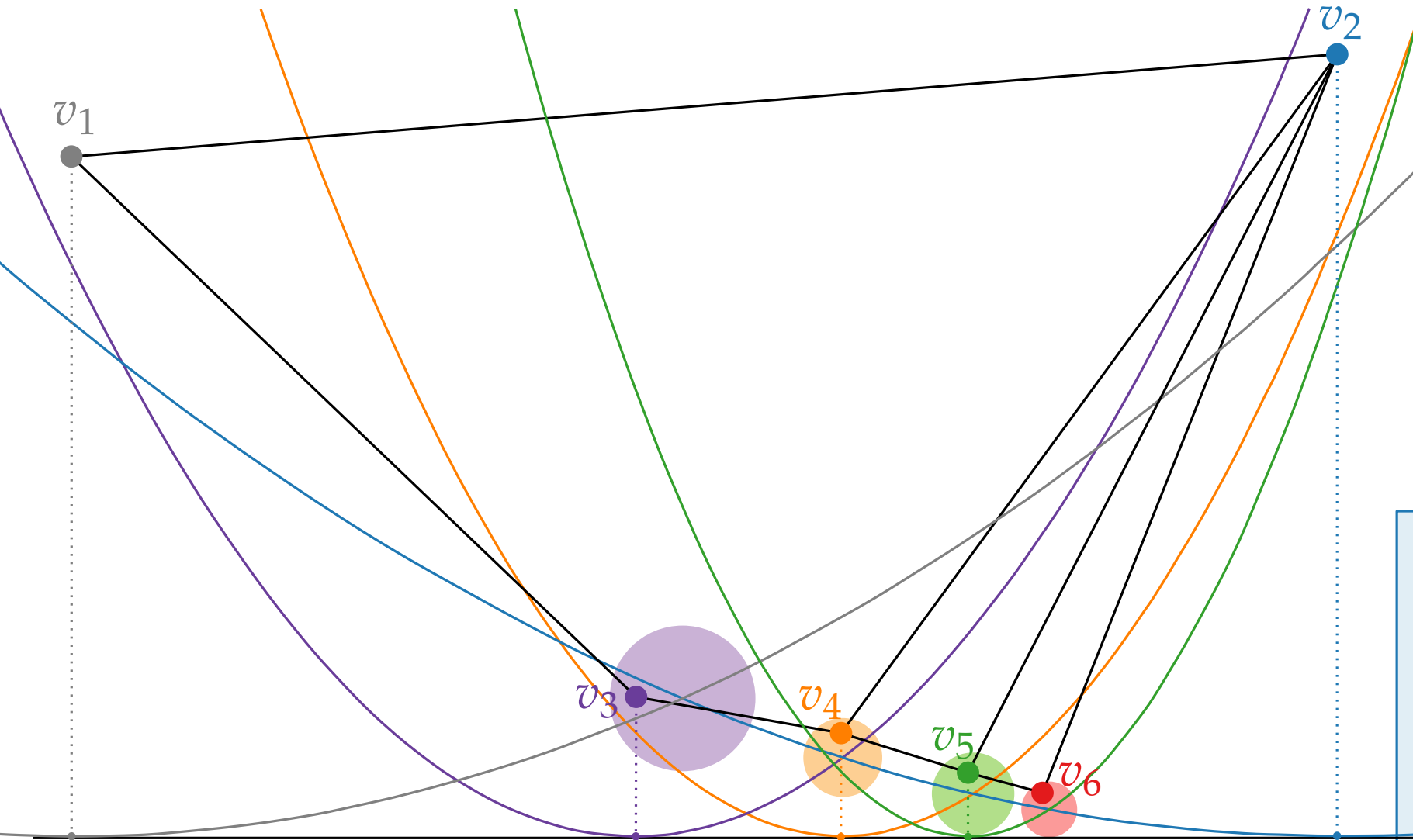
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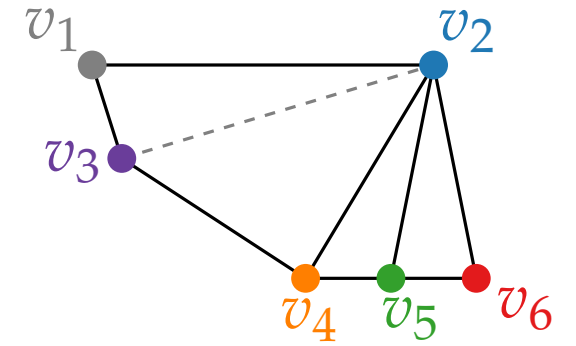
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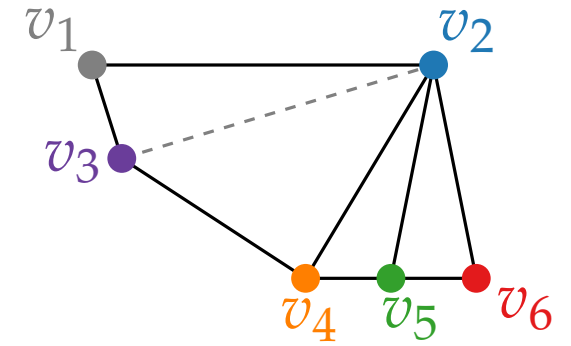
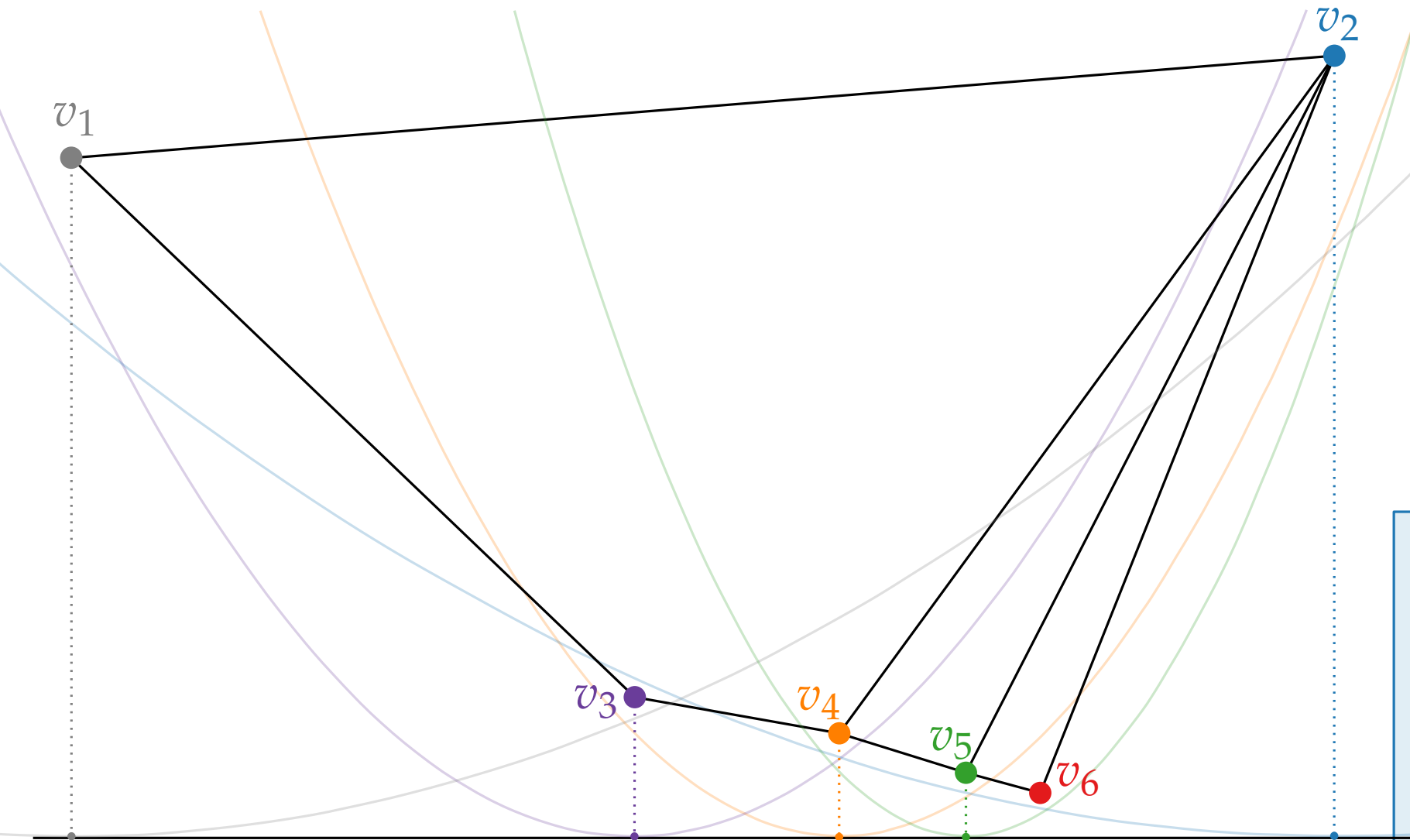
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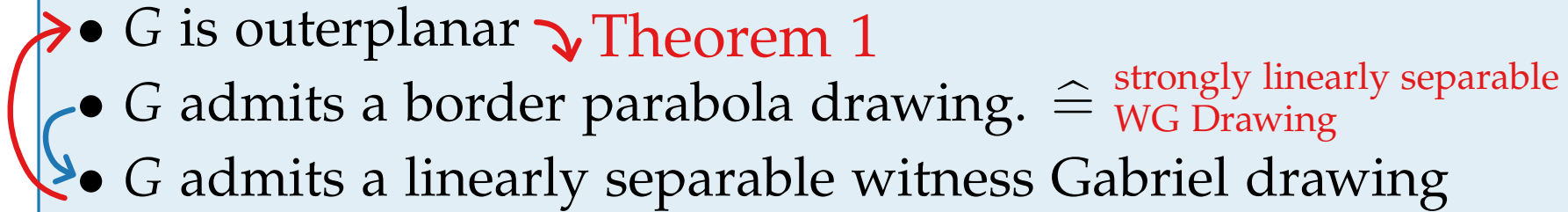
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Proof of Main Theorem

Main Theorem

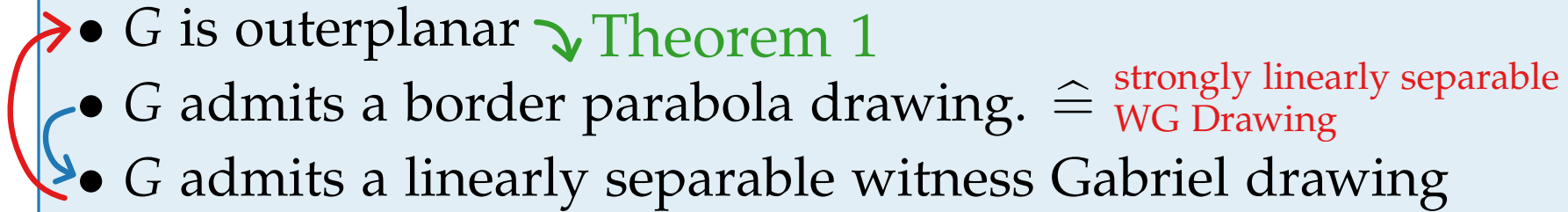
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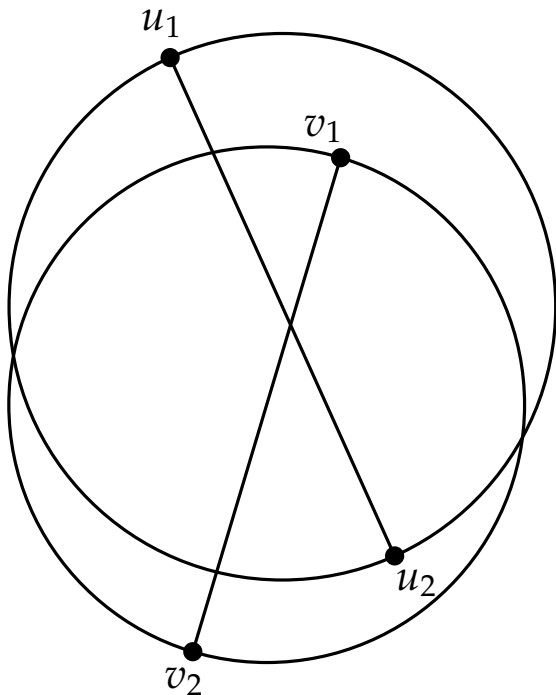
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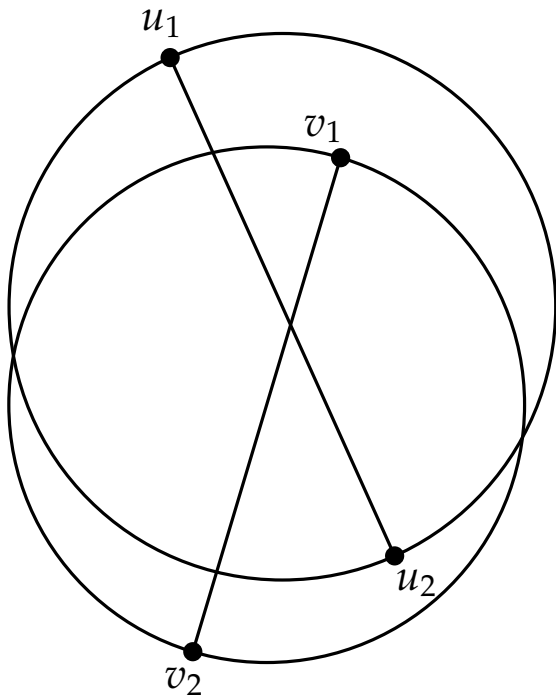


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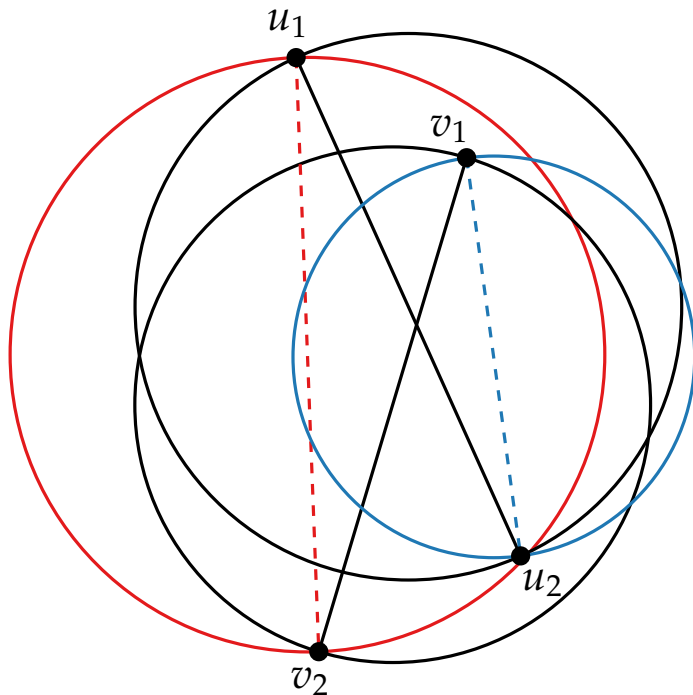


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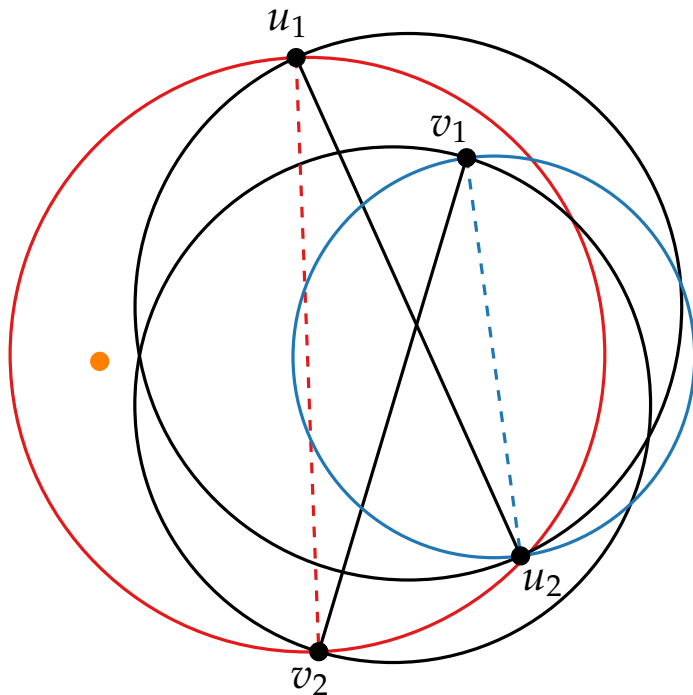


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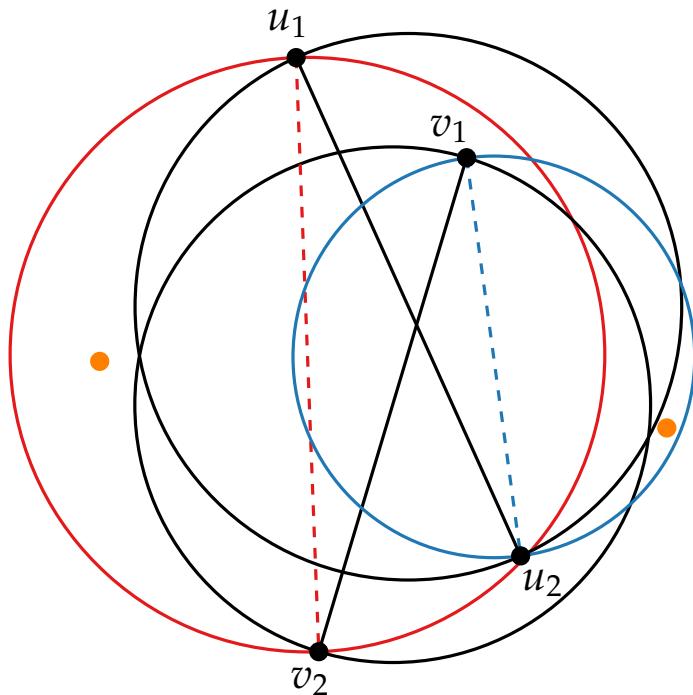


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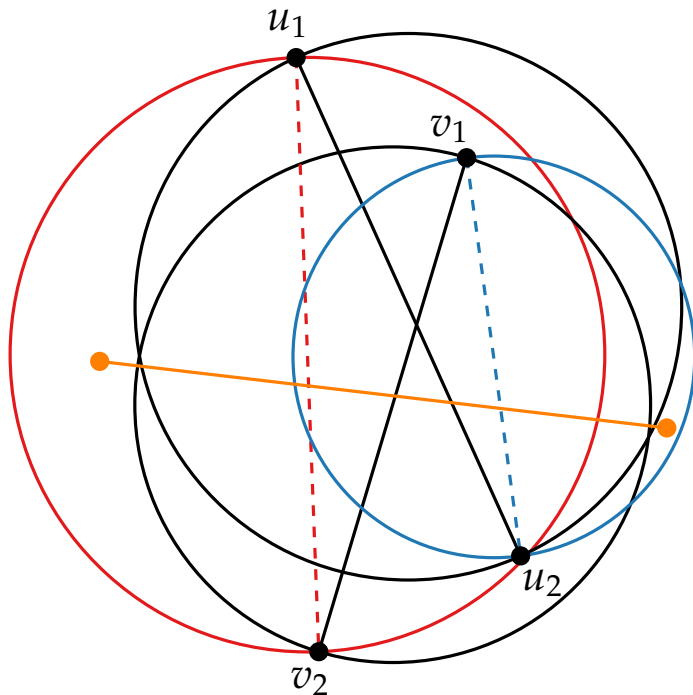


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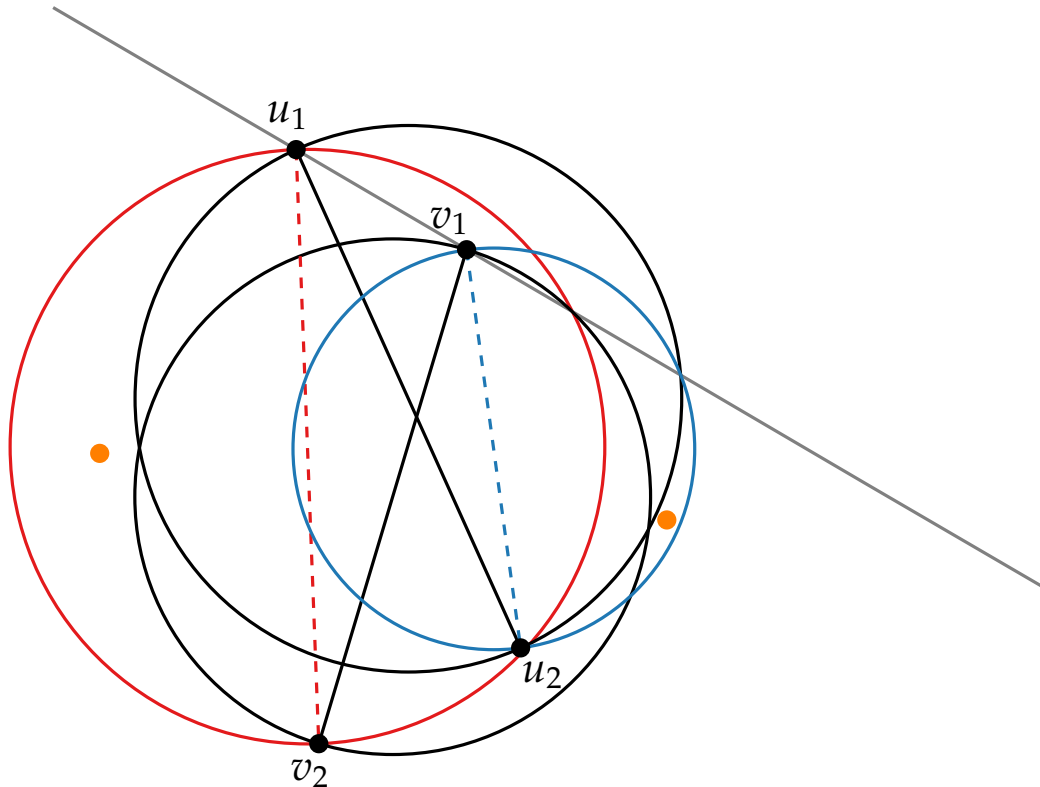


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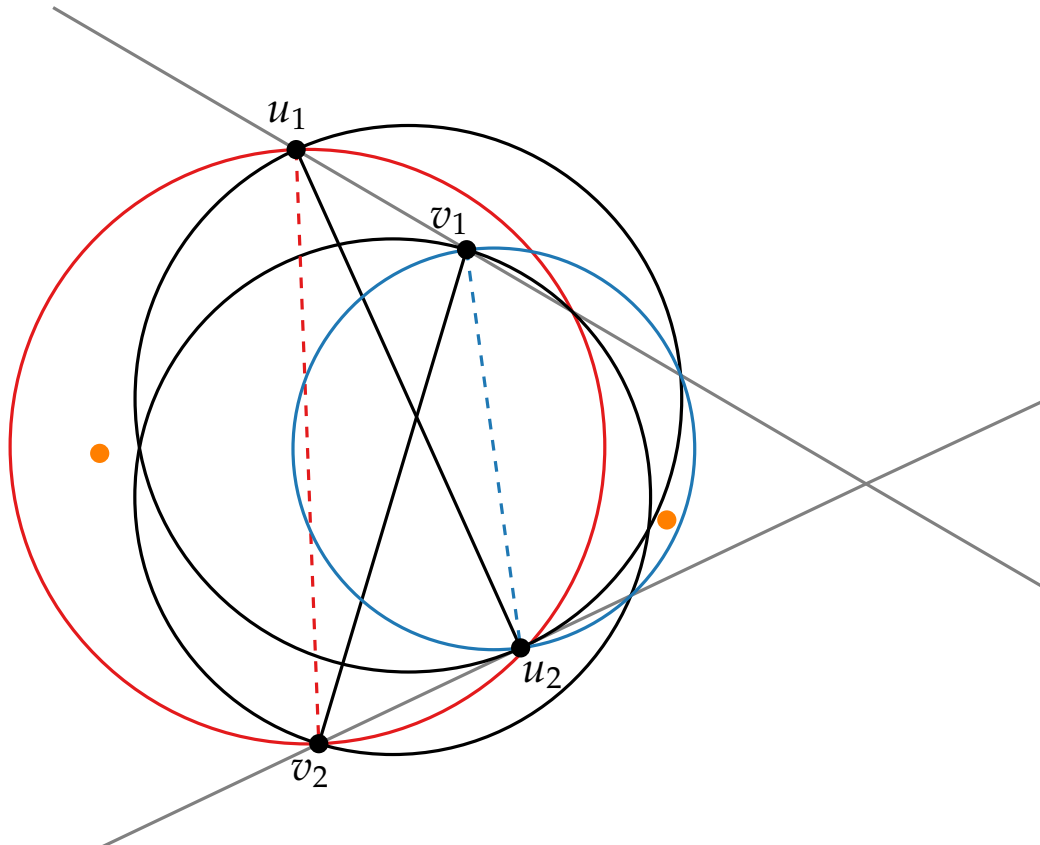


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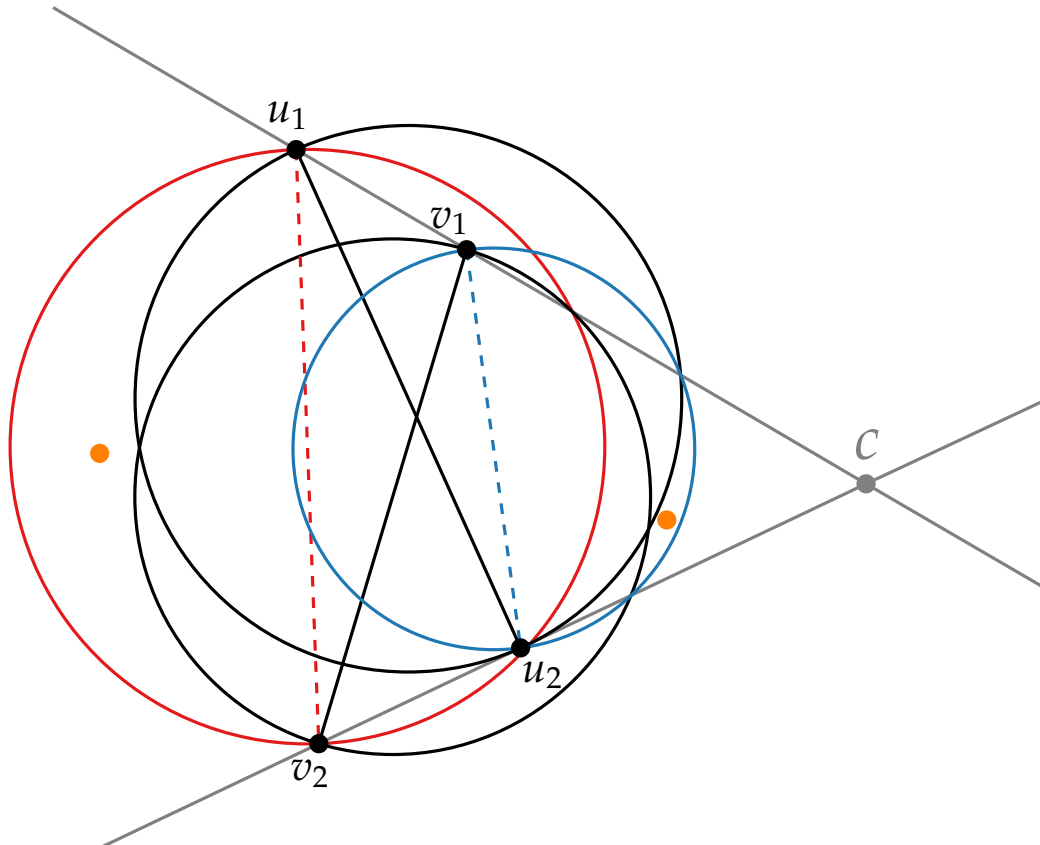


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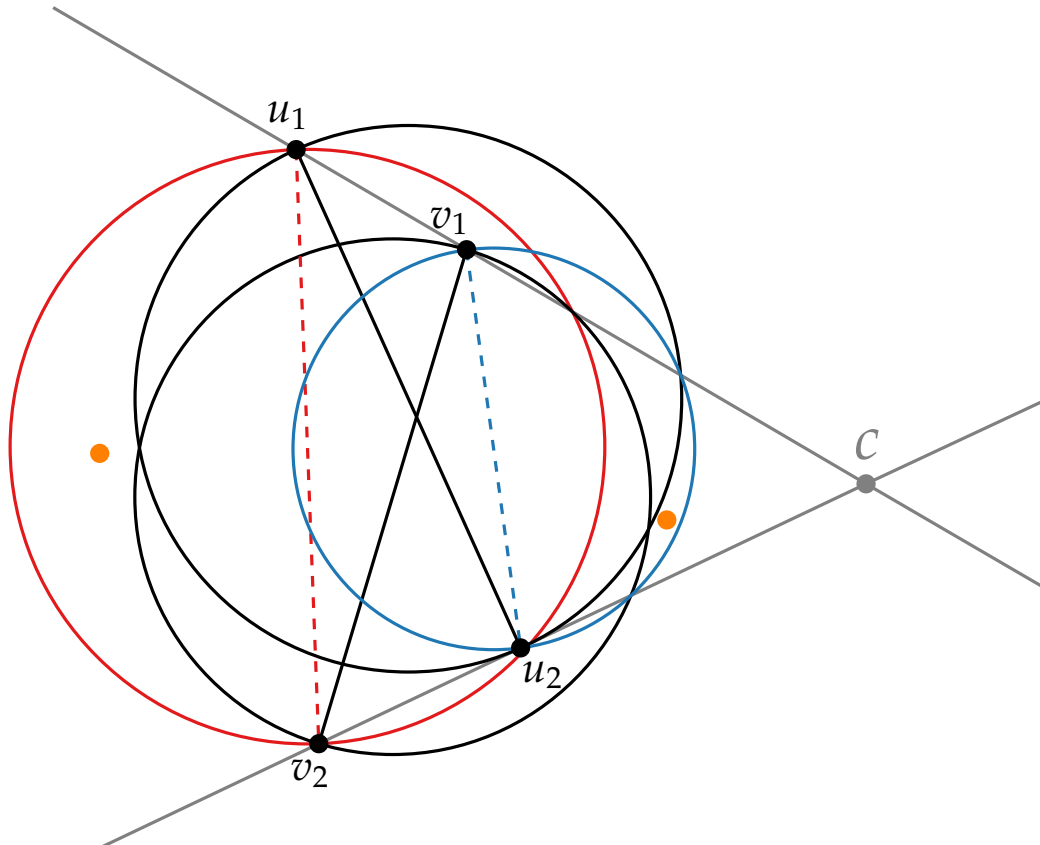


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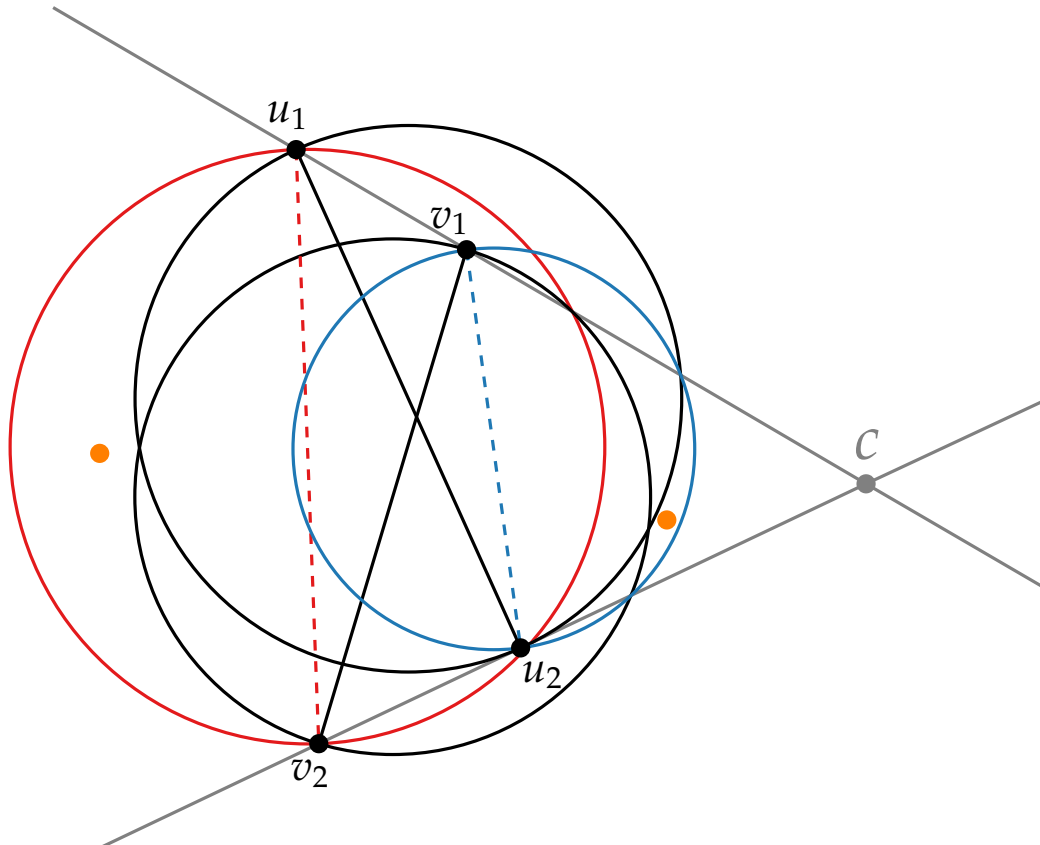
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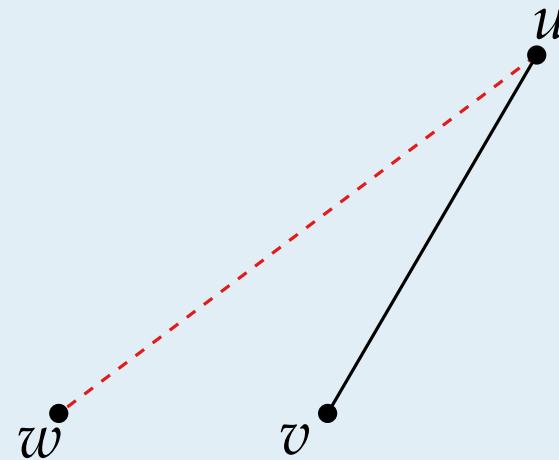
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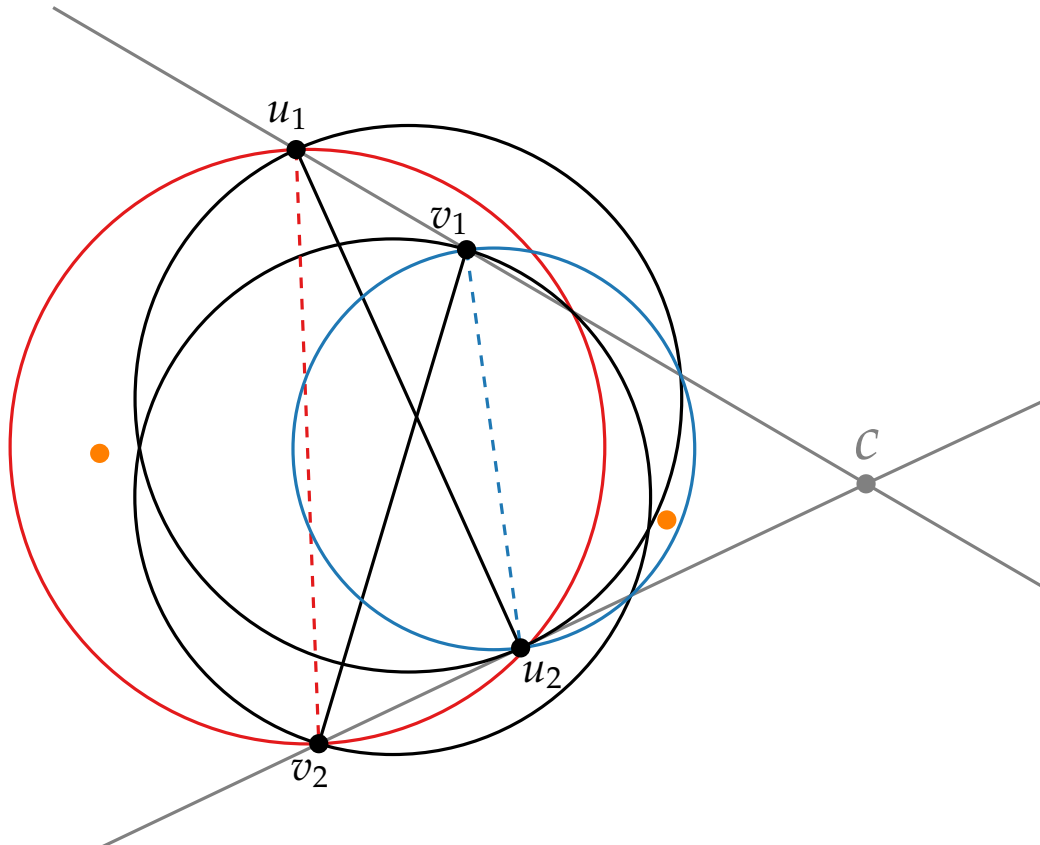


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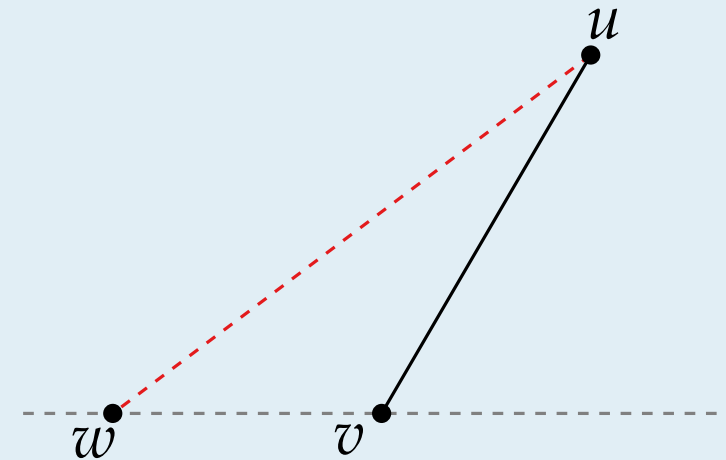
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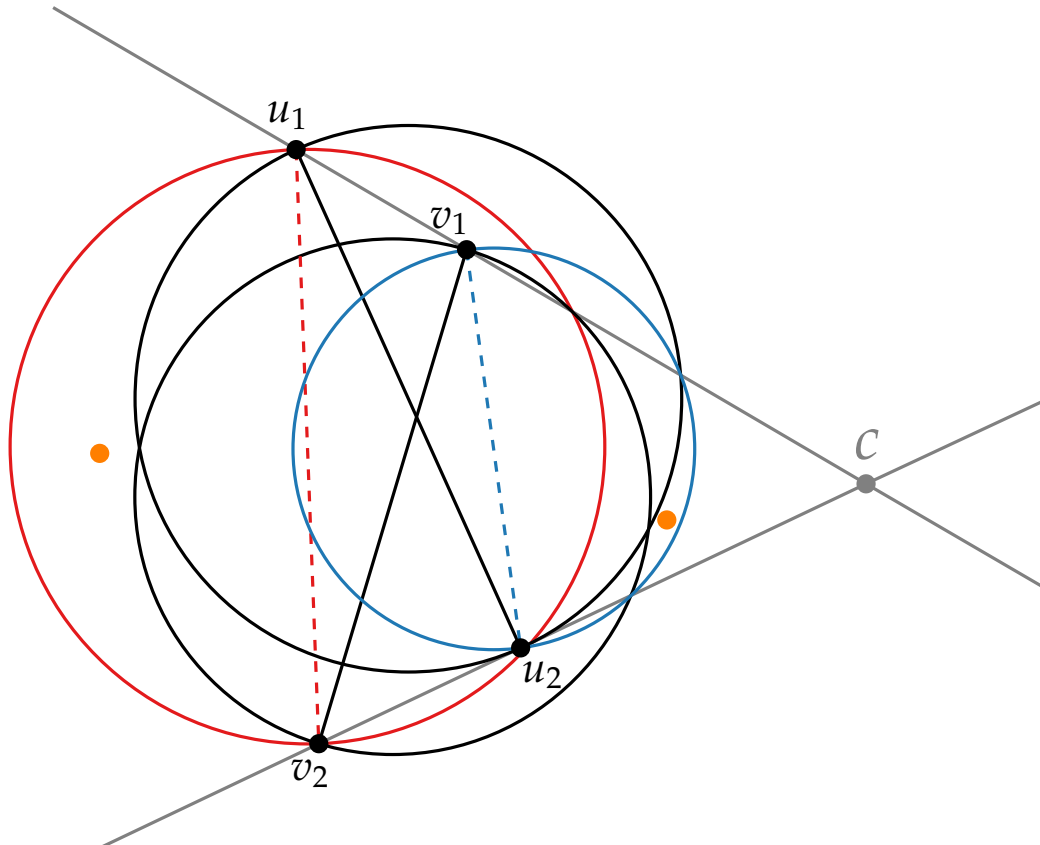


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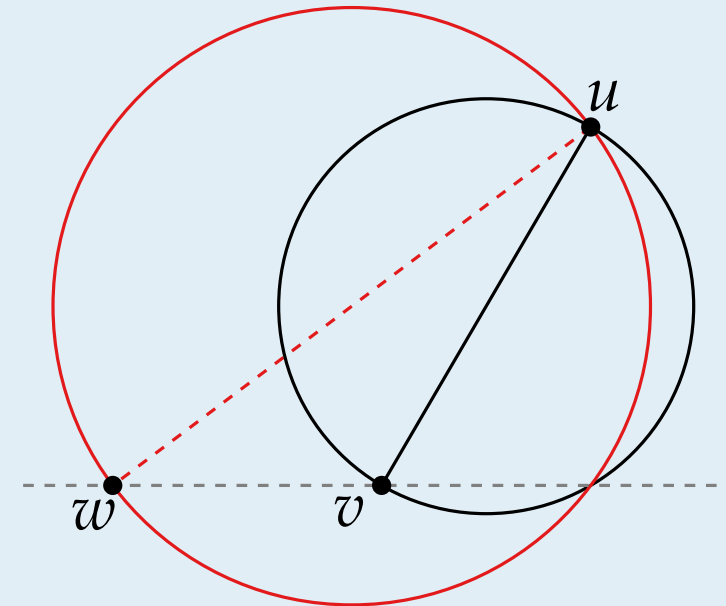
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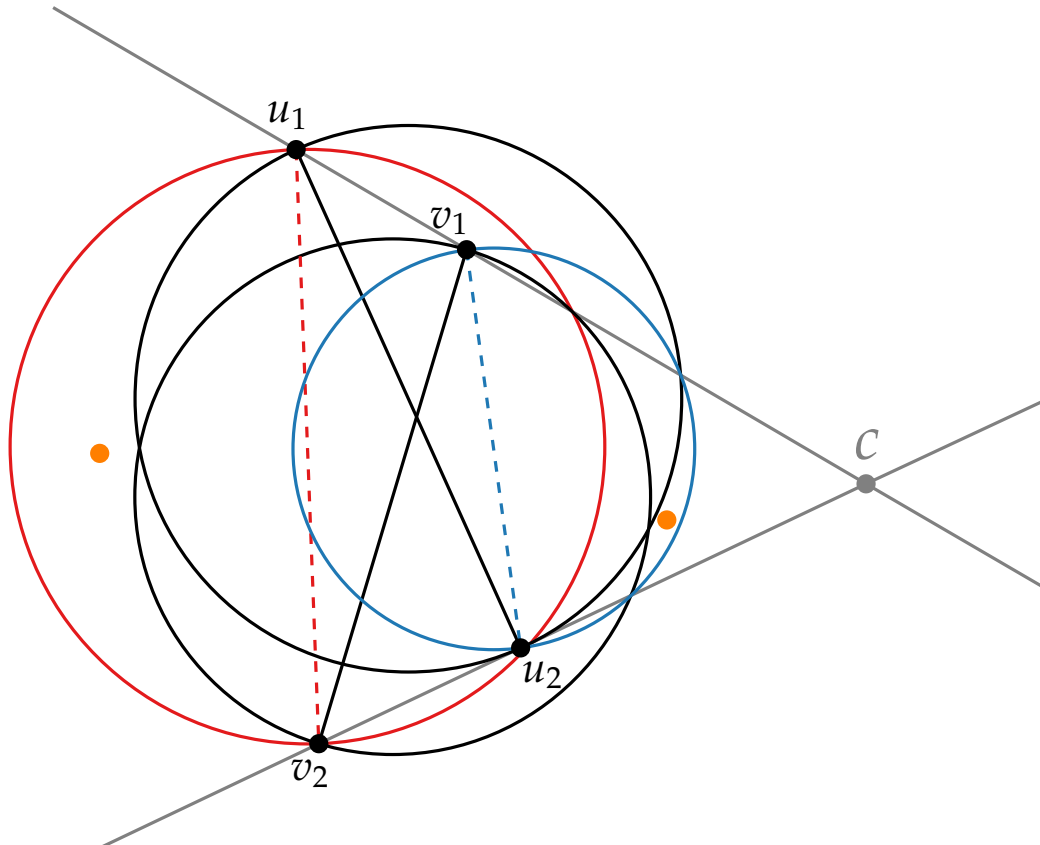


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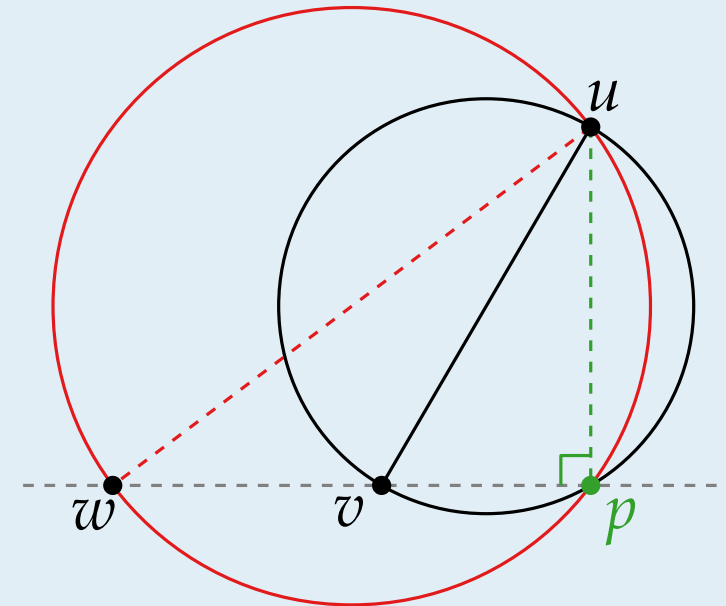
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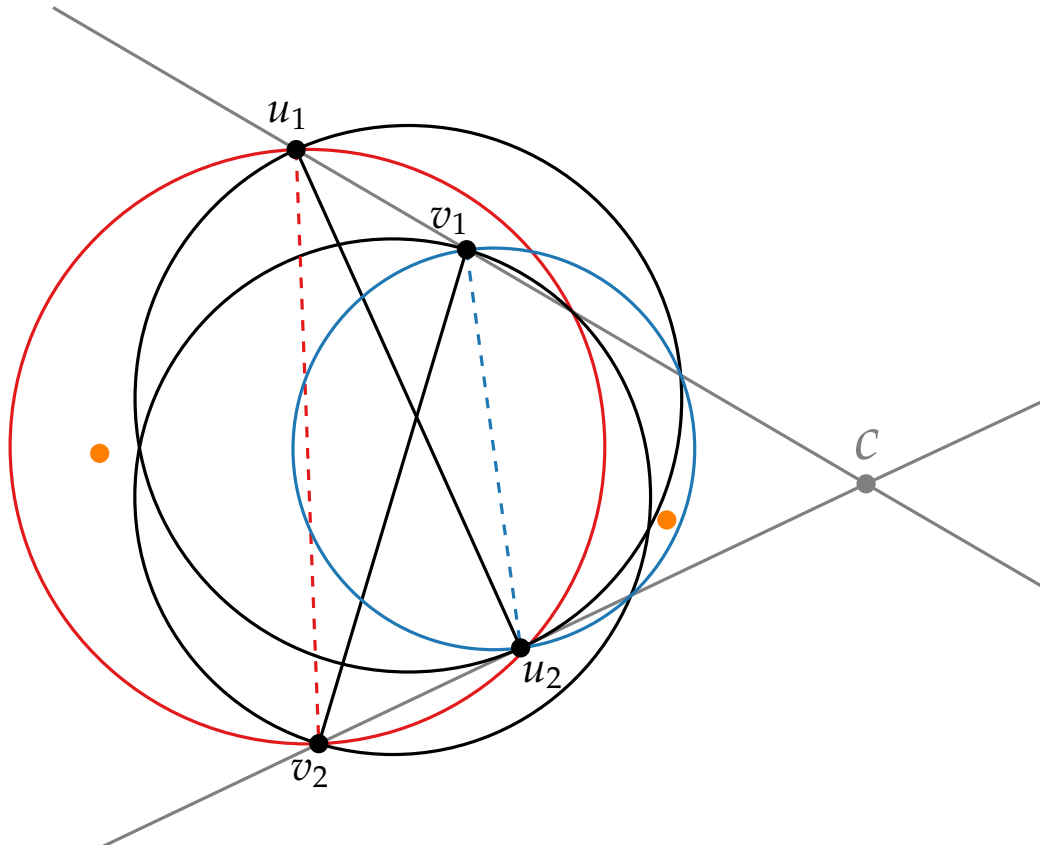


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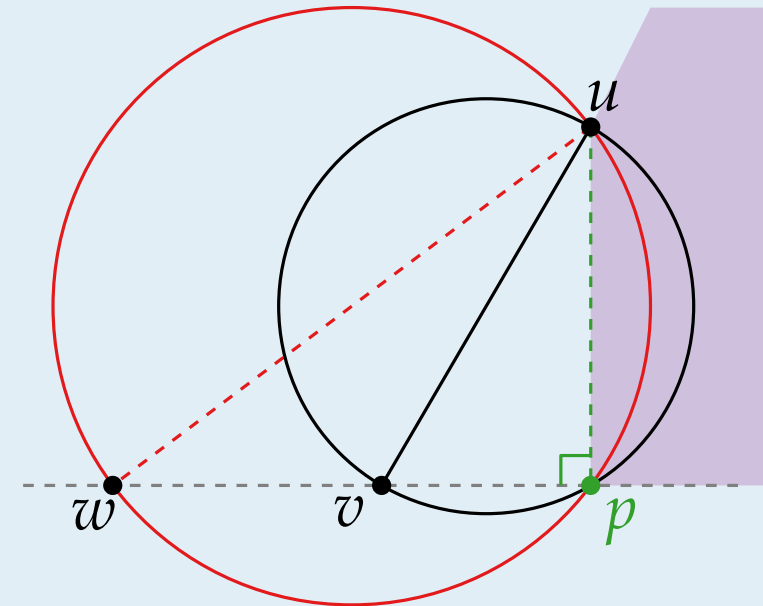
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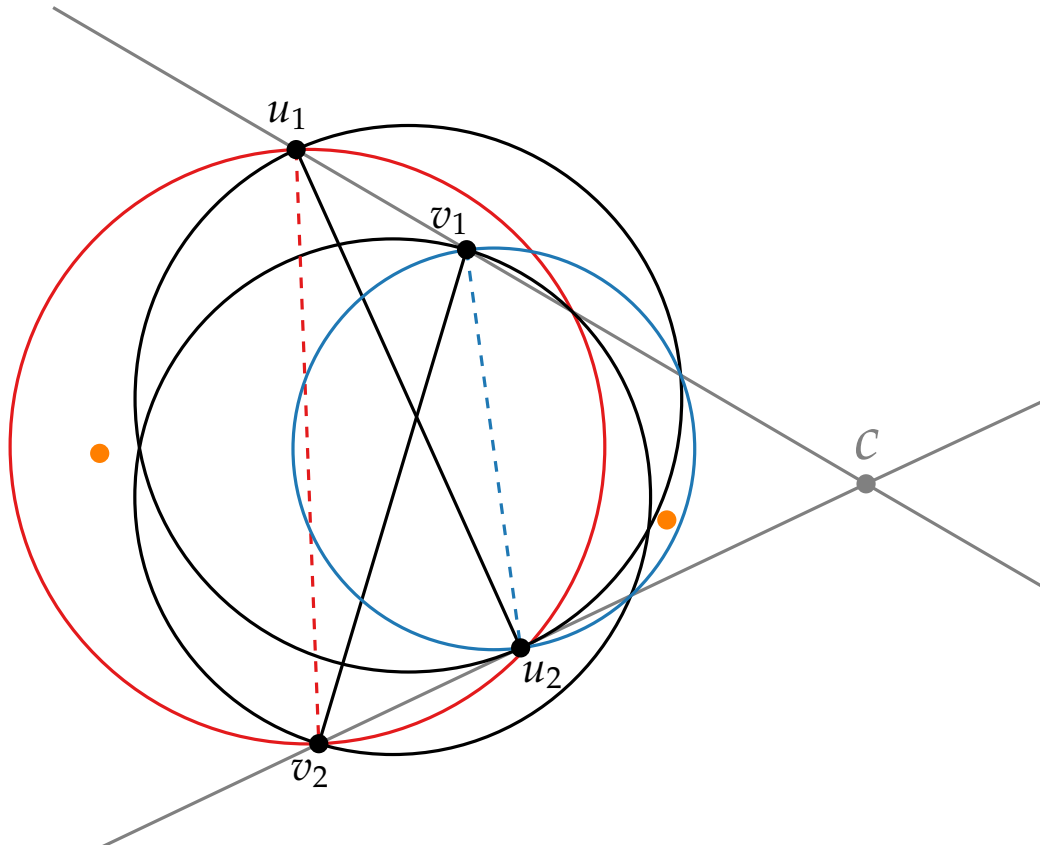


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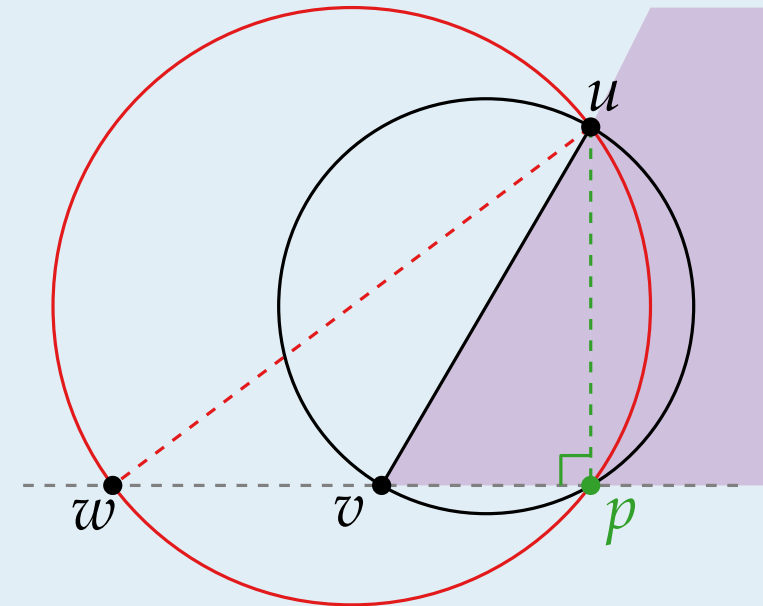
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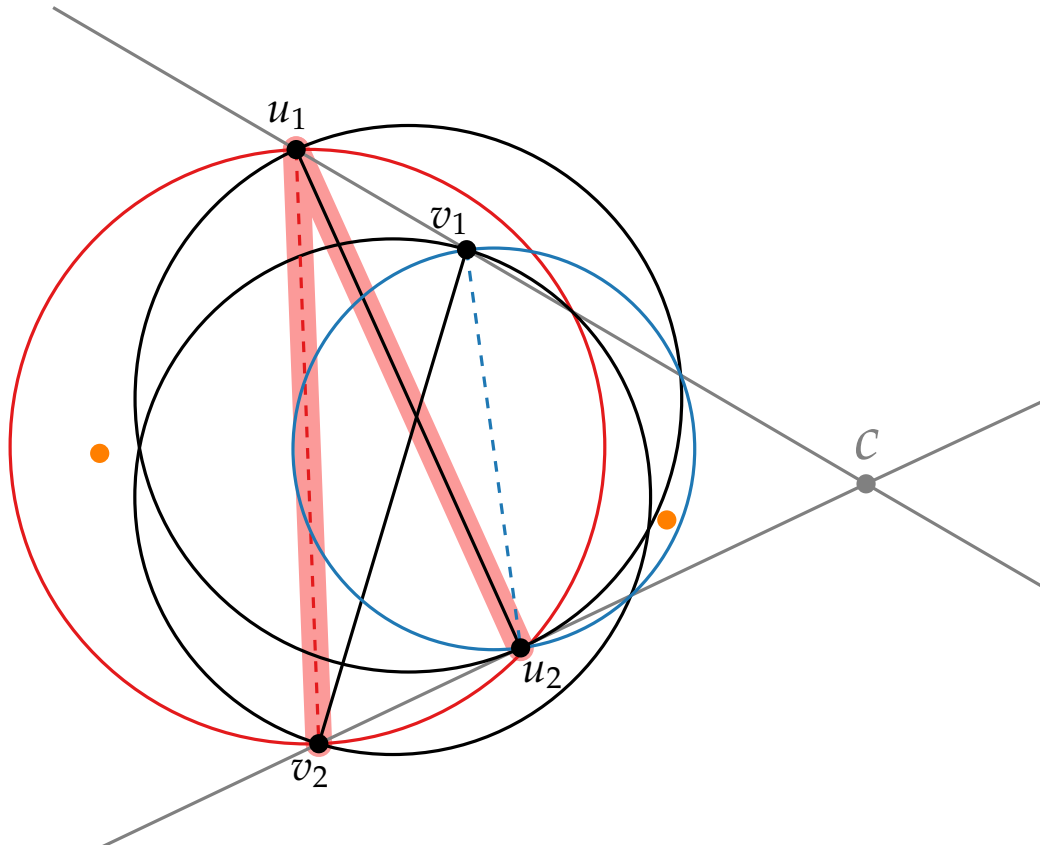


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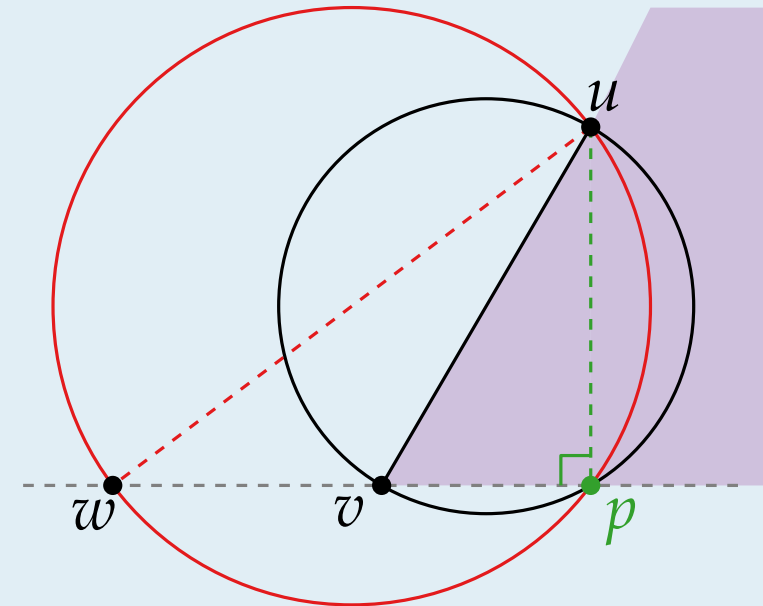
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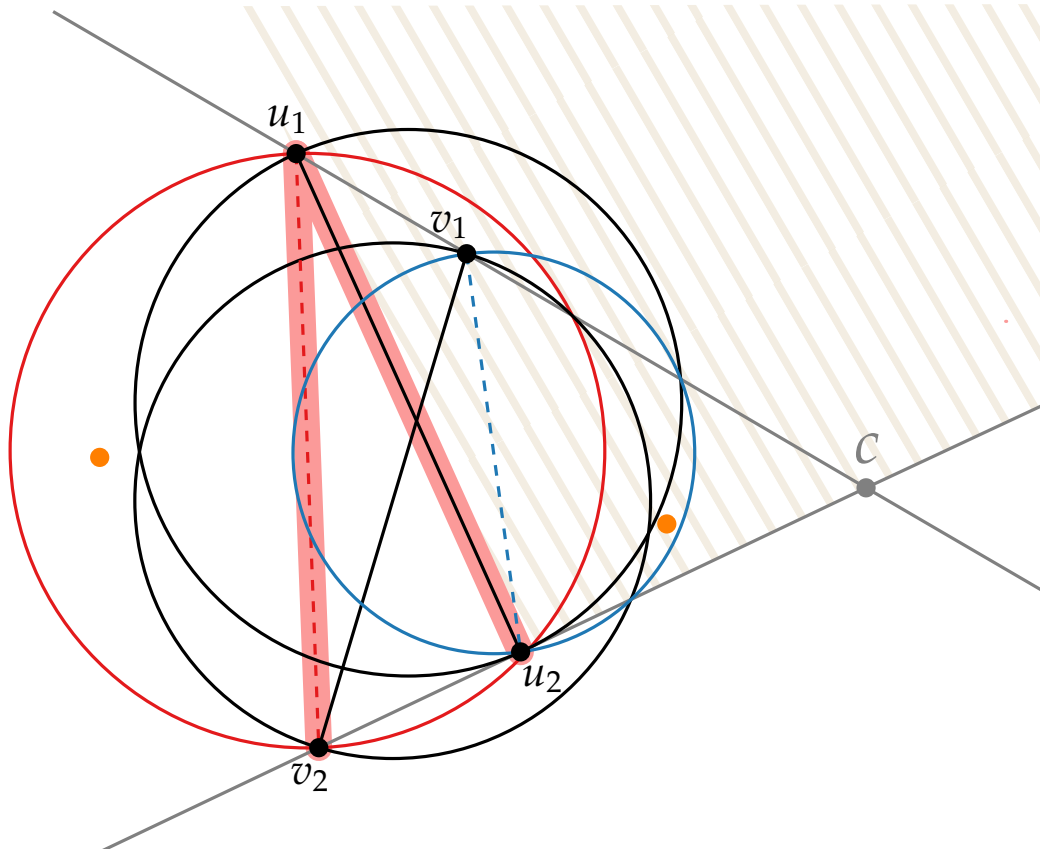


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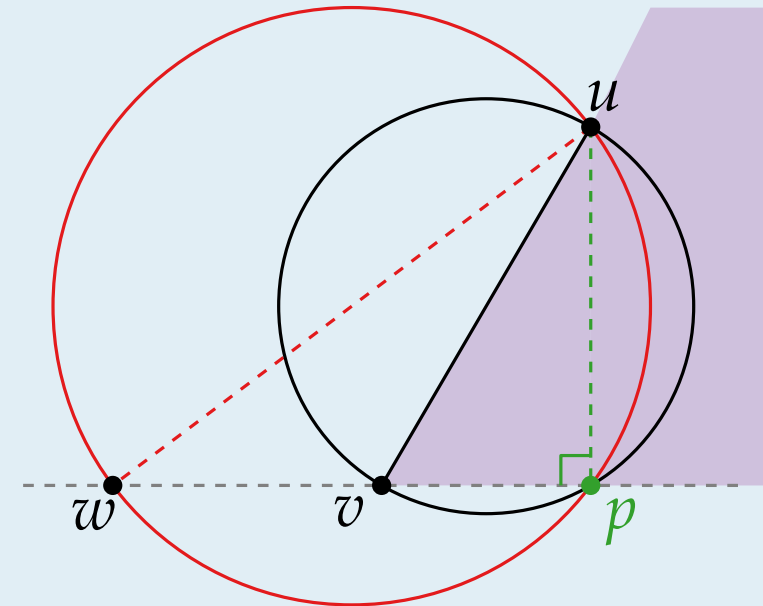
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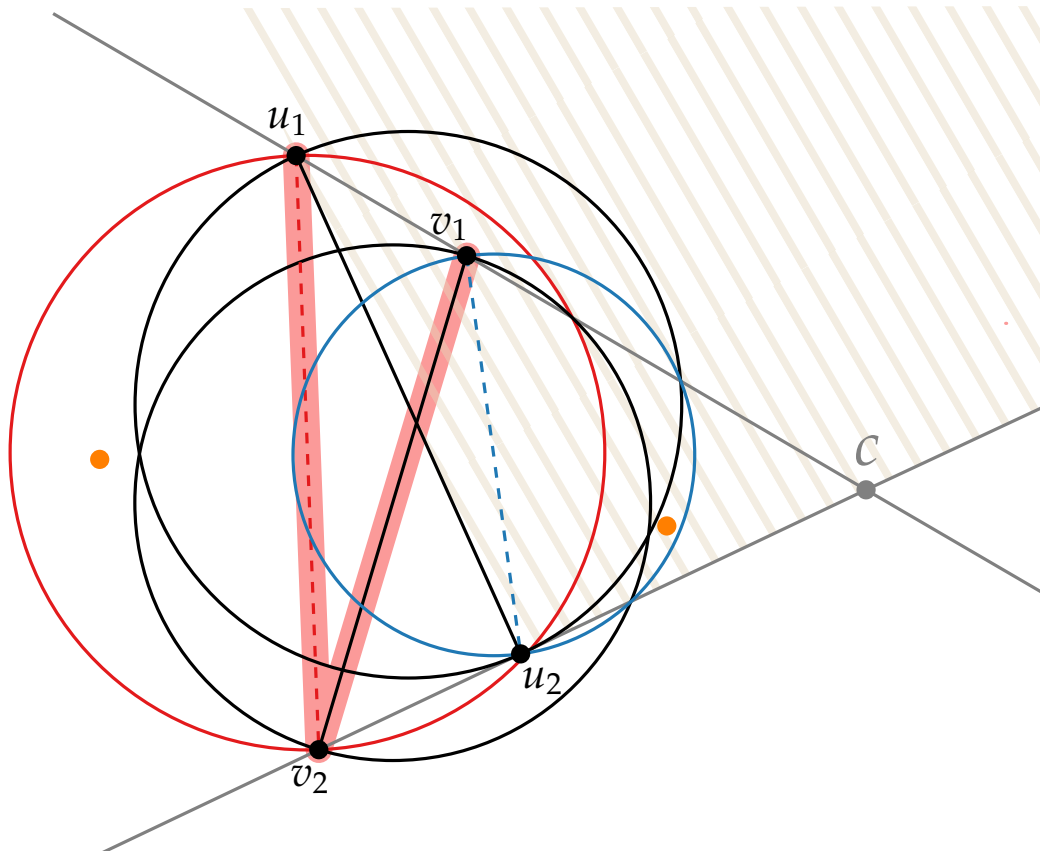


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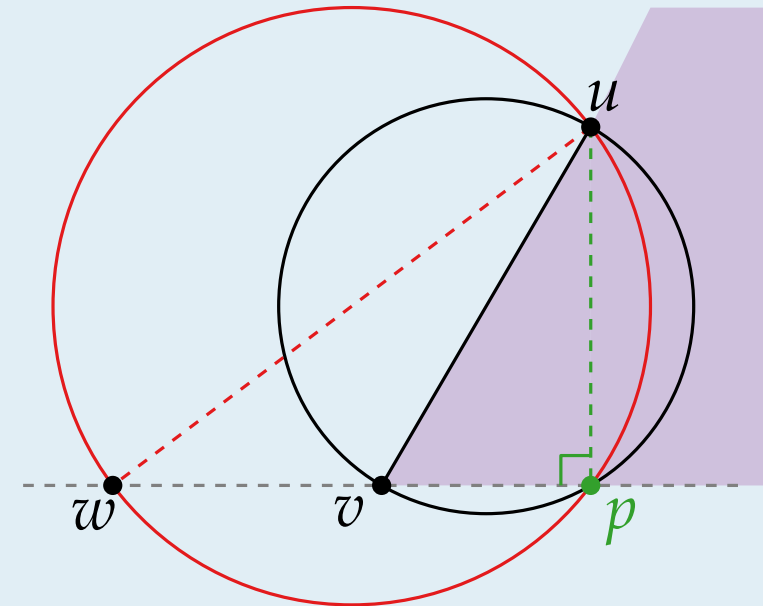
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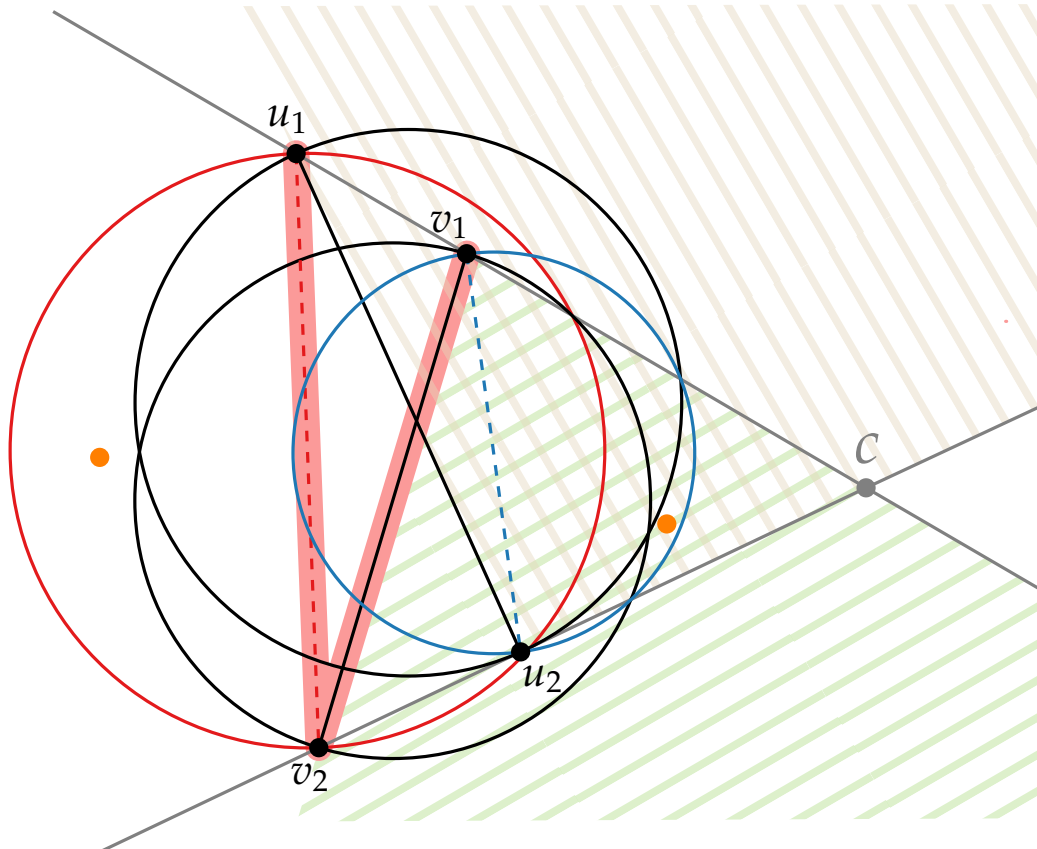


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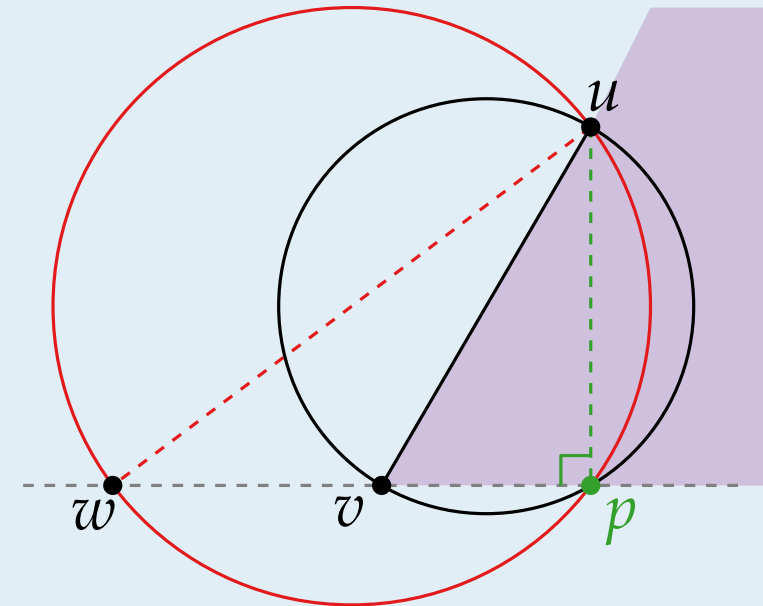
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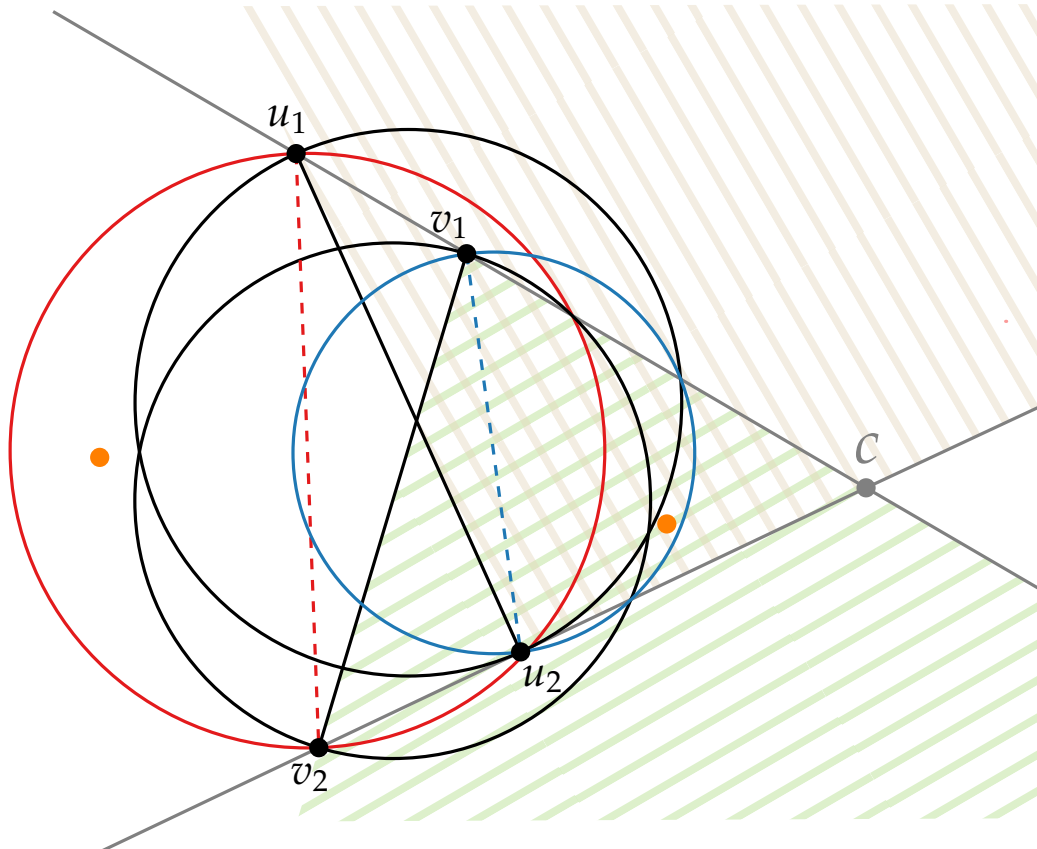


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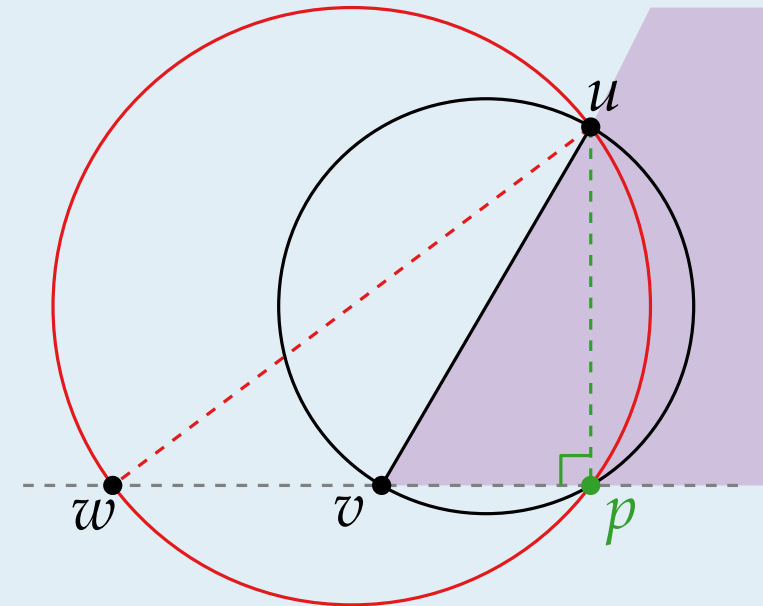
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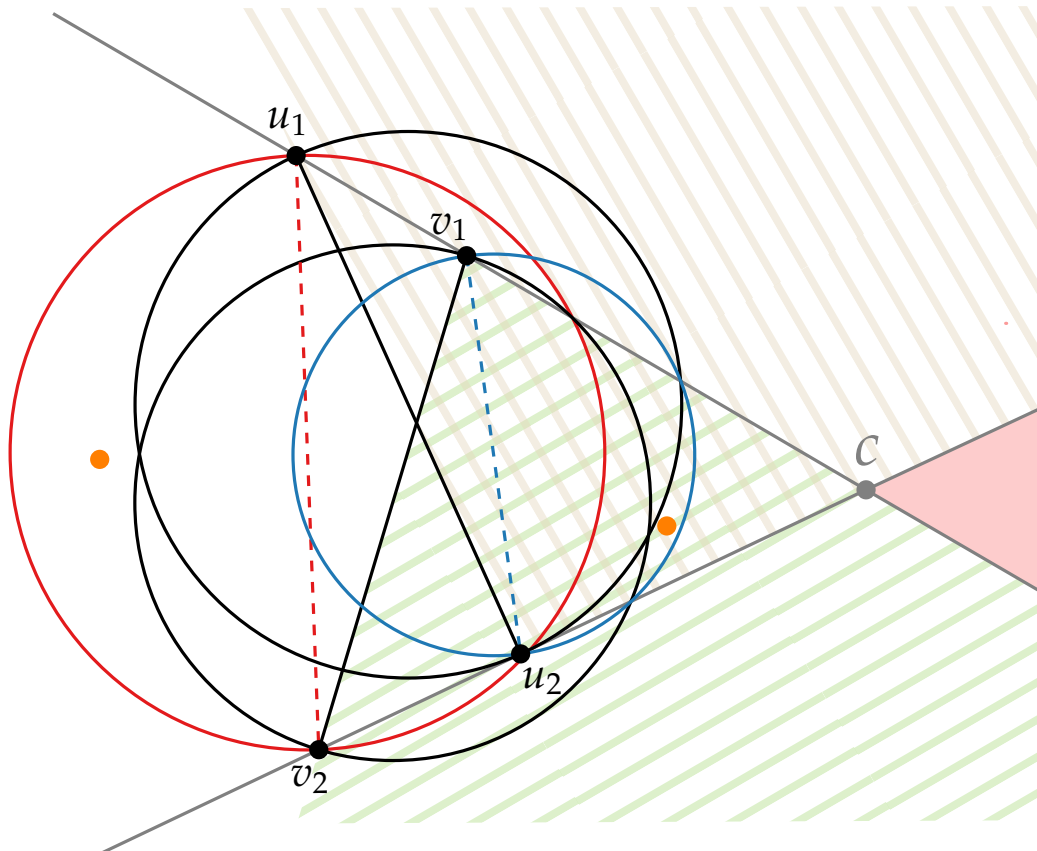


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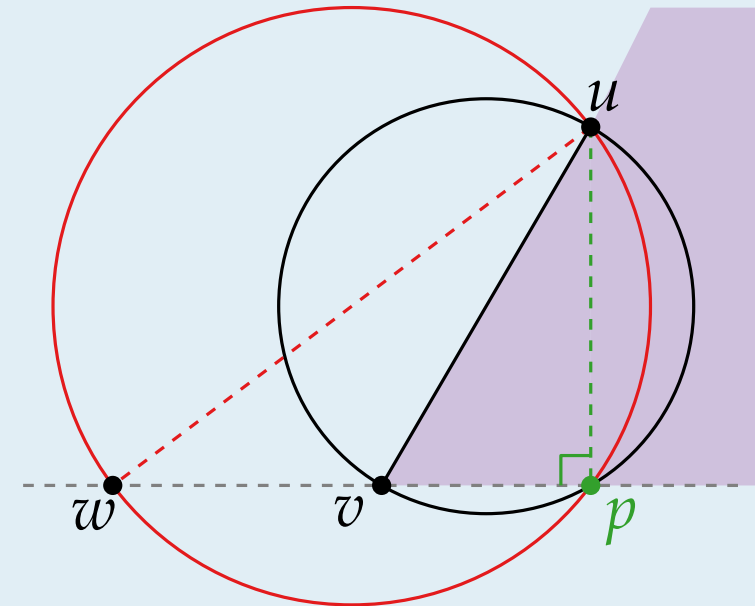
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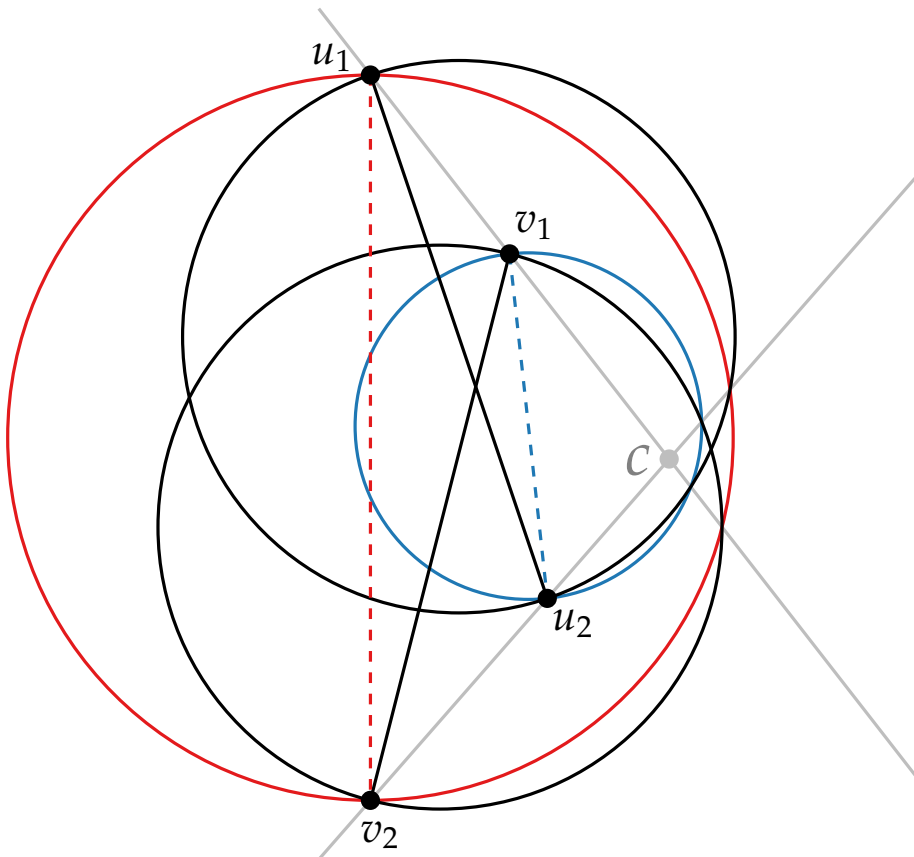


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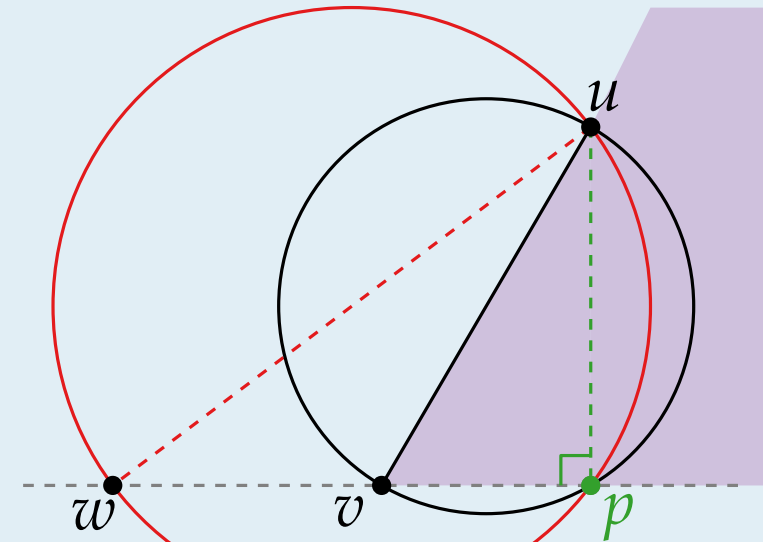
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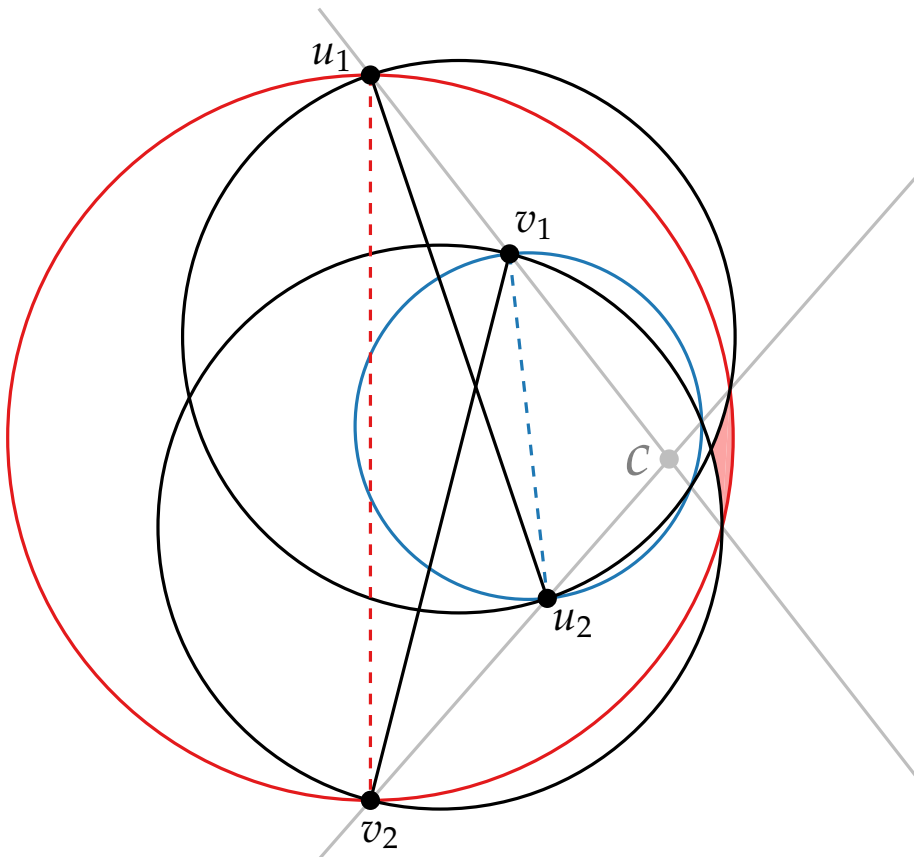


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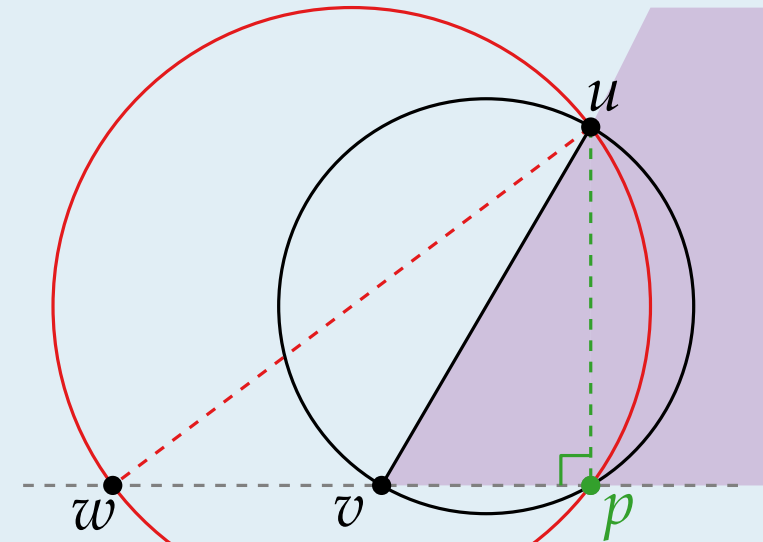
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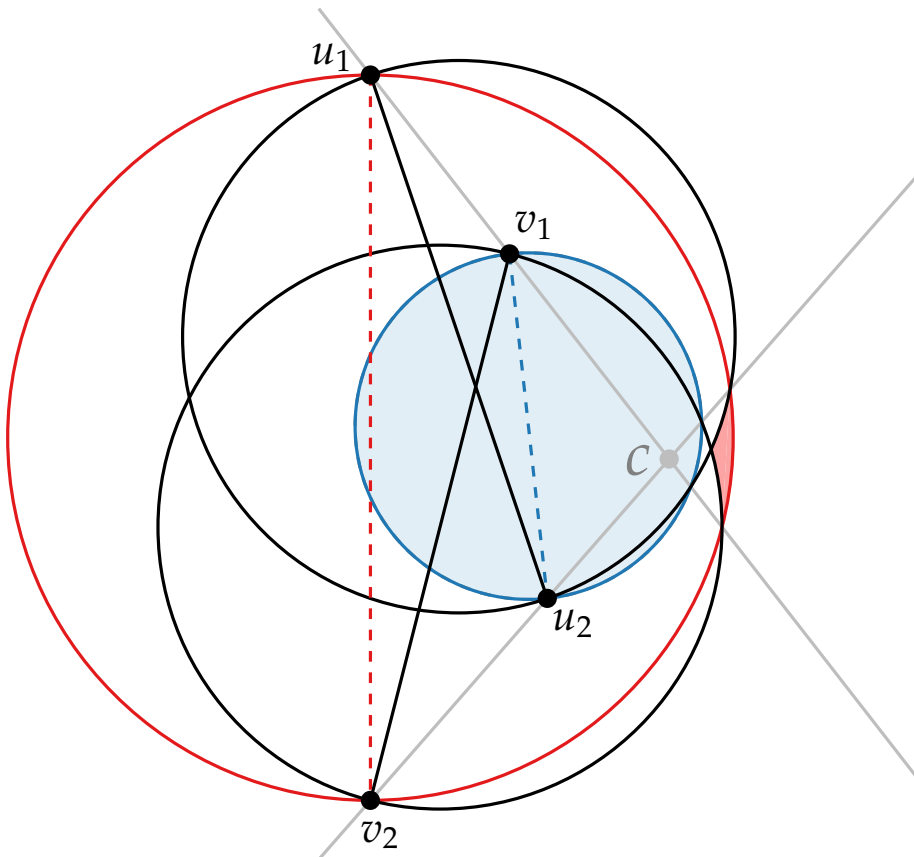


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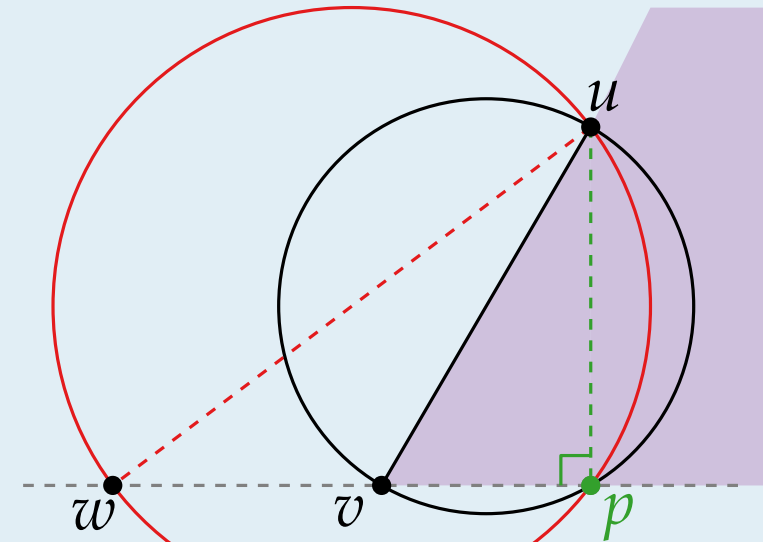
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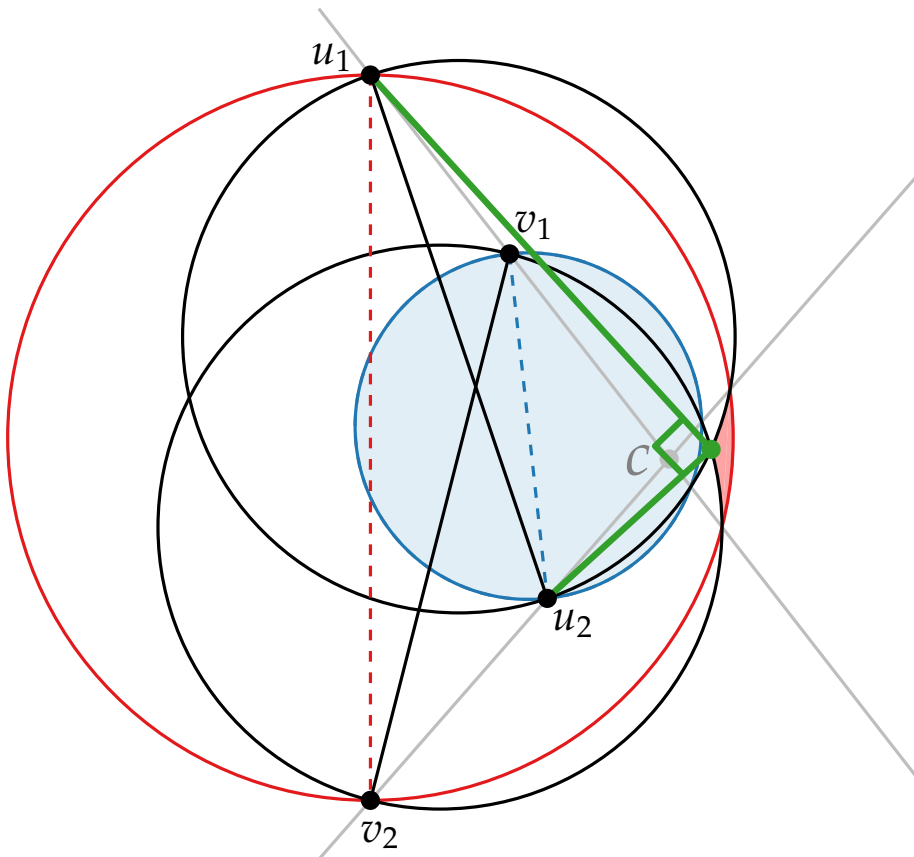


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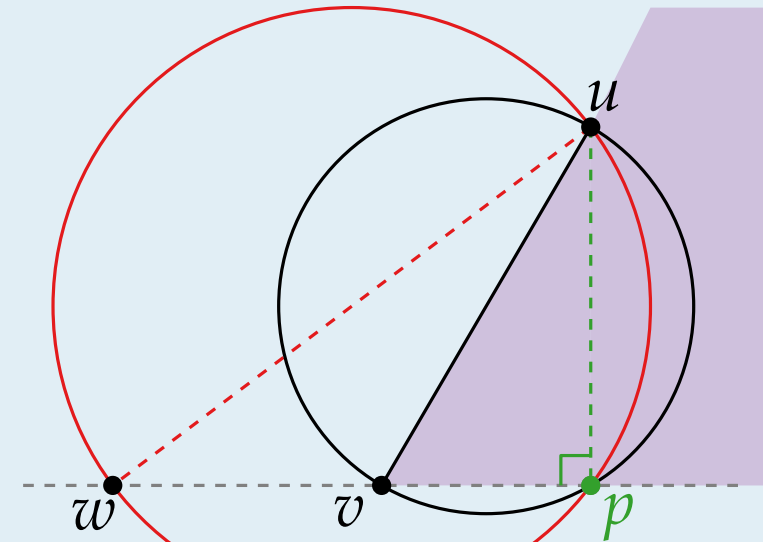
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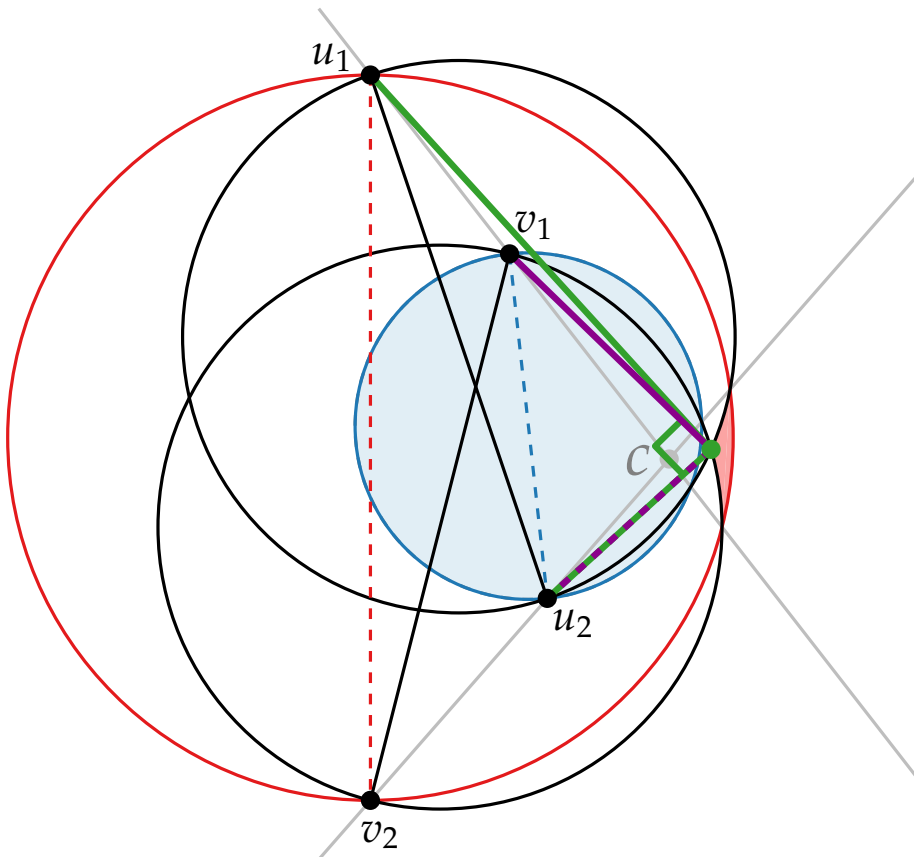


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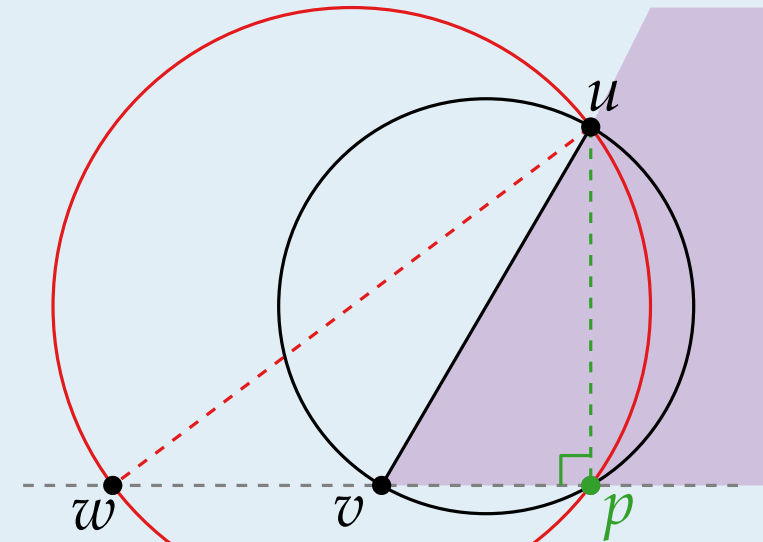
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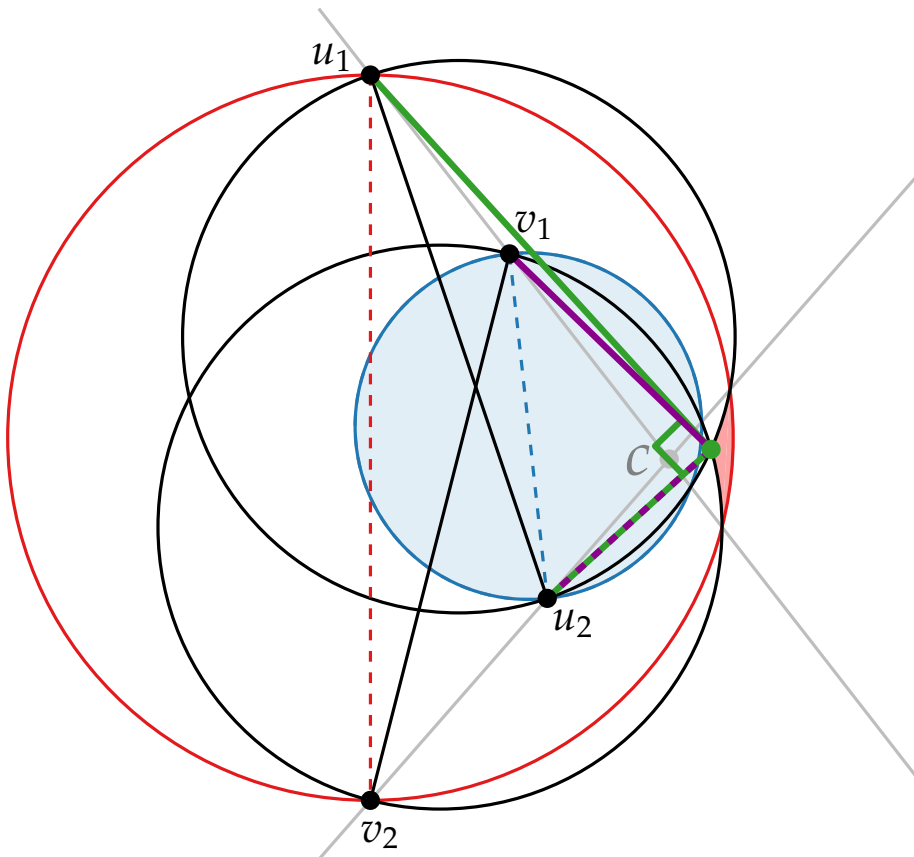


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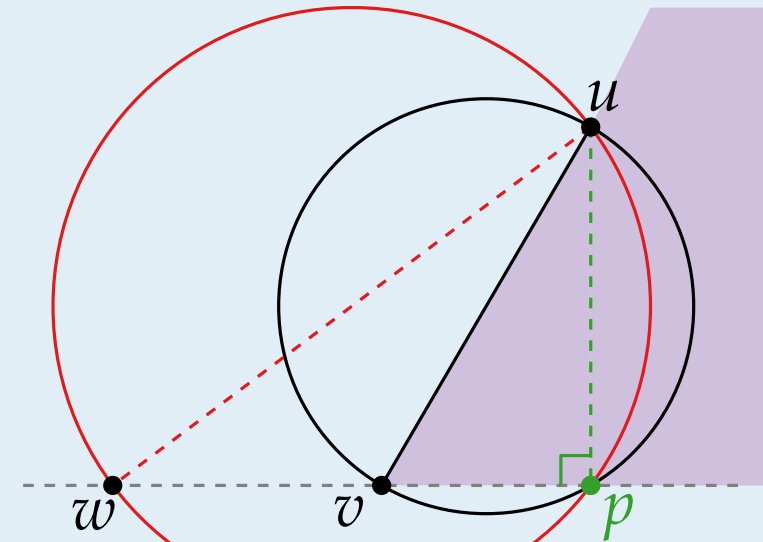
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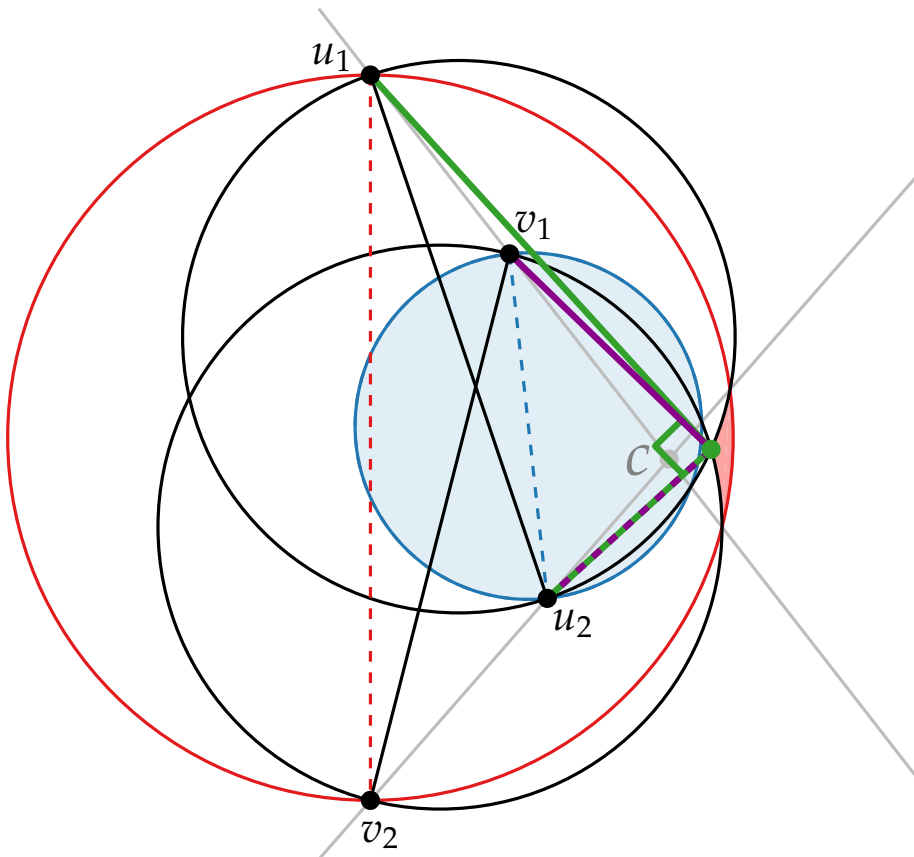


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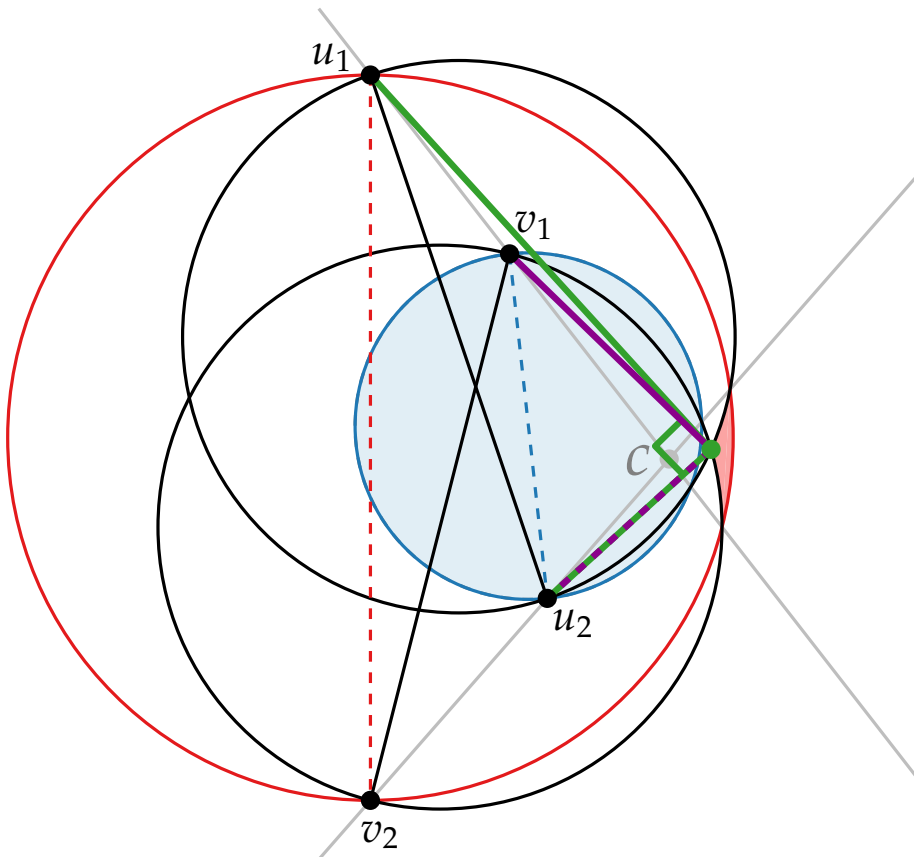
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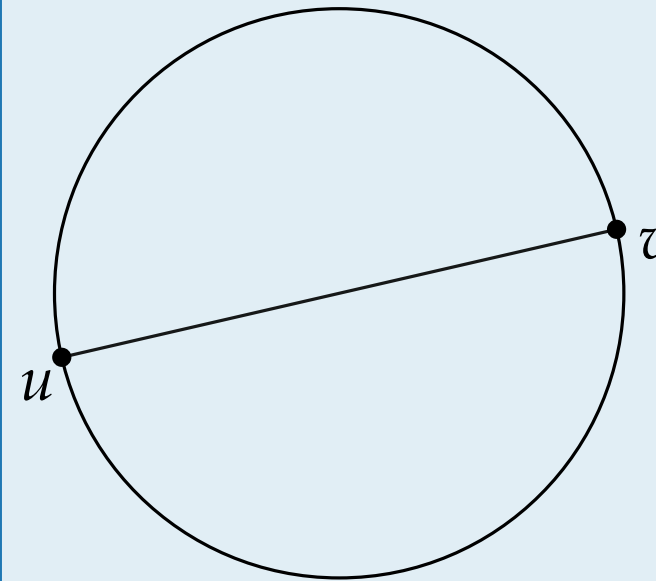
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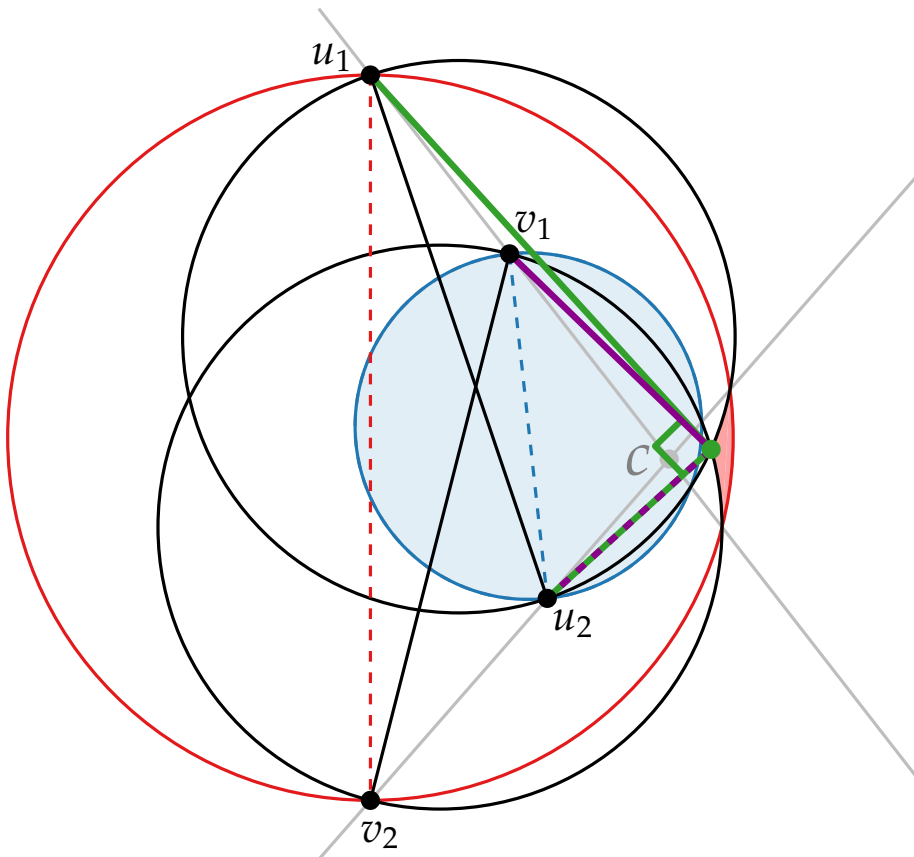


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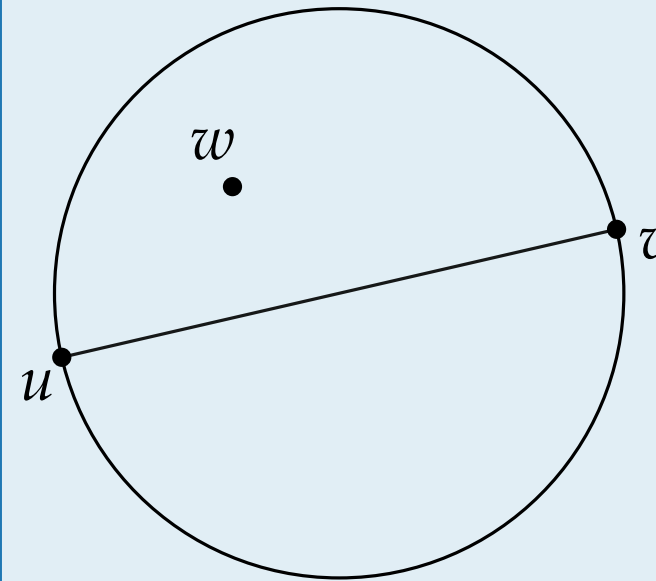
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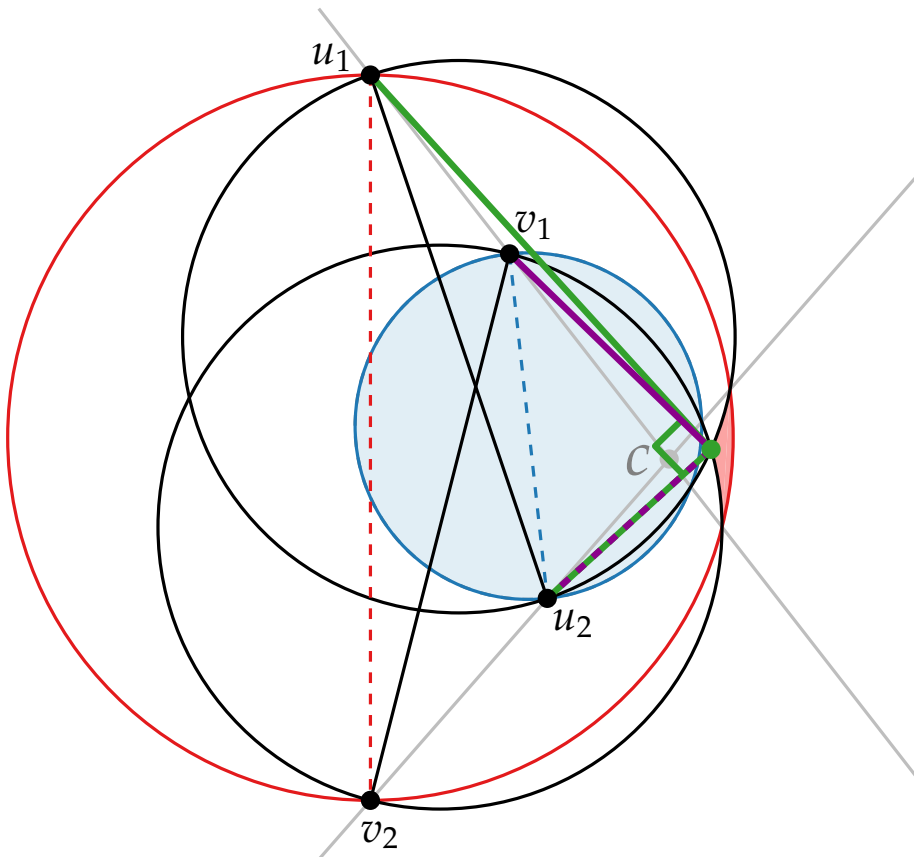


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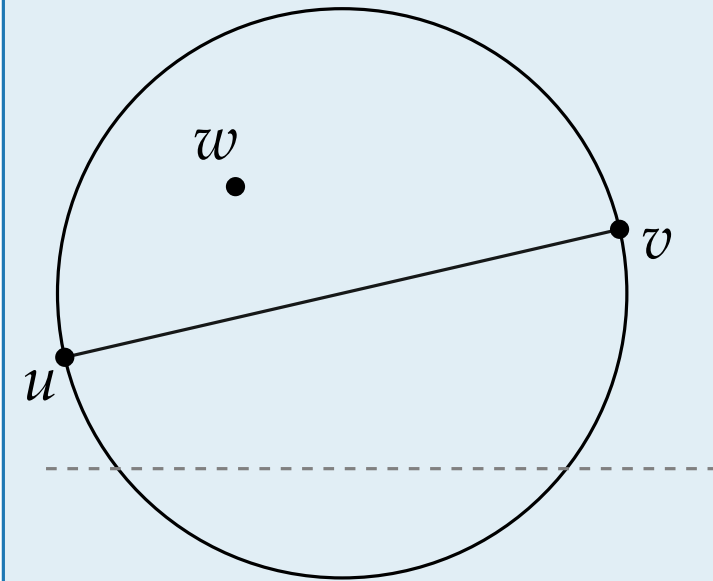
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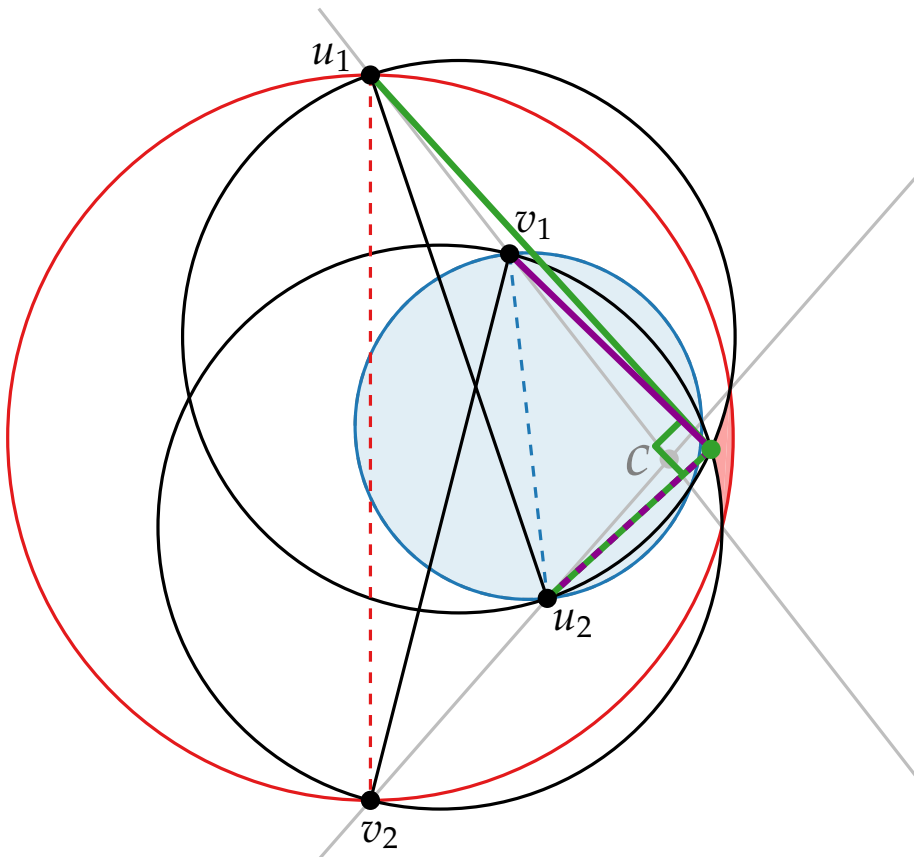


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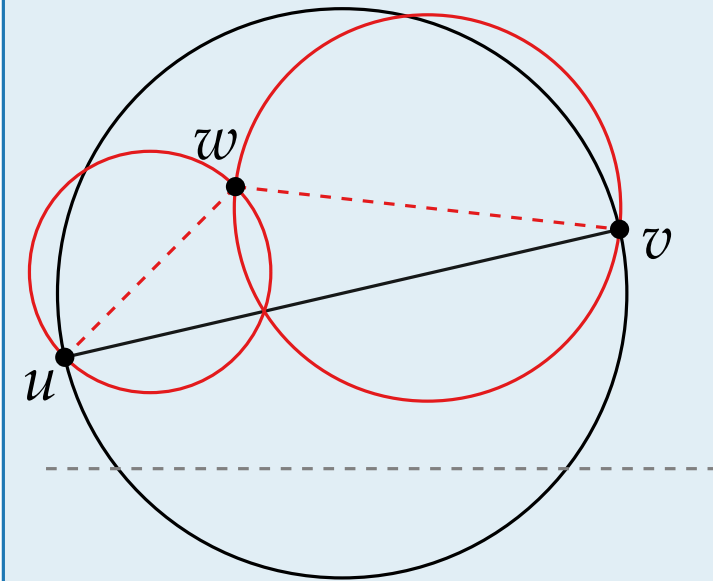
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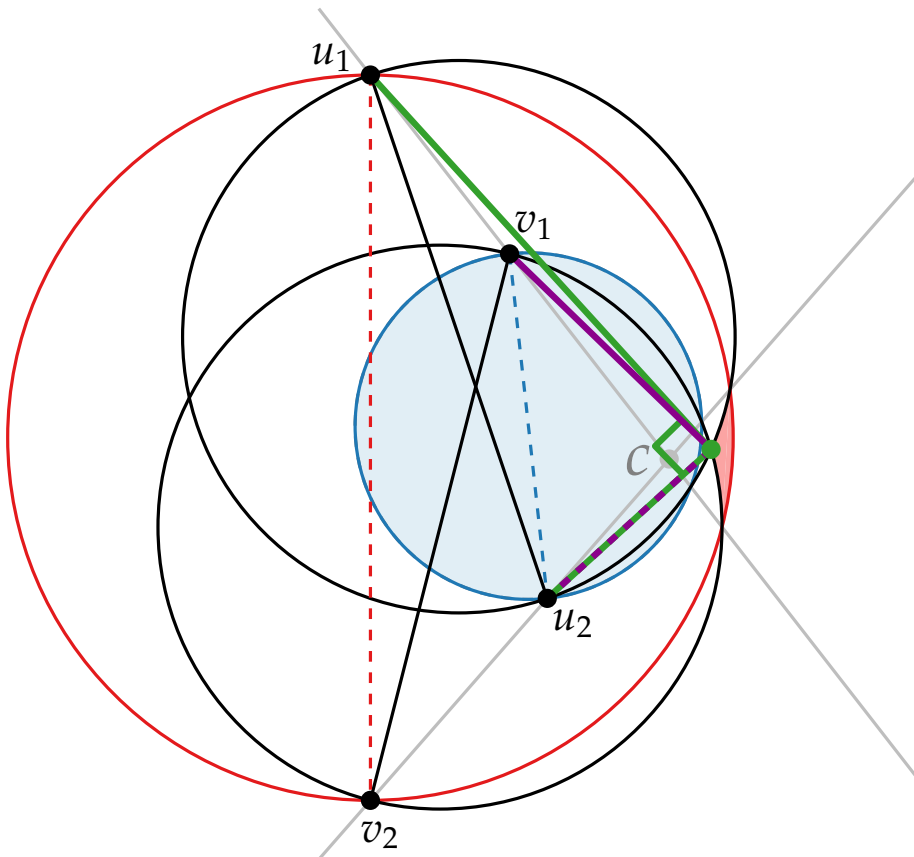


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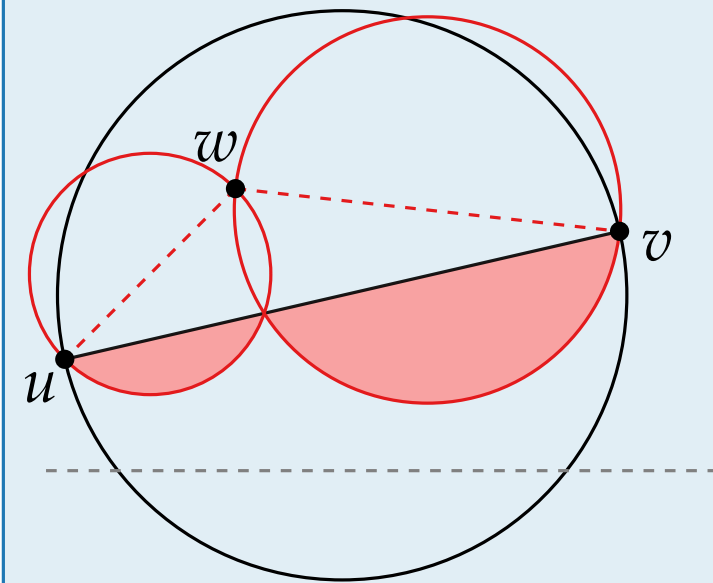
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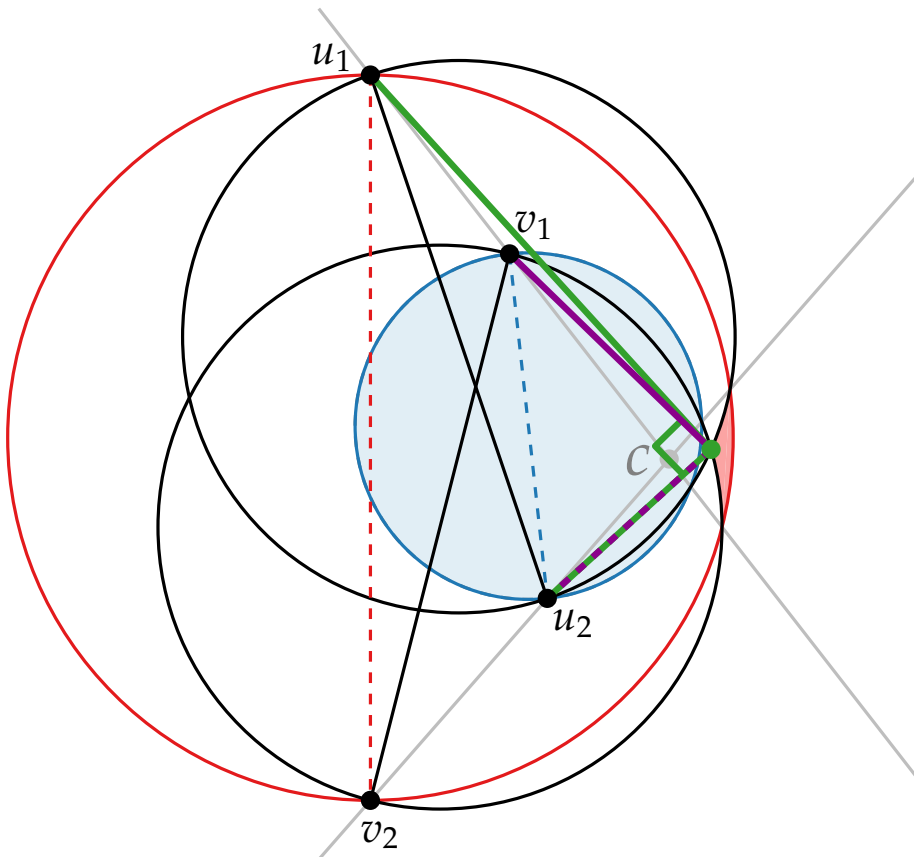


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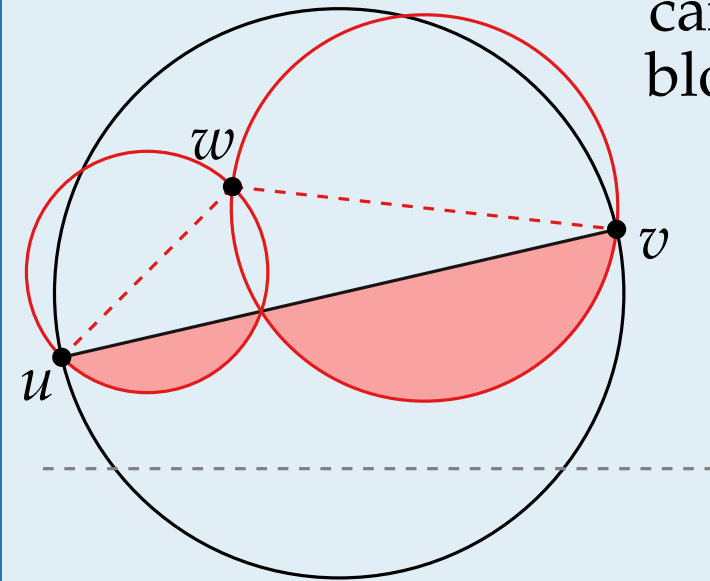
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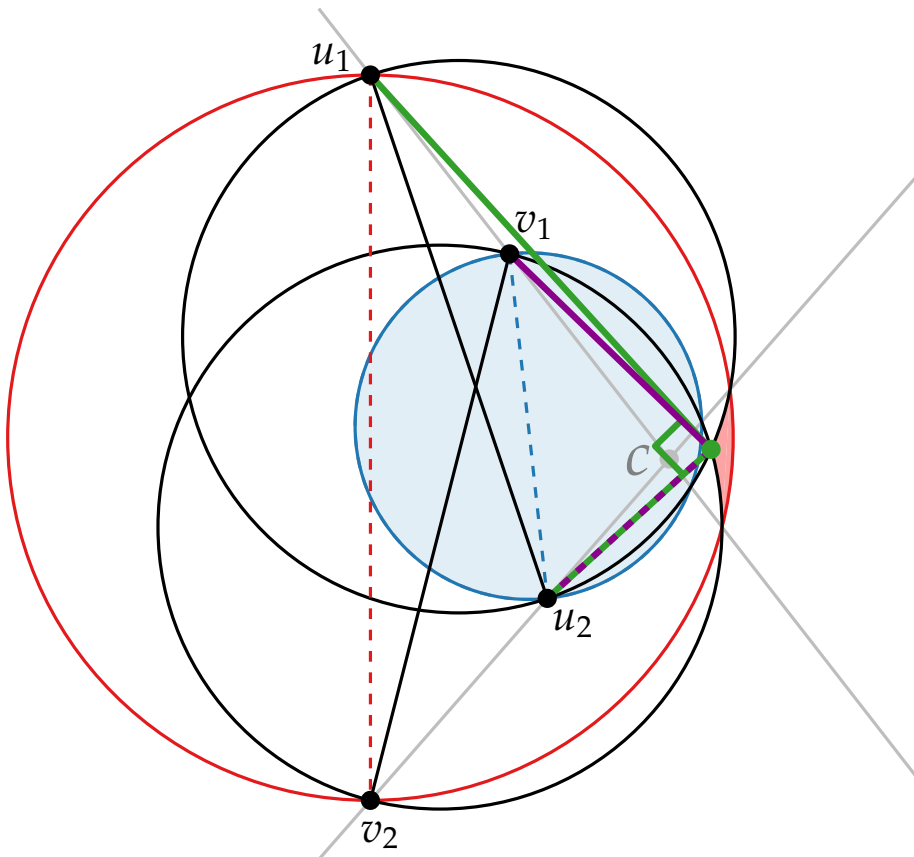


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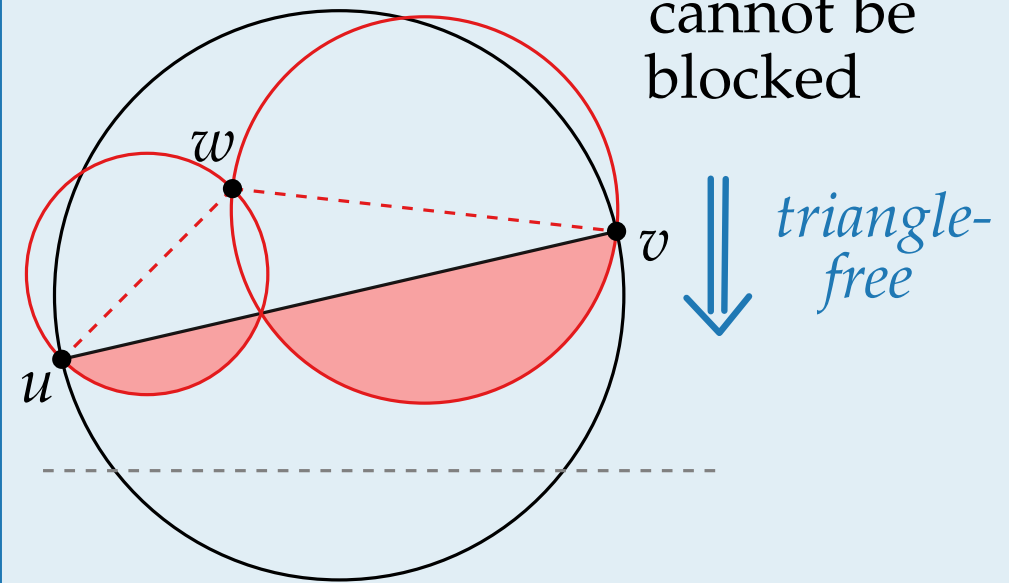
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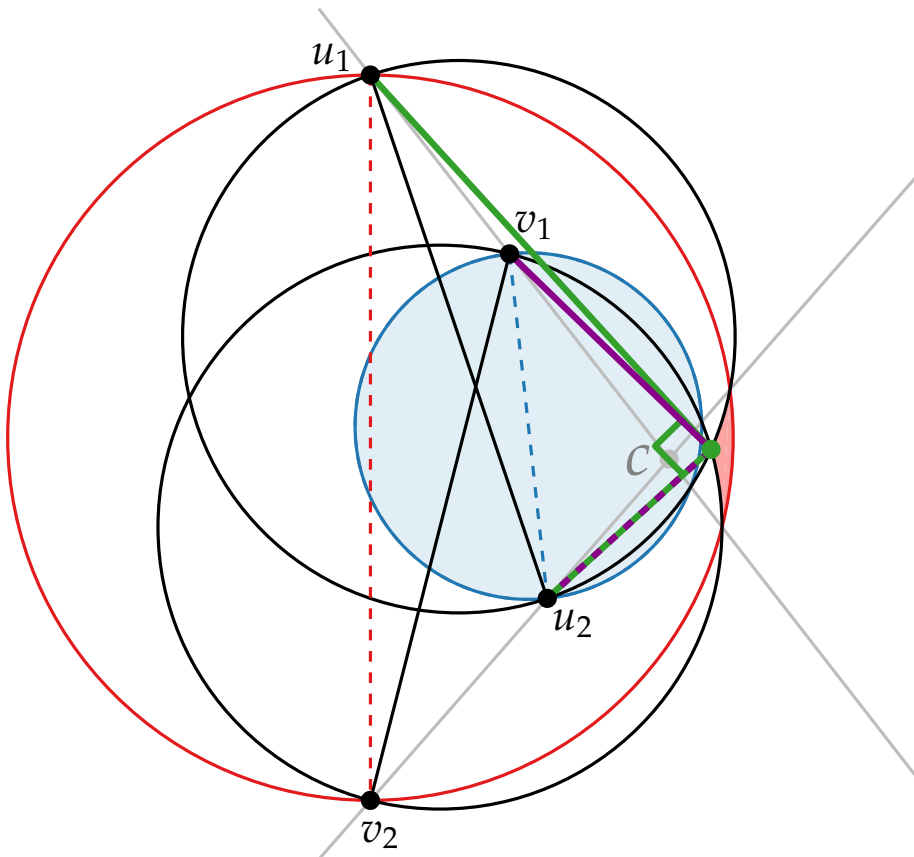


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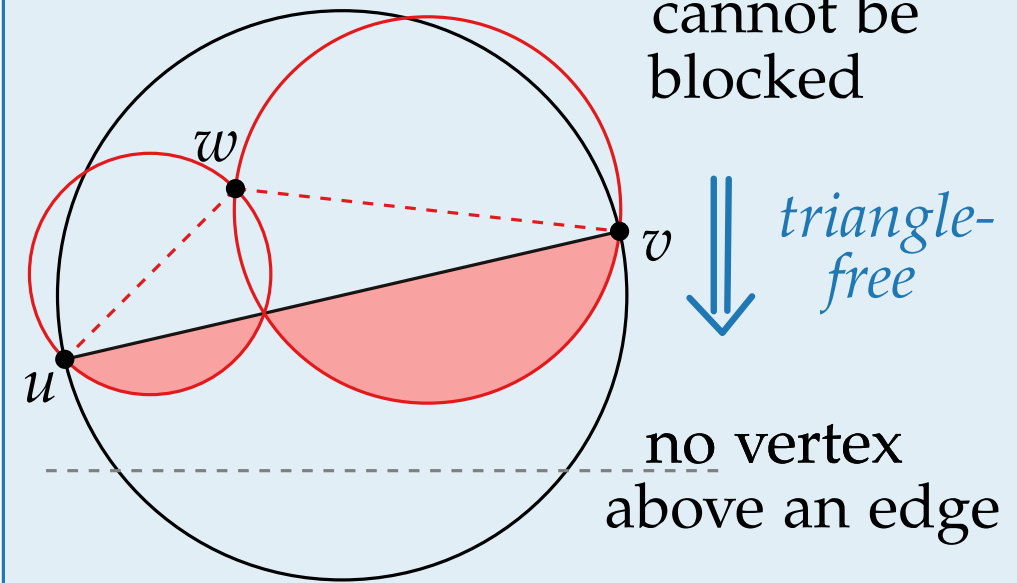
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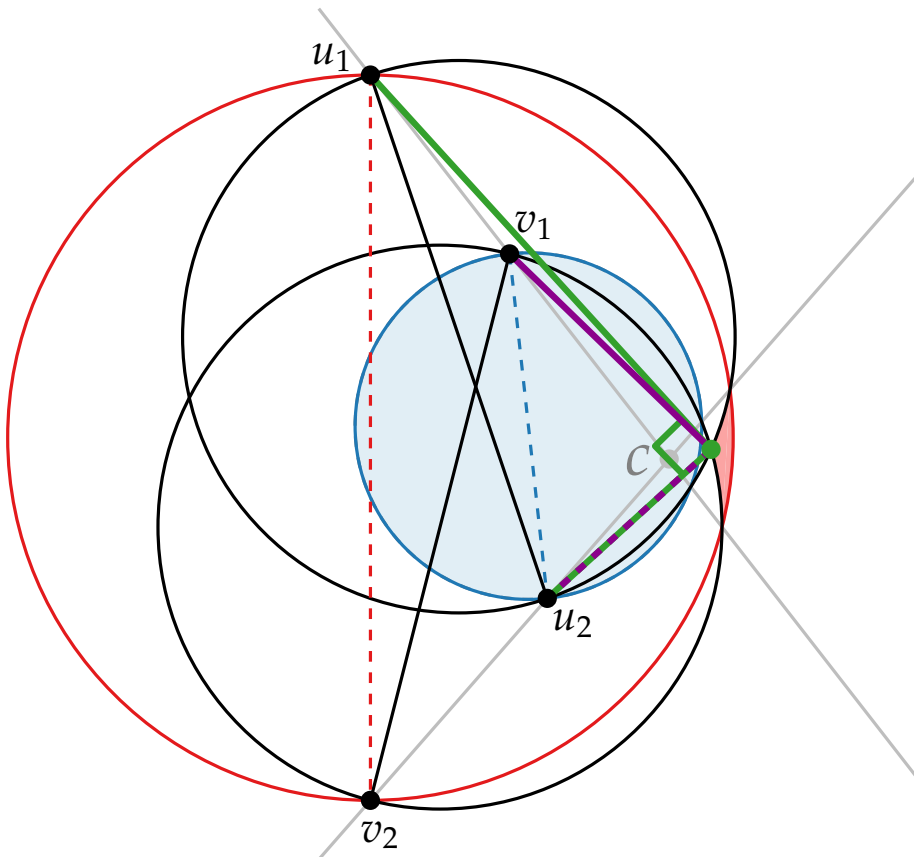


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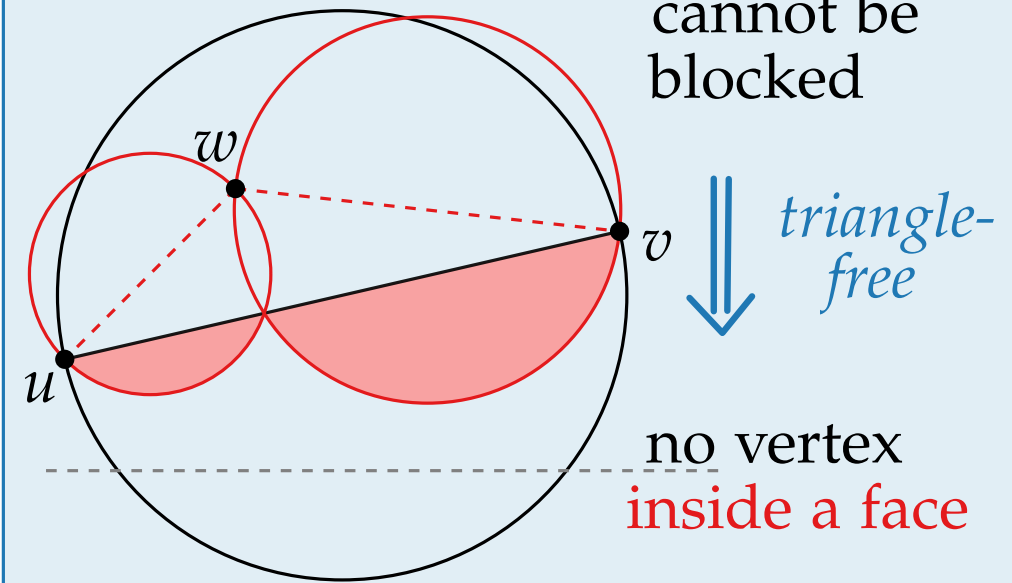
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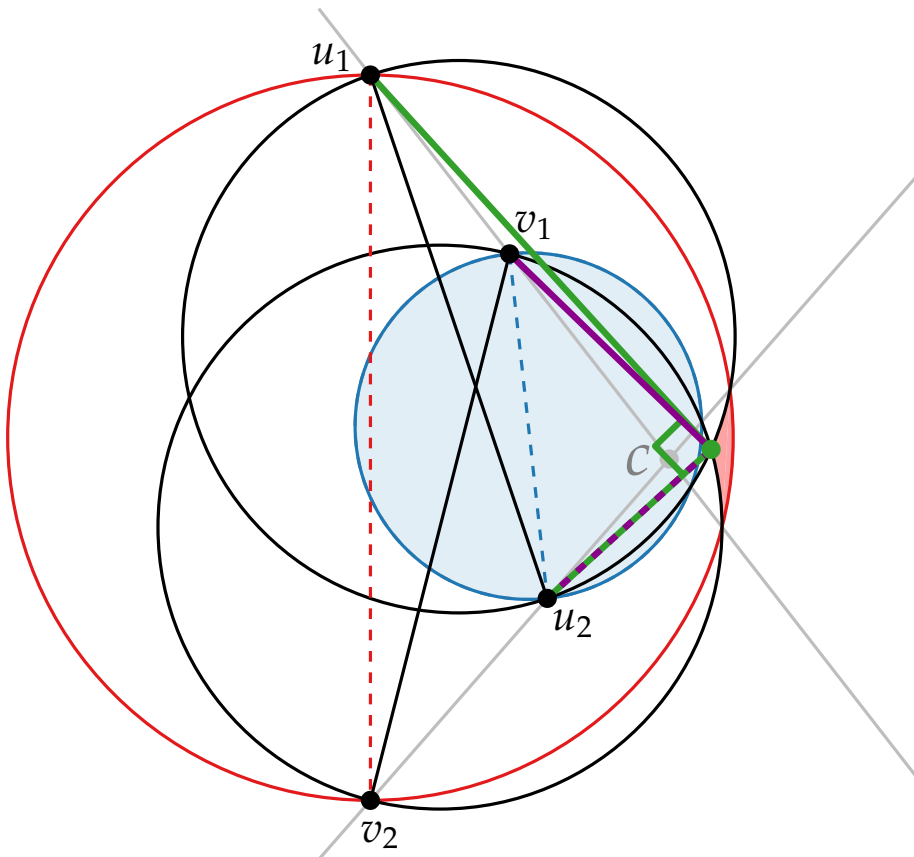
no vertex
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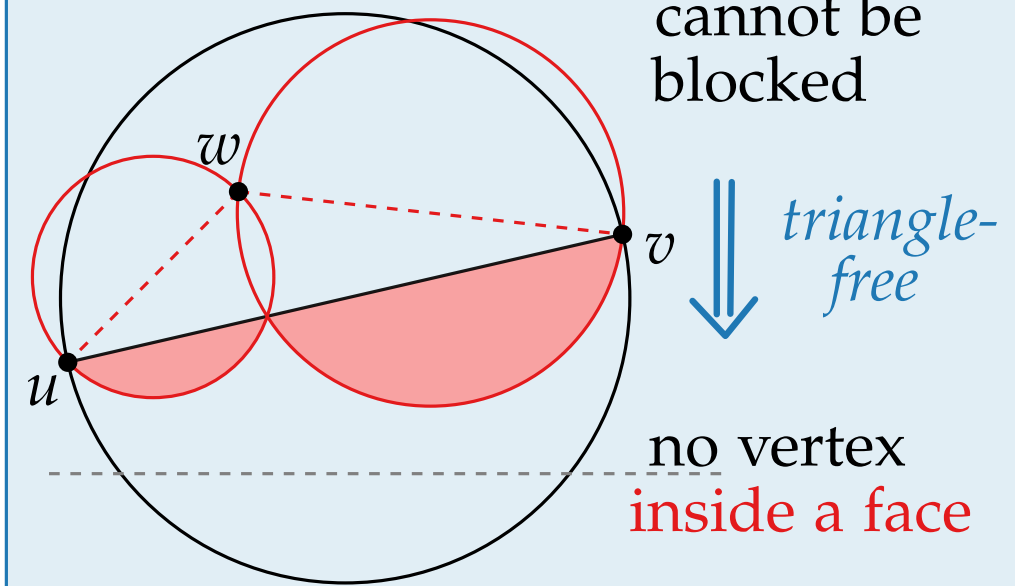
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Convexly Separable Witness Gabriel Drawings

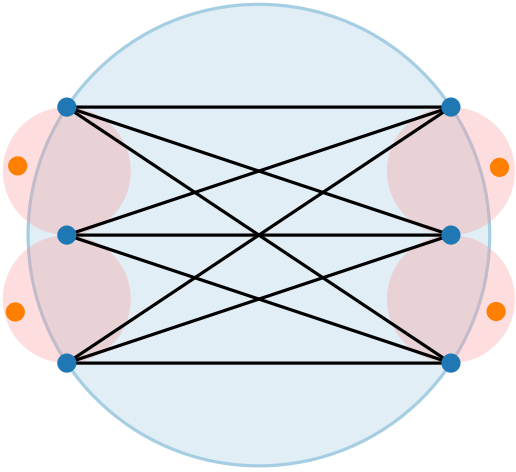
Convexly Separable Witness Gabriel Drawings



no witnesses in the interior of the convex hull of the vertex set

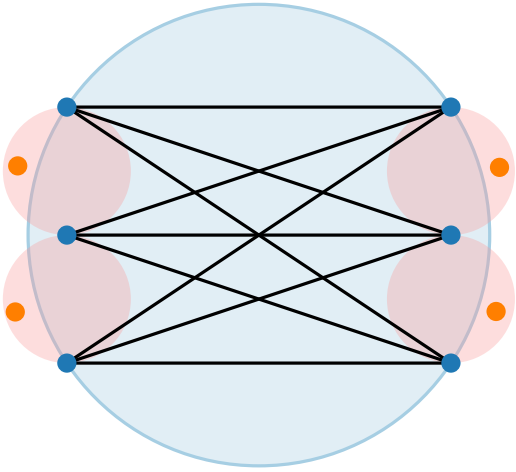
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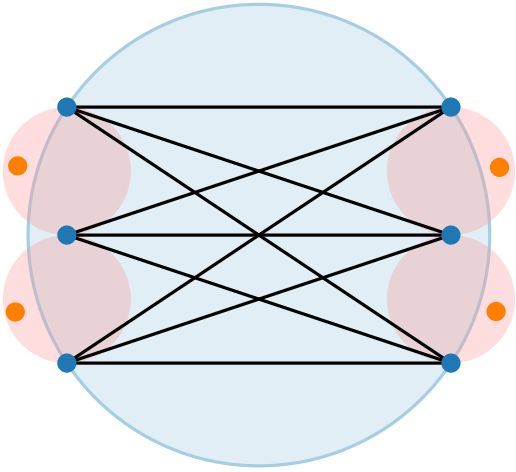
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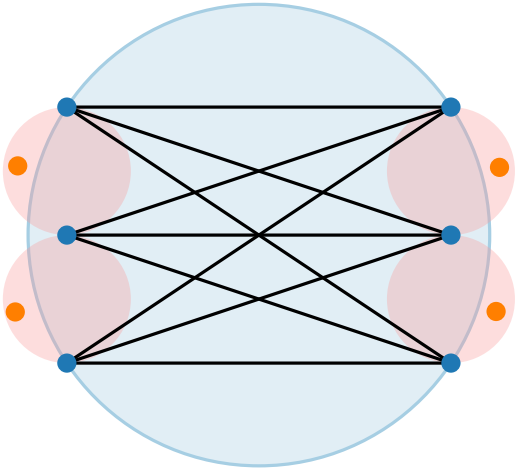
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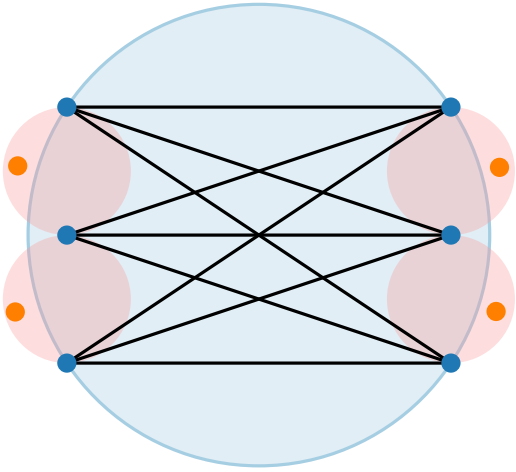
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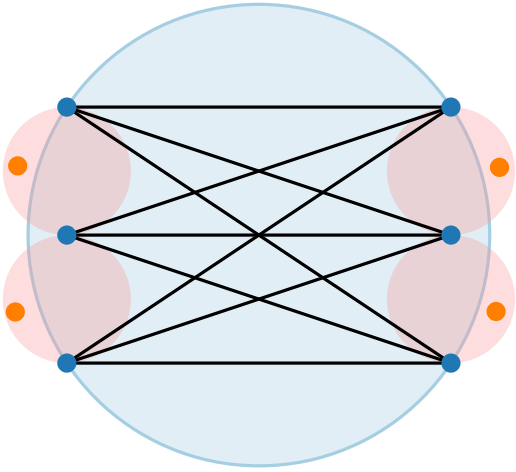
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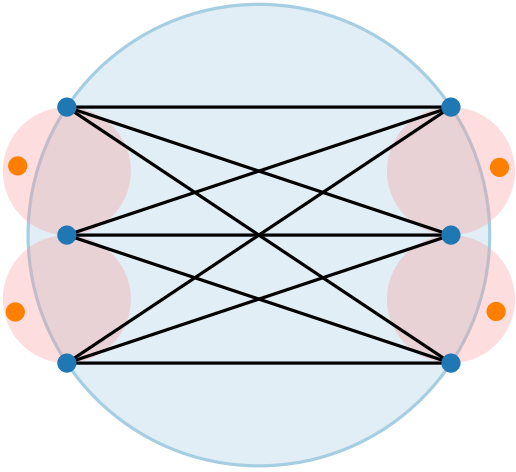
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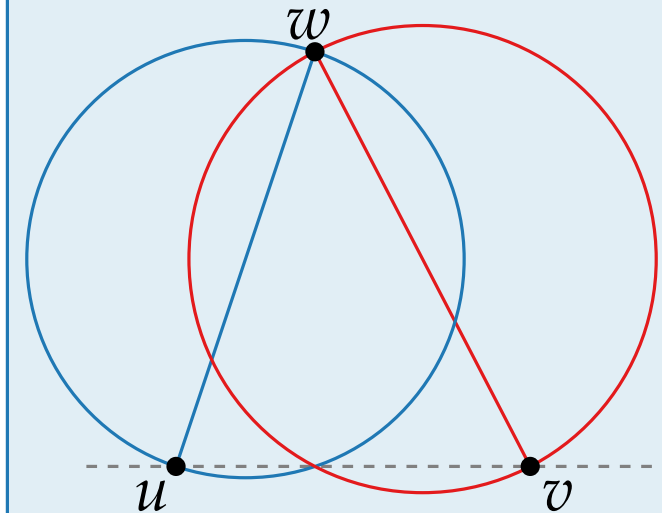
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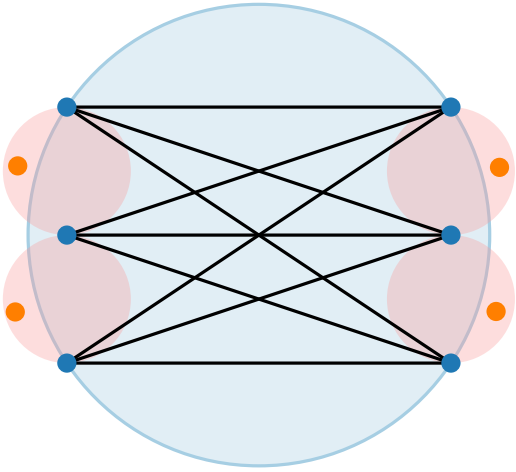
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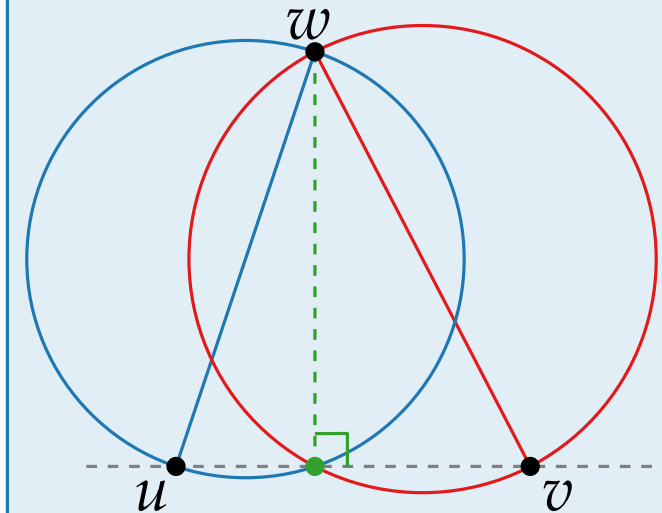
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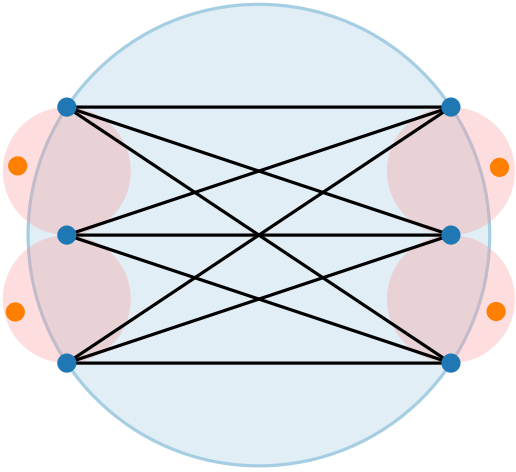
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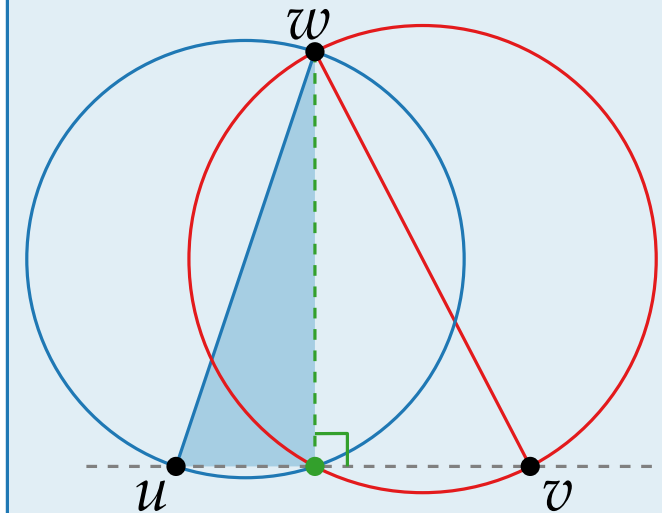
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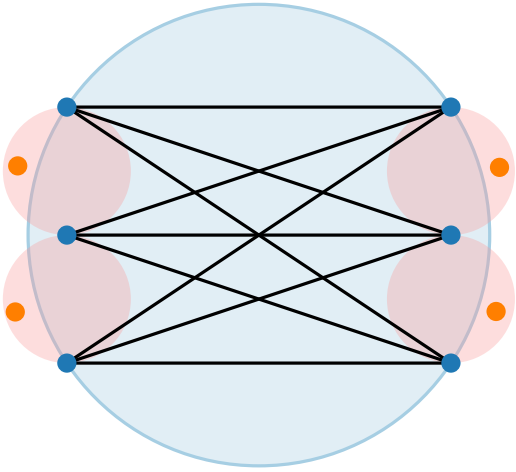
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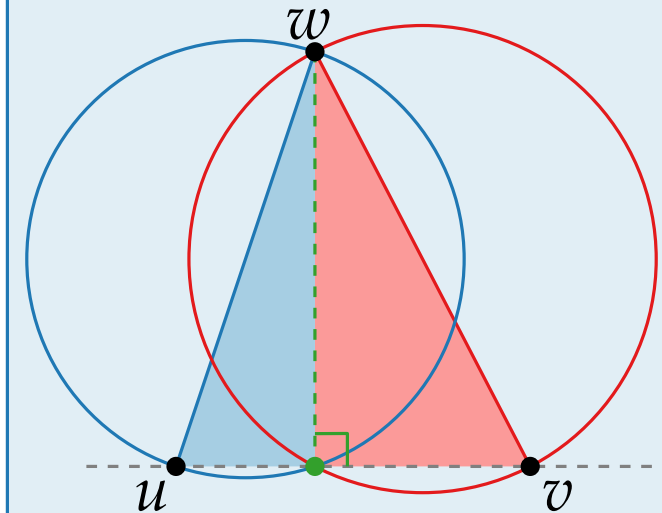
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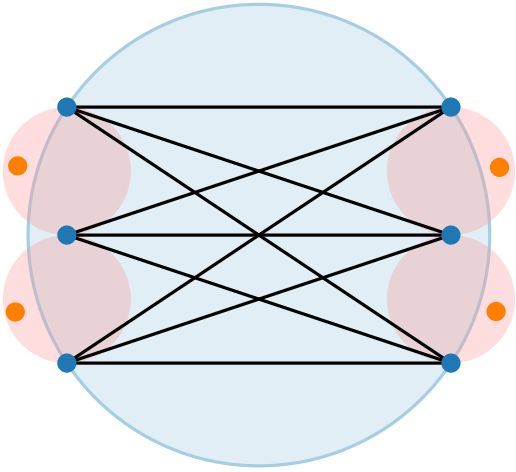
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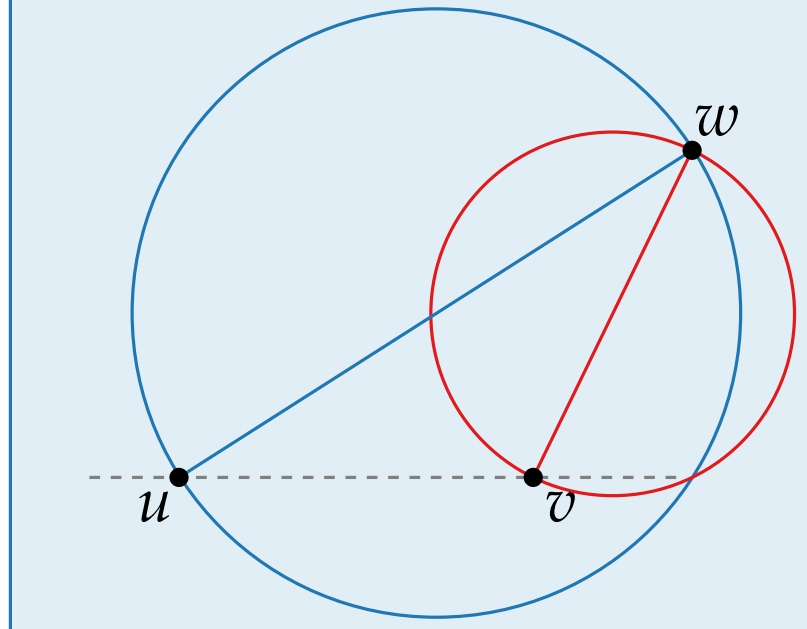
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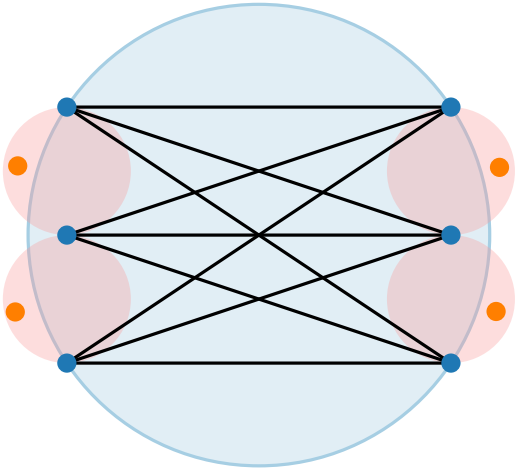
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Convexly Separable Witness Gabriel Drawings

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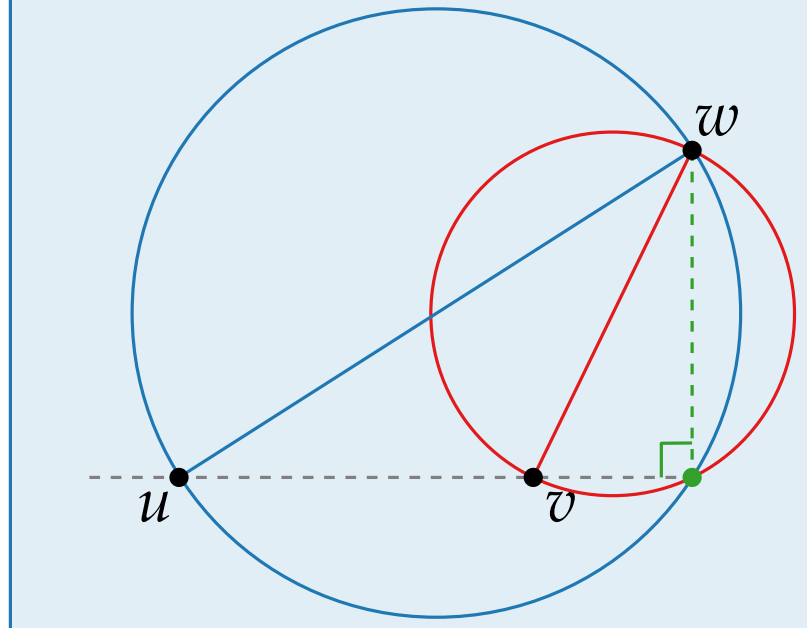
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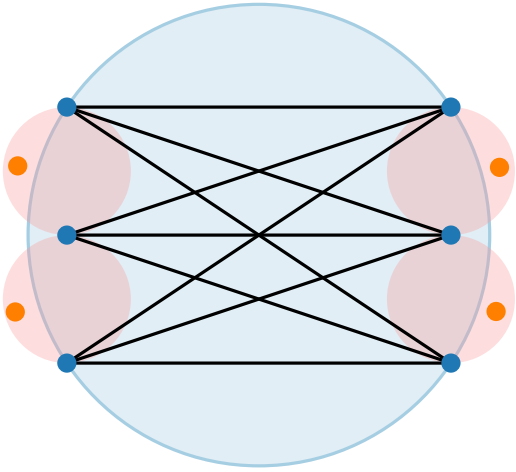
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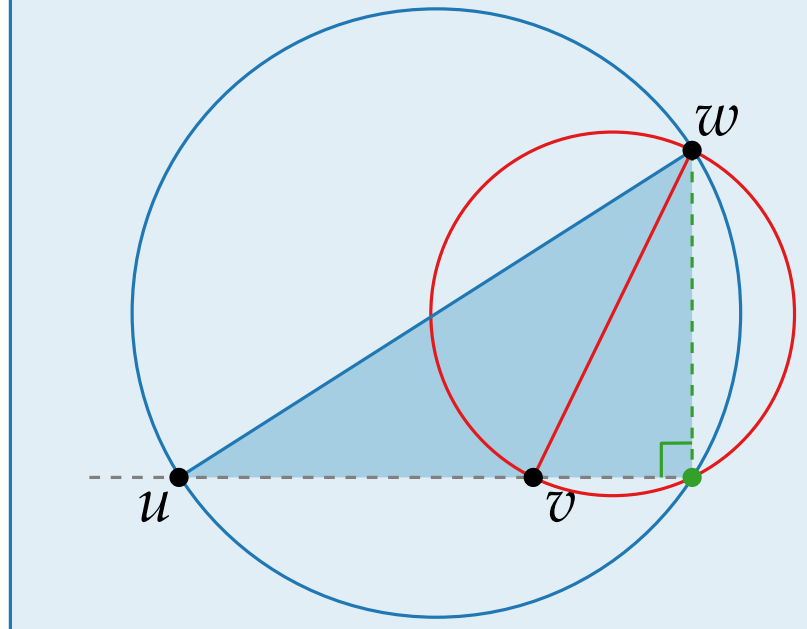
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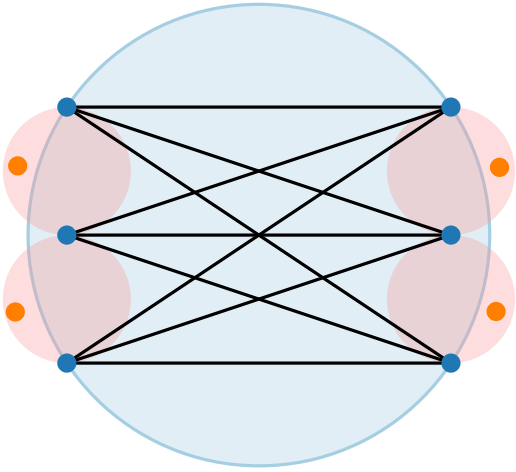
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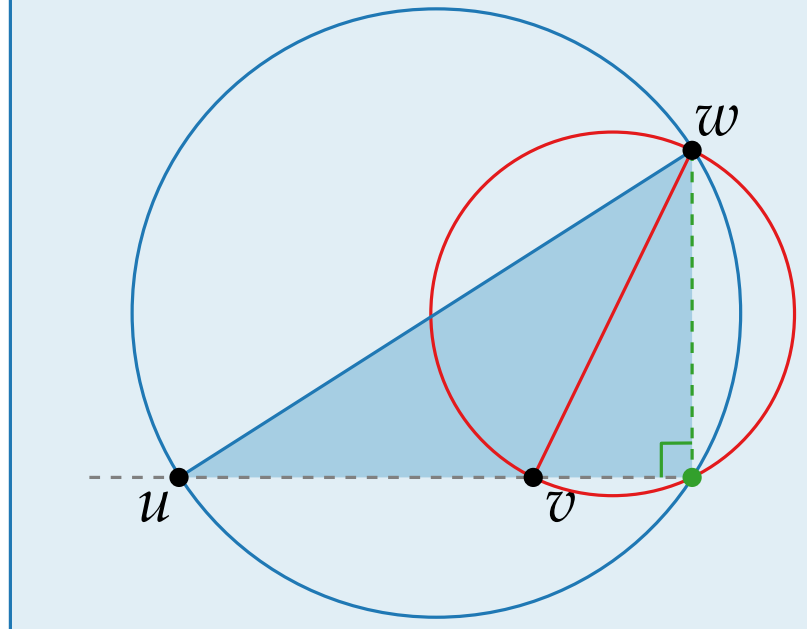
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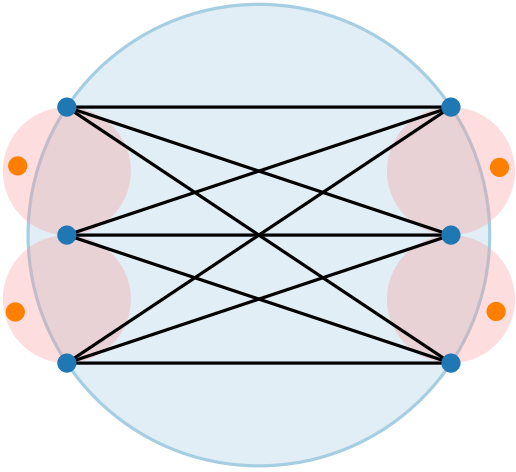
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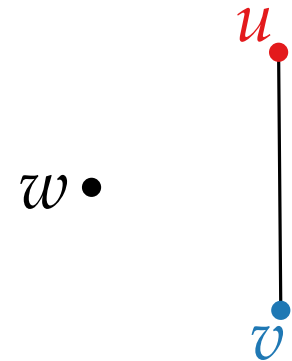
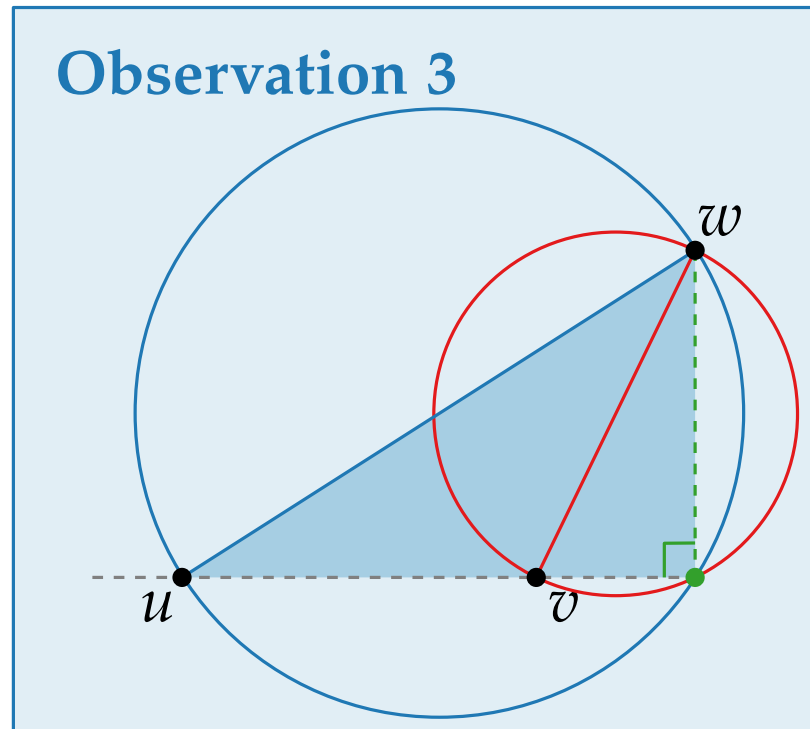
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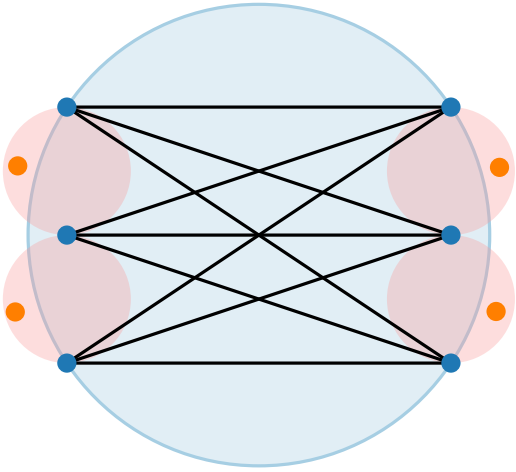
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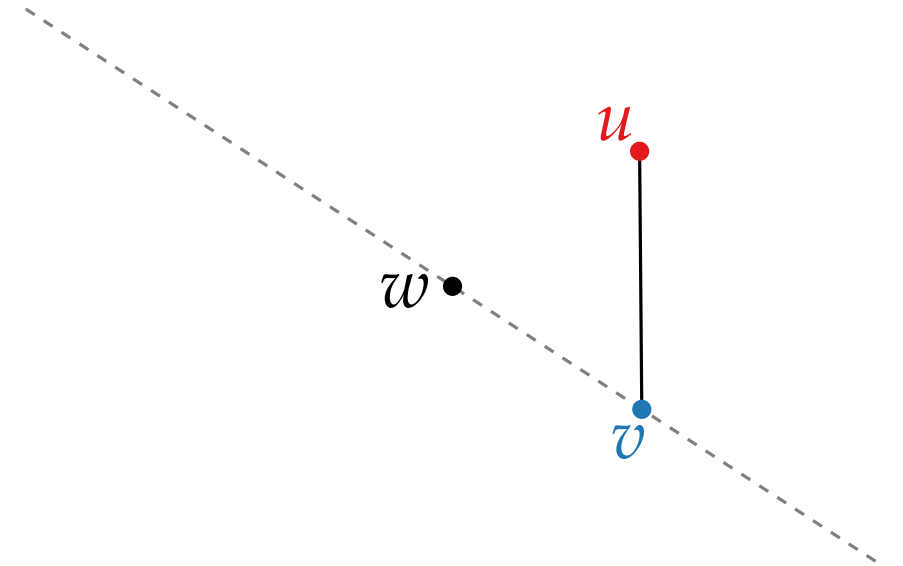
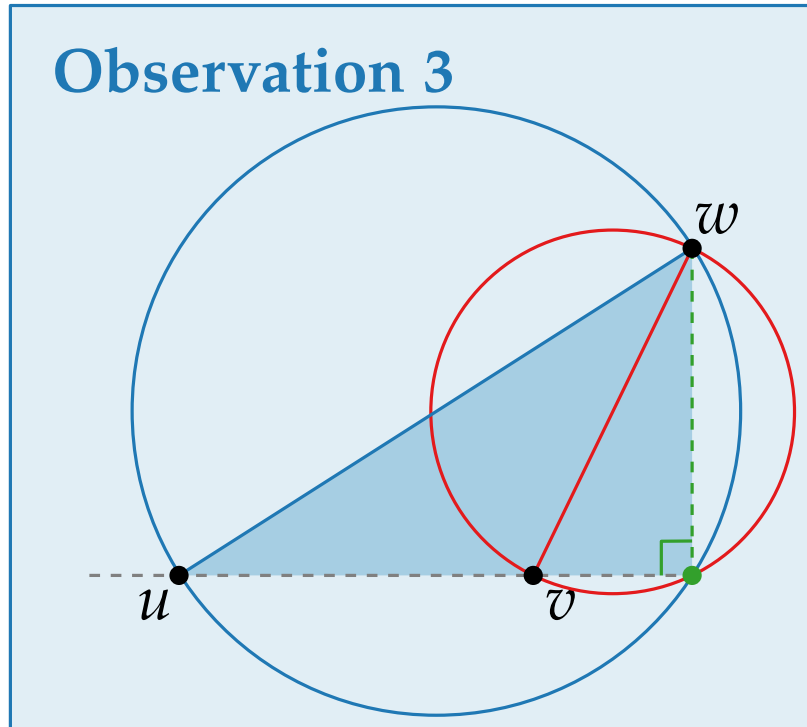
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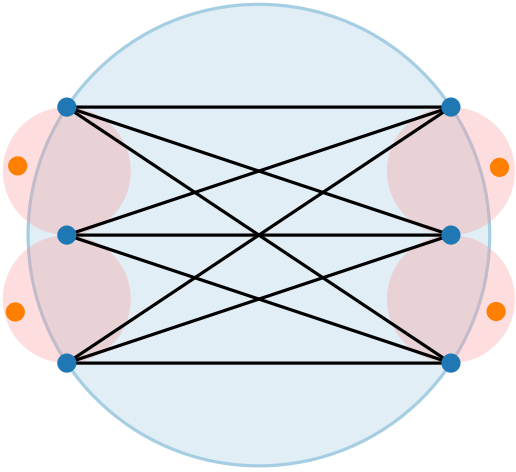
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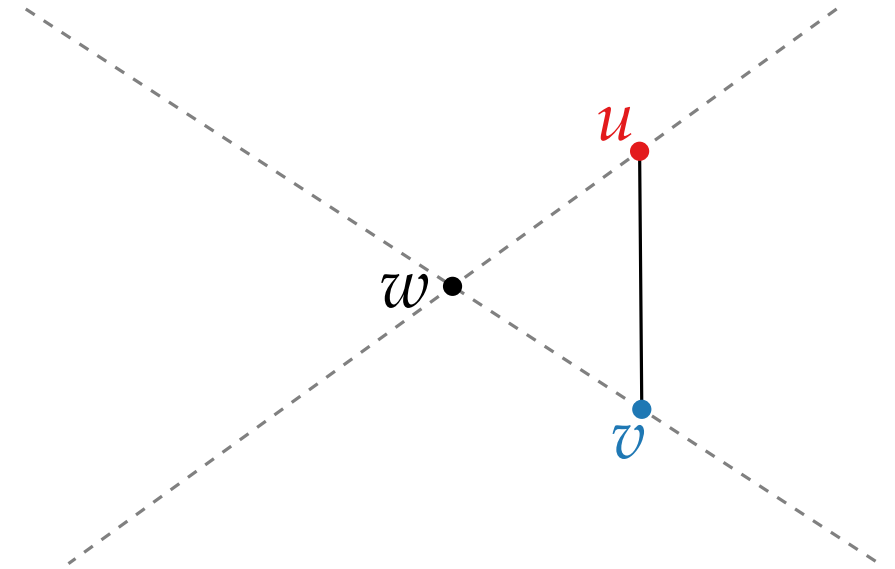
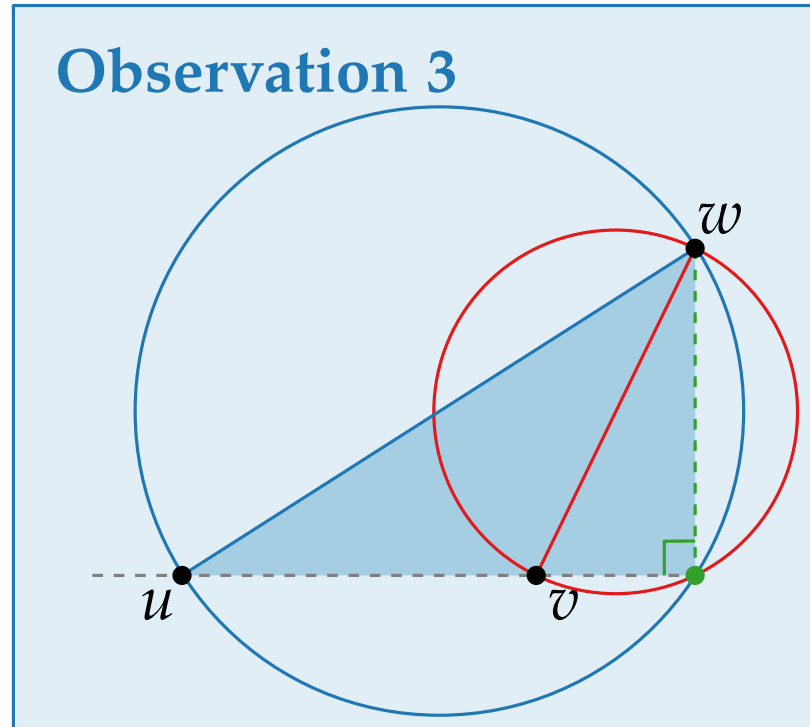
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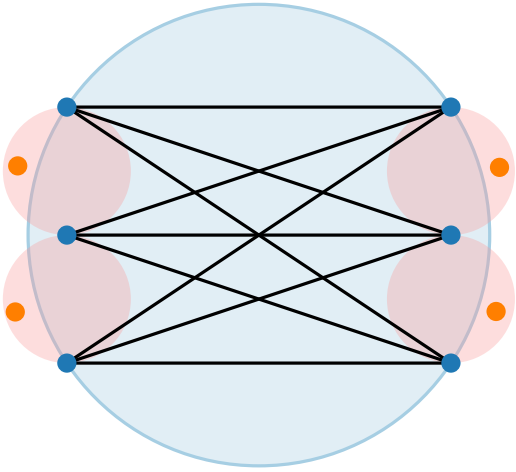
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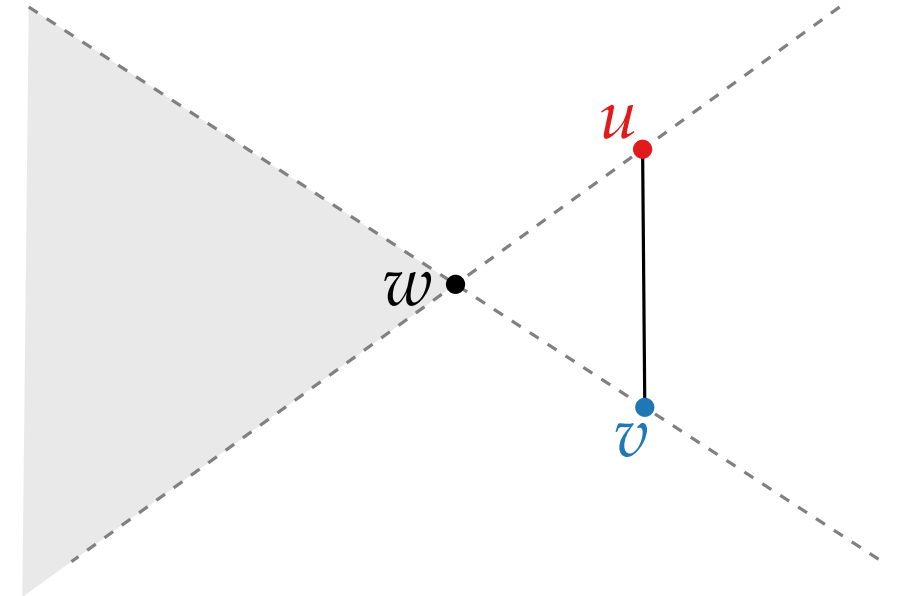
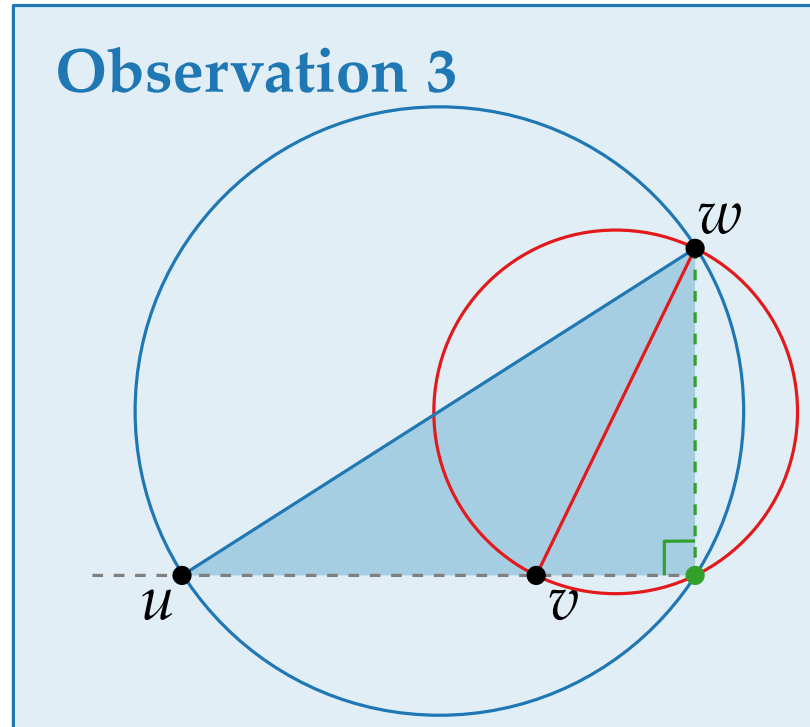
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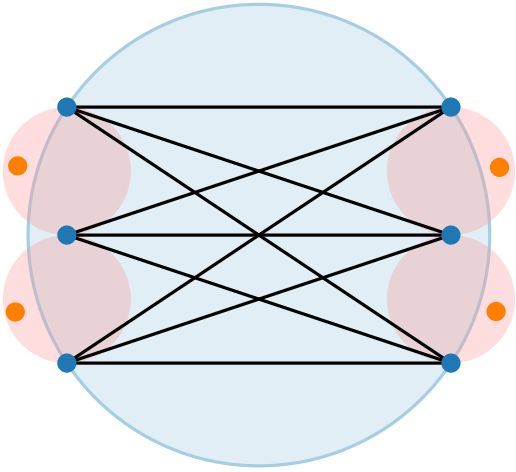
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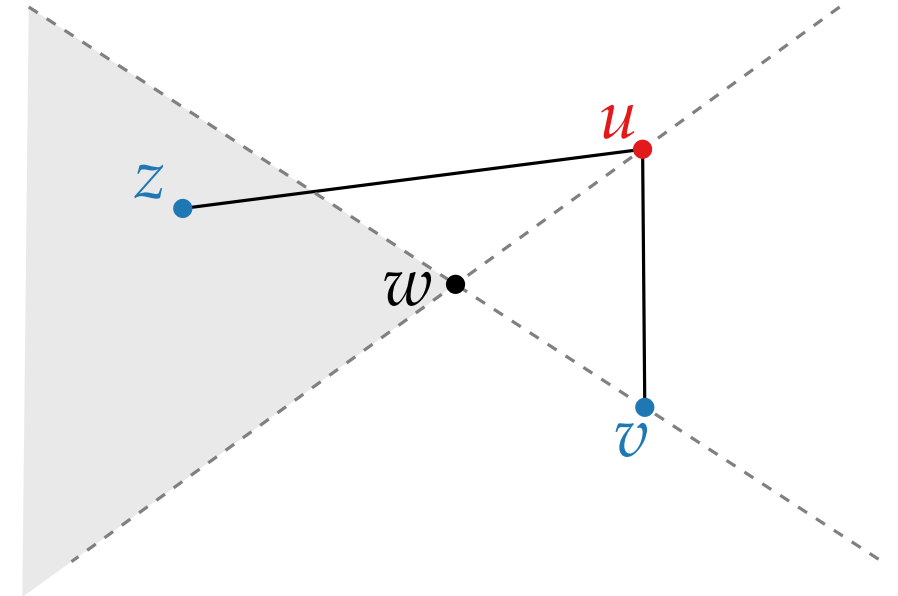
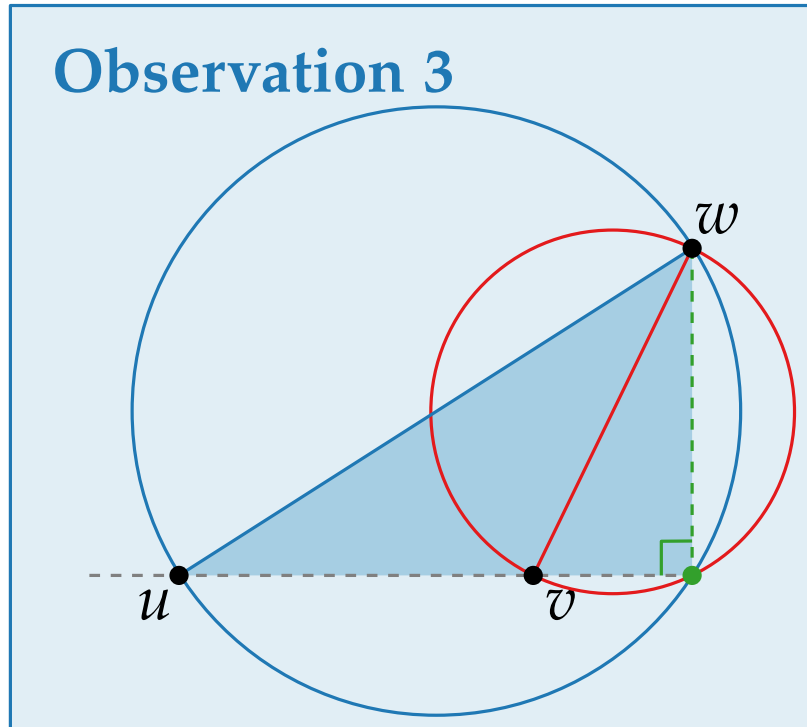
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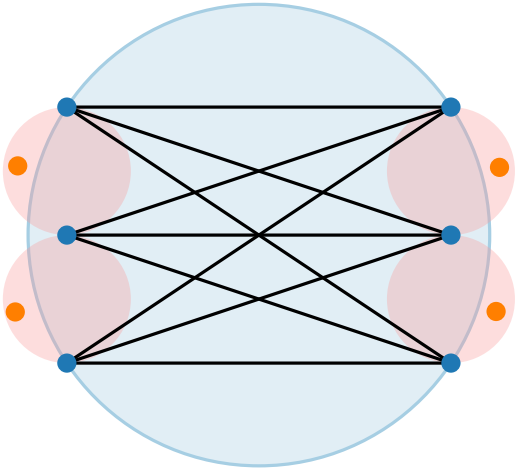
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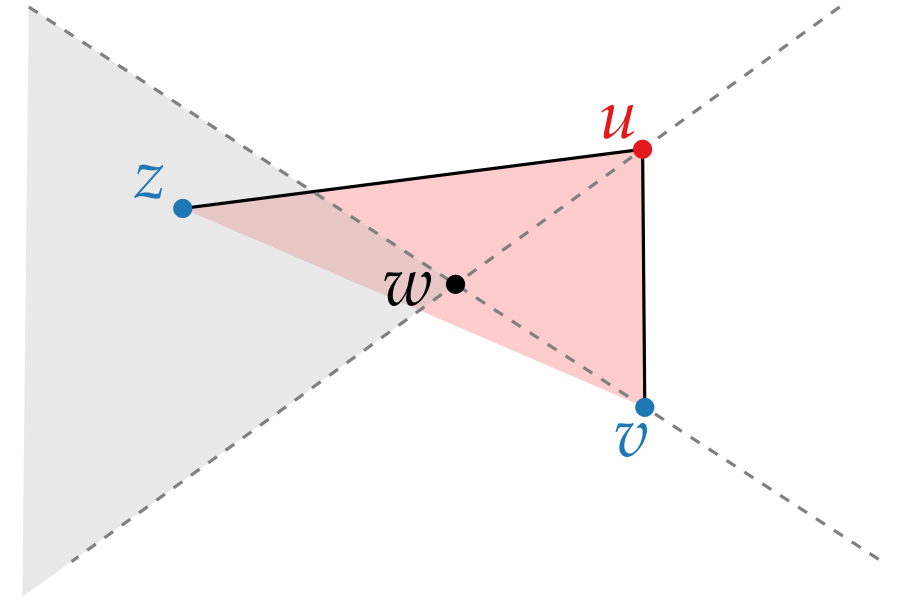
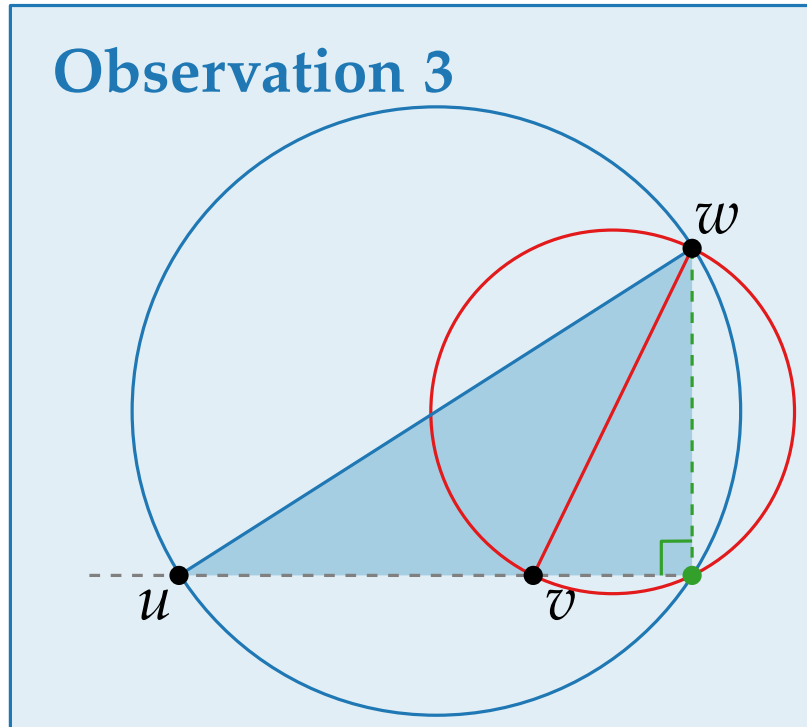
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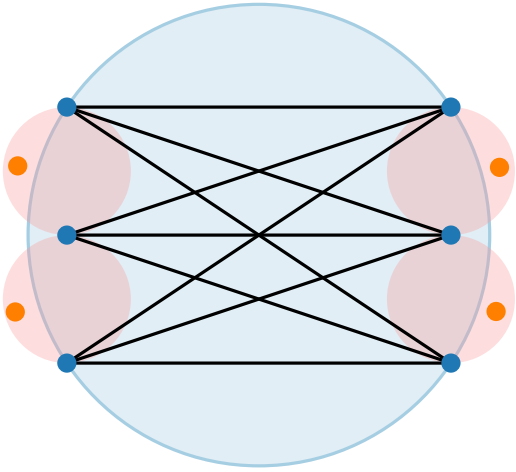
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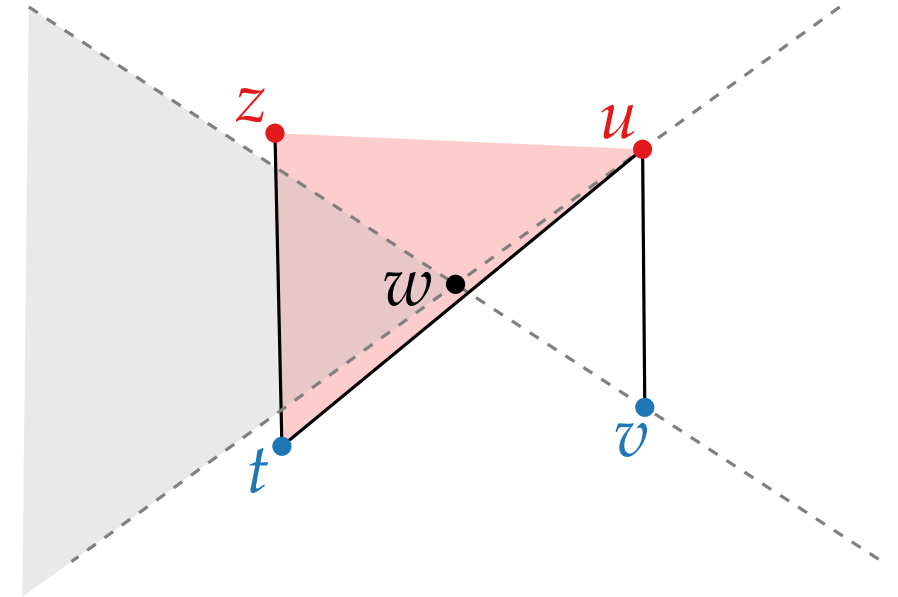
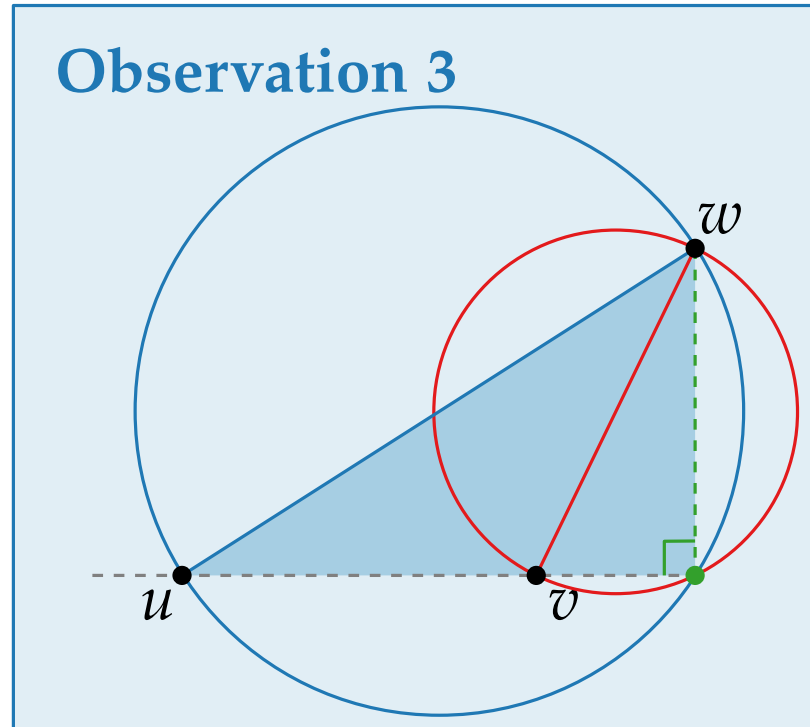
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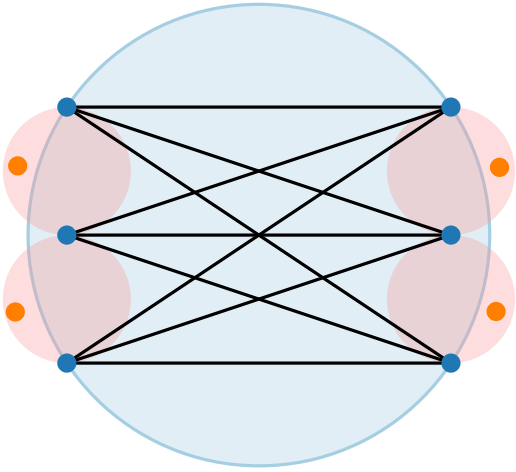
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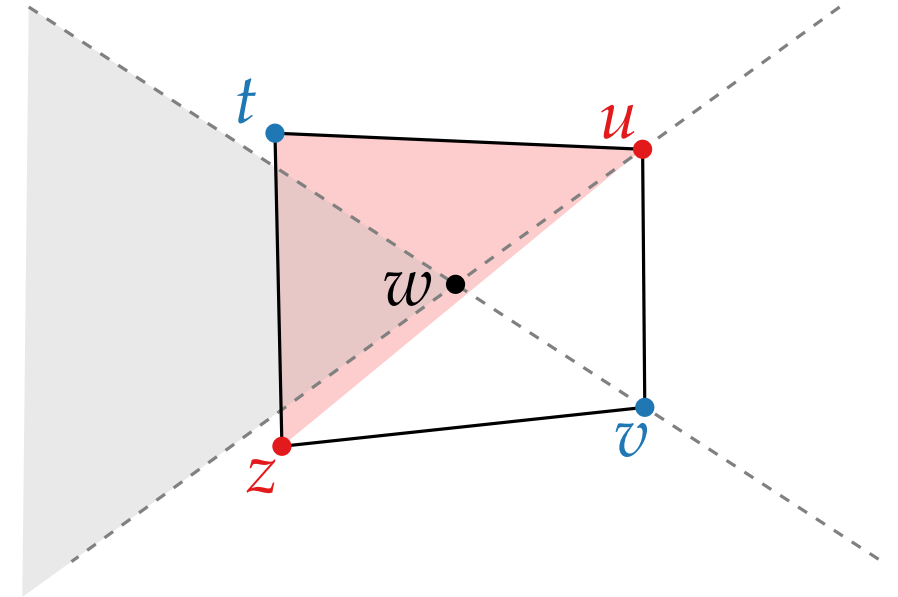
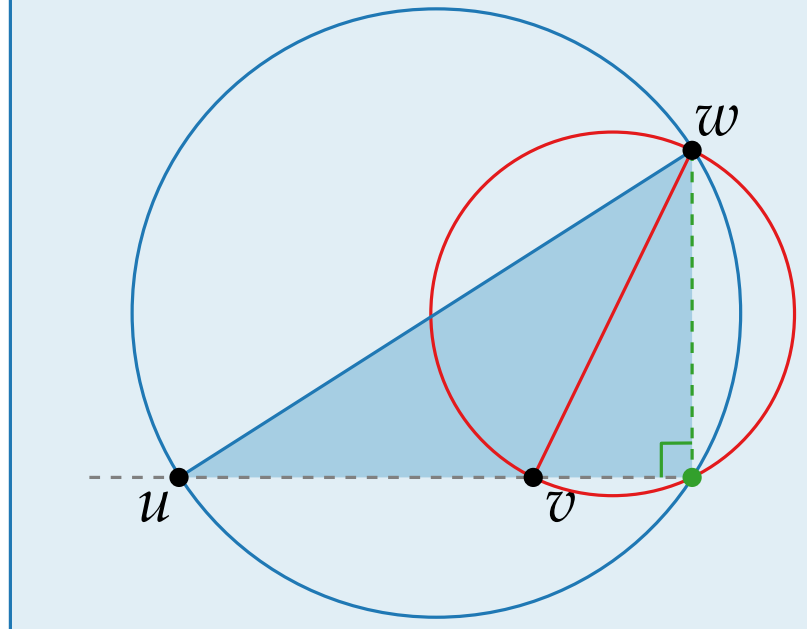
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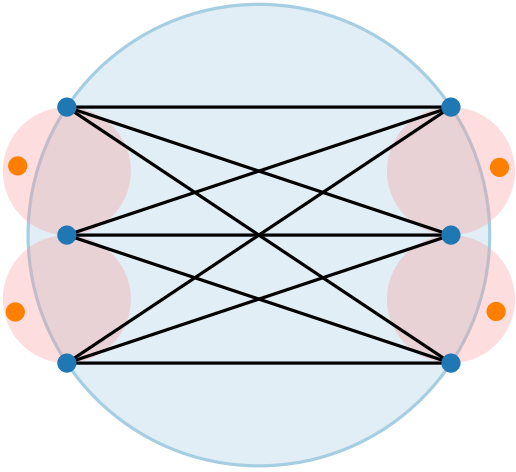
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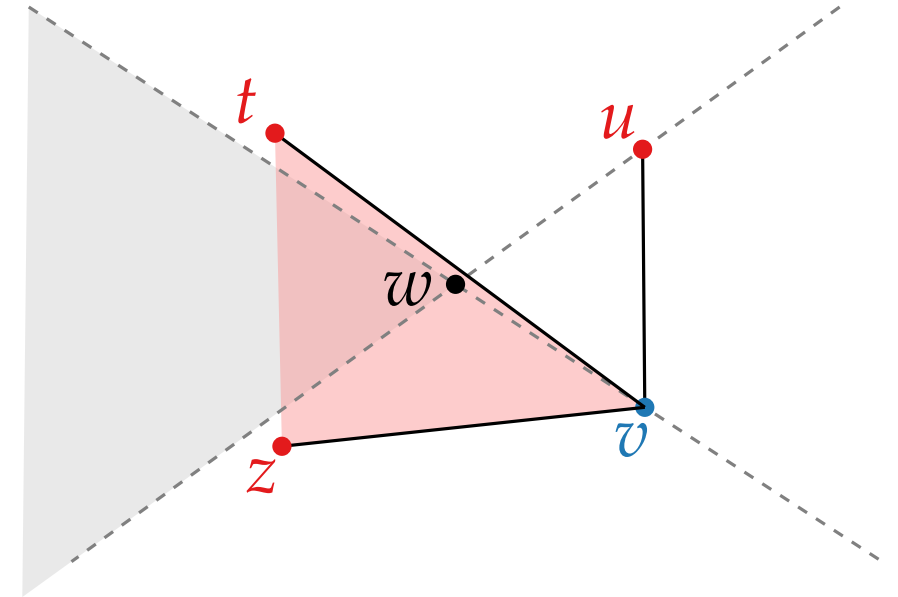
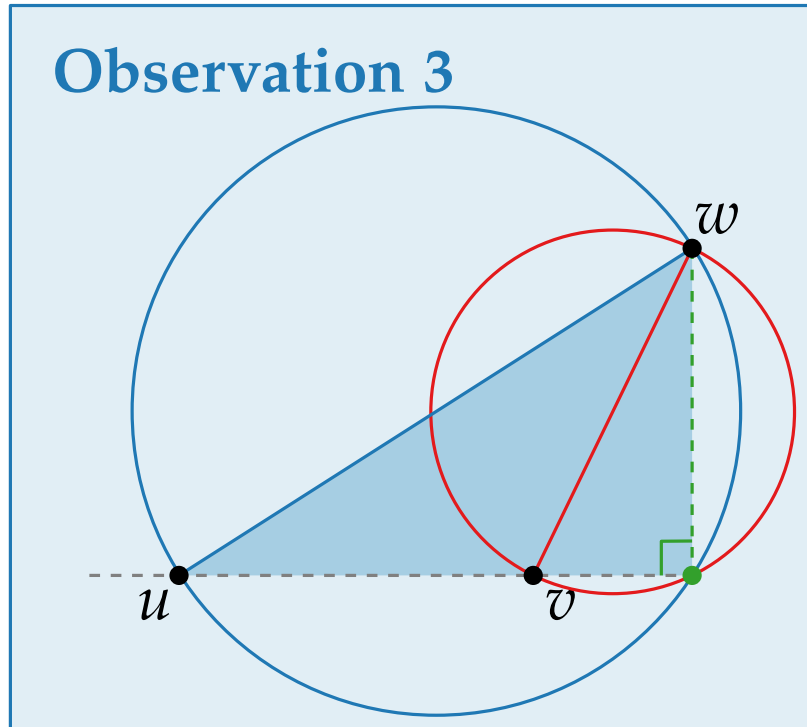
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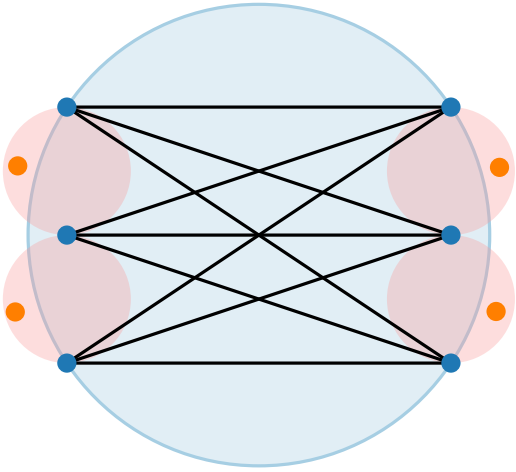
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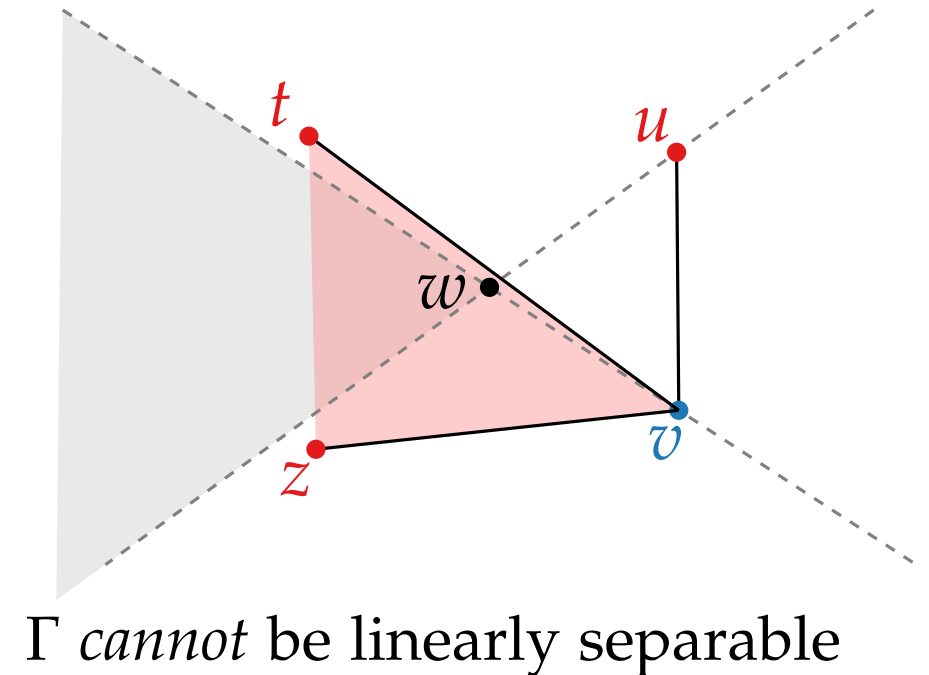
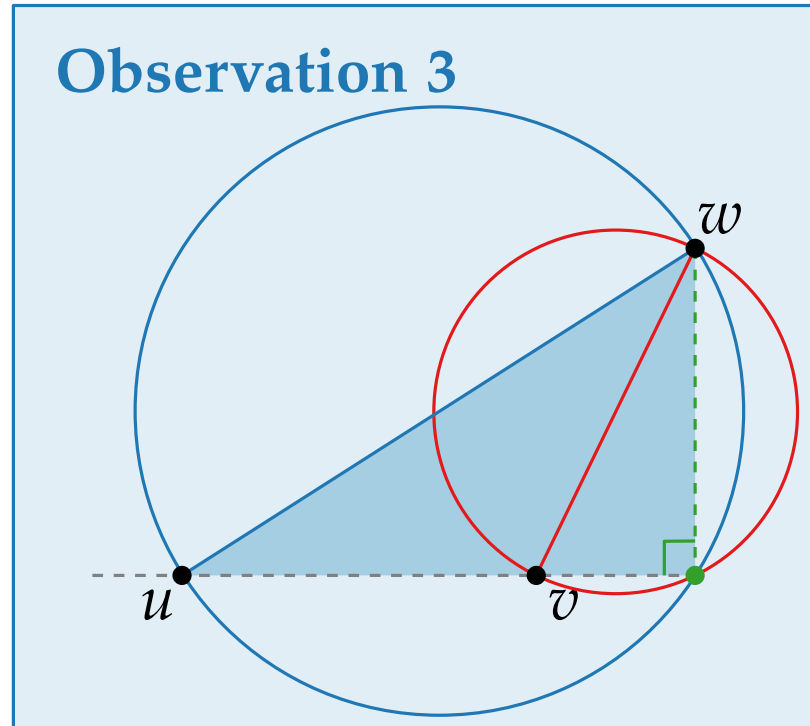
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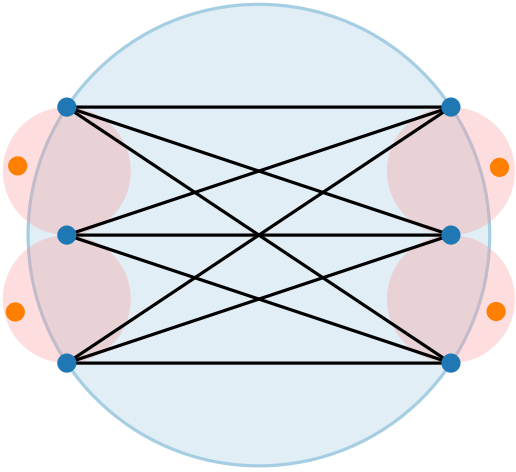
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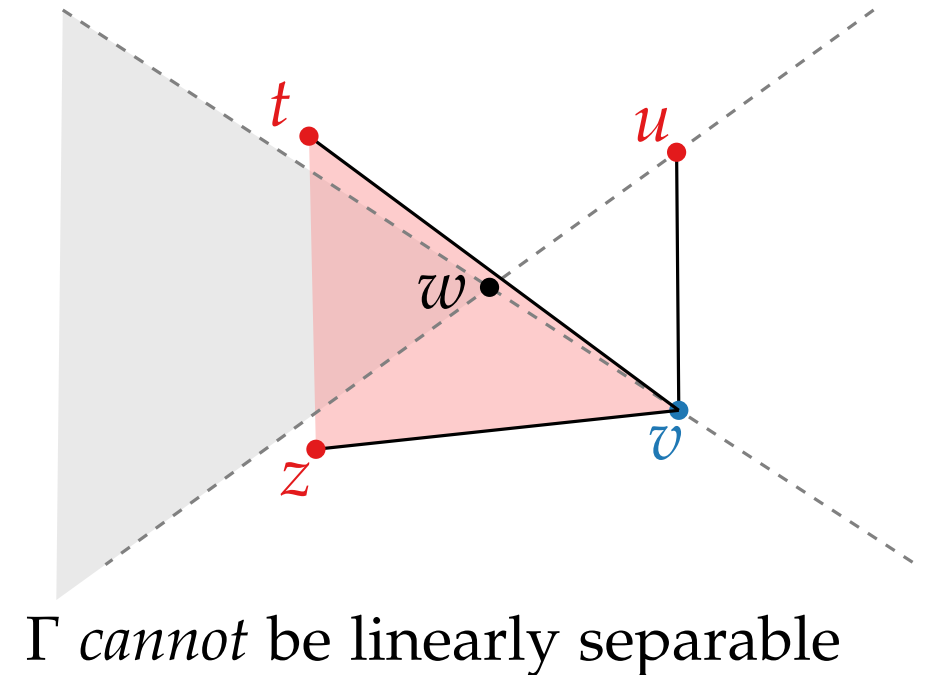
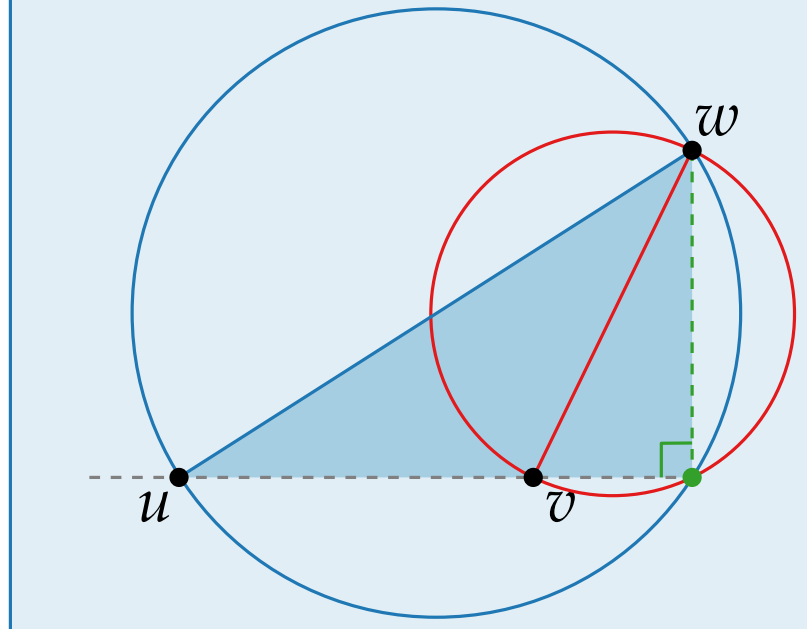
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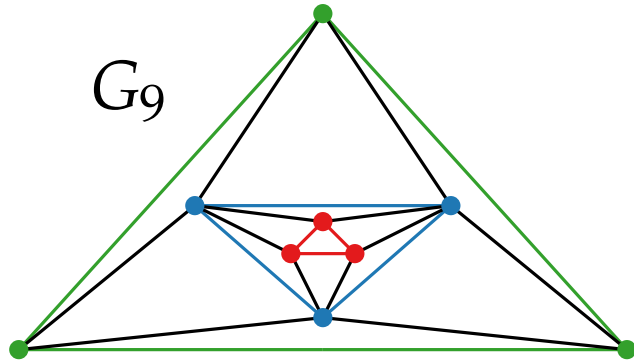
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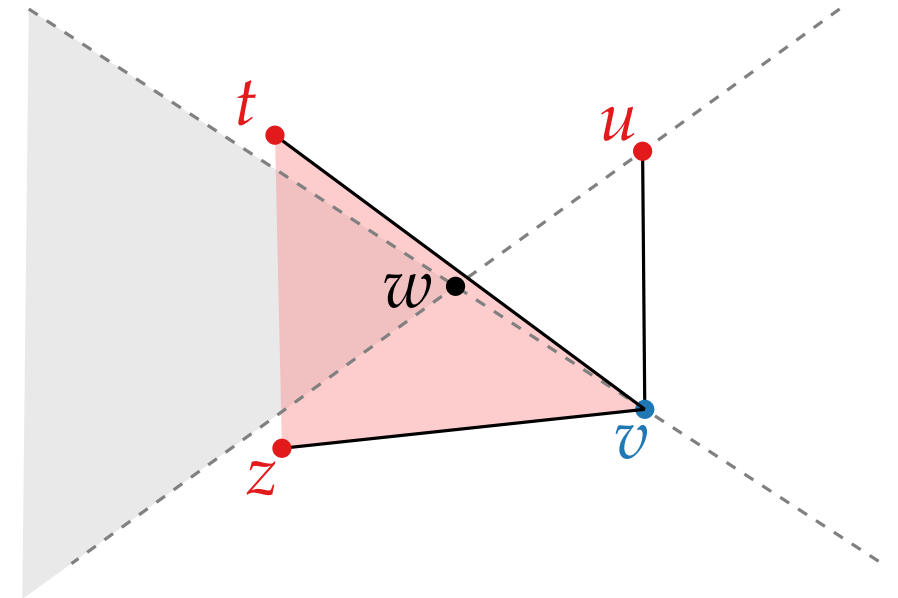
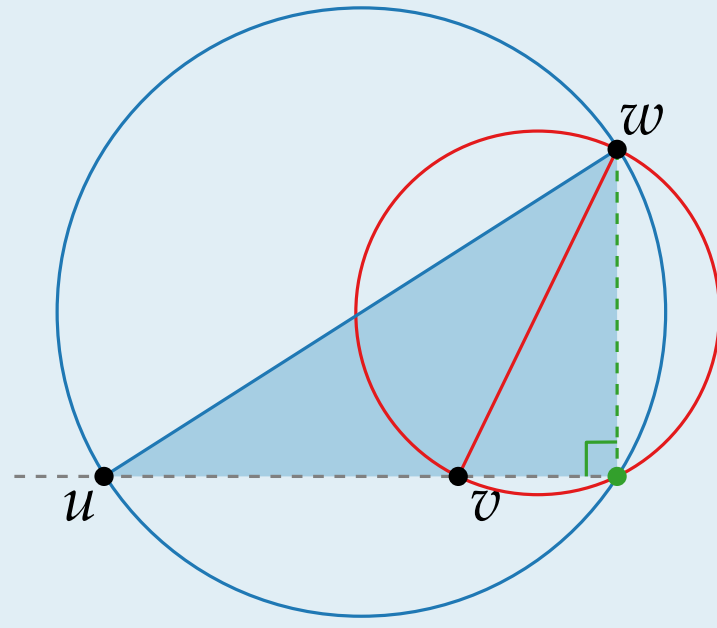
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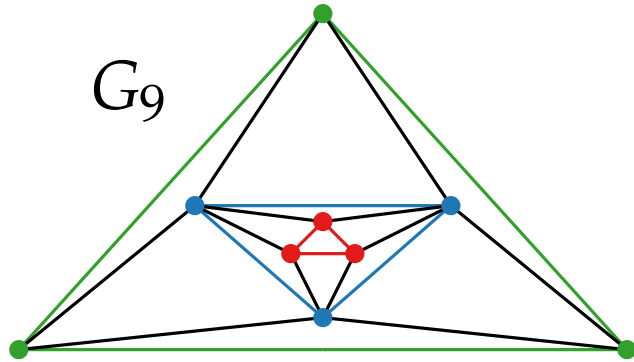
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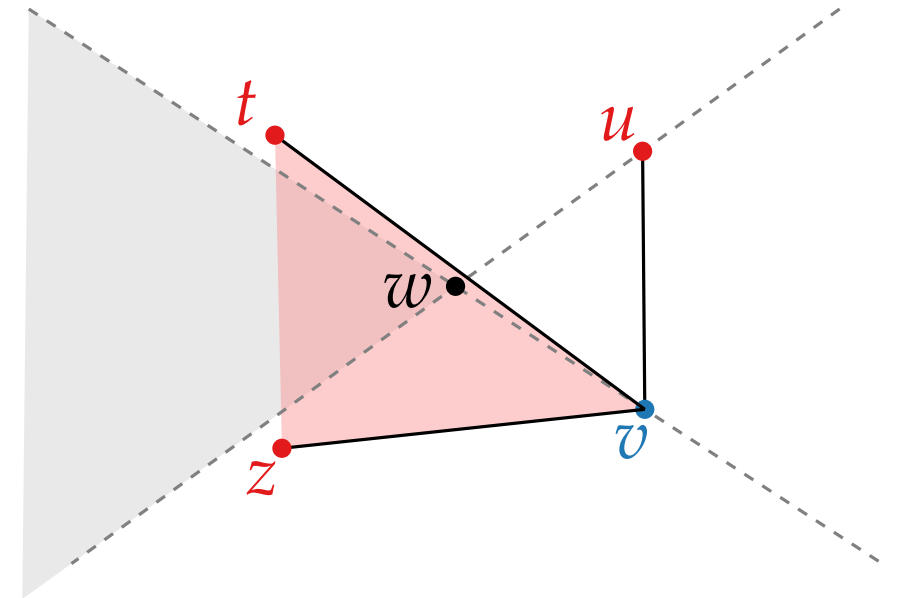
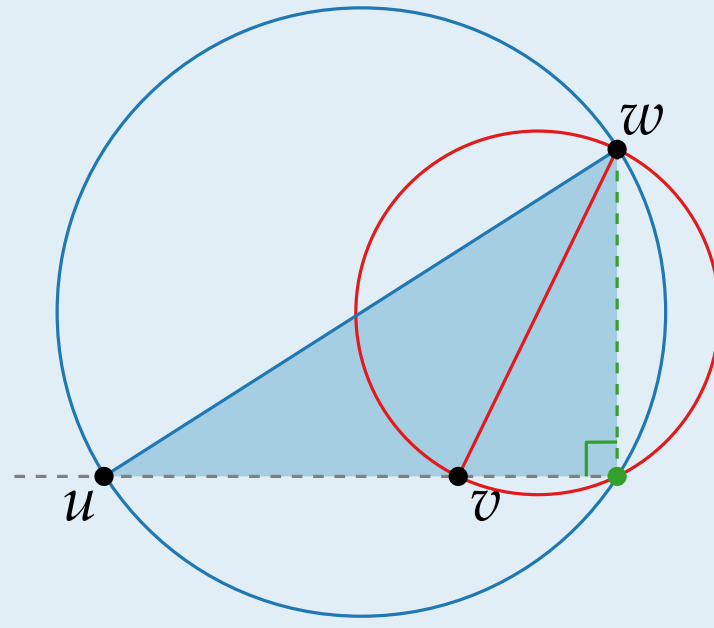
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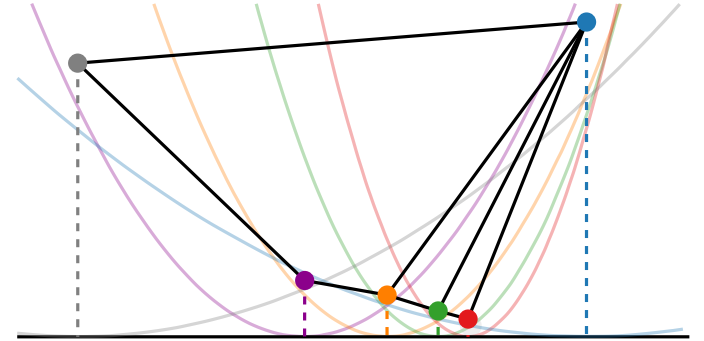


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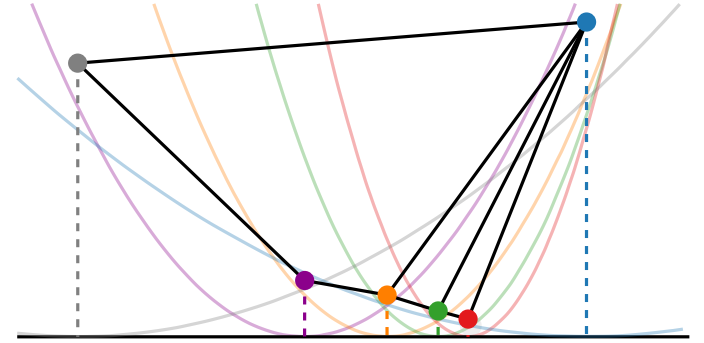
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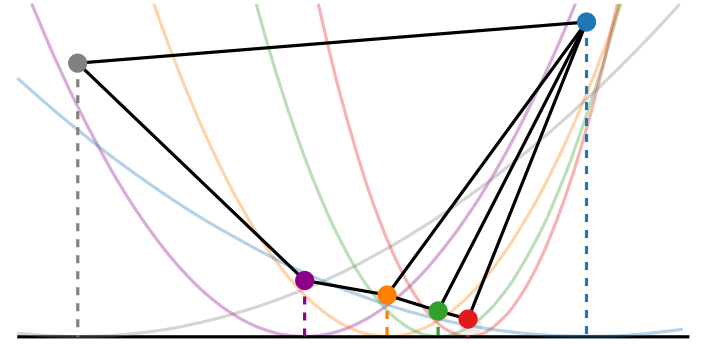
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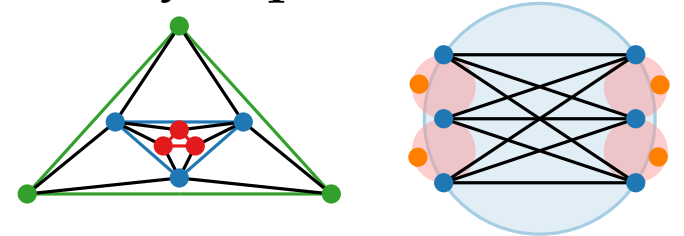
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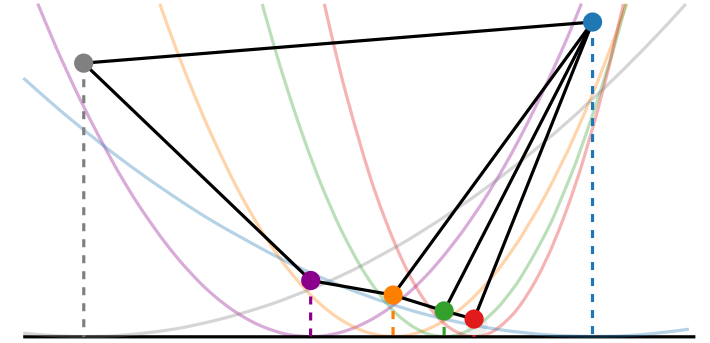
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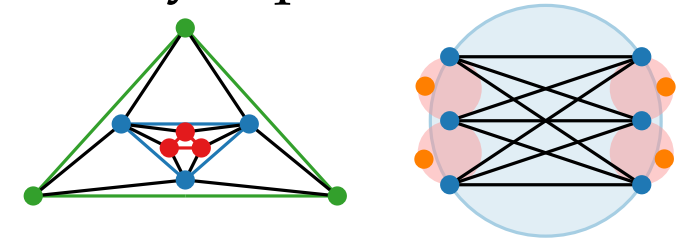
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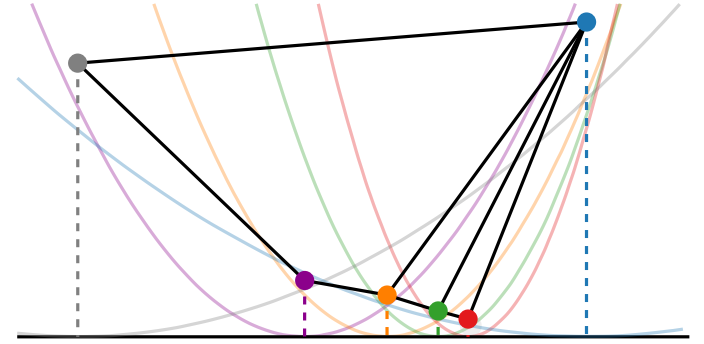
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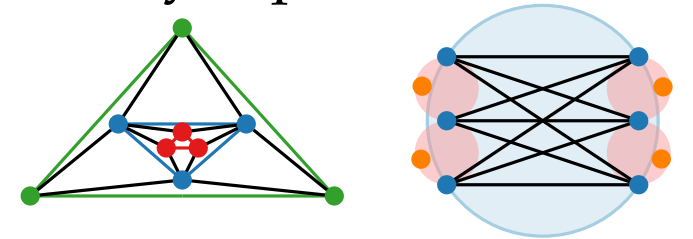
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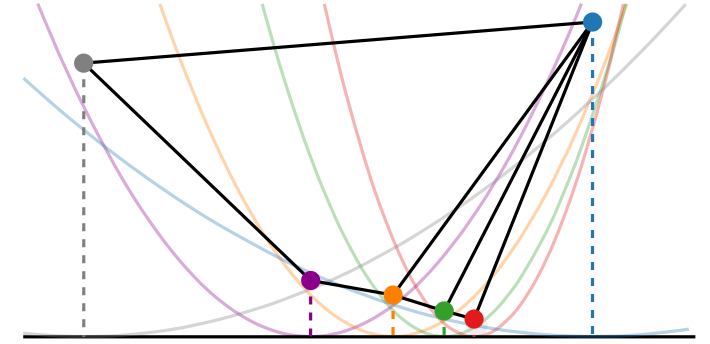
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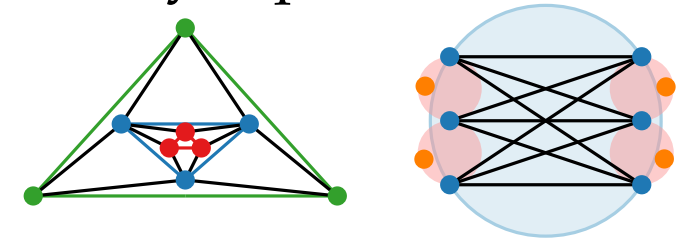
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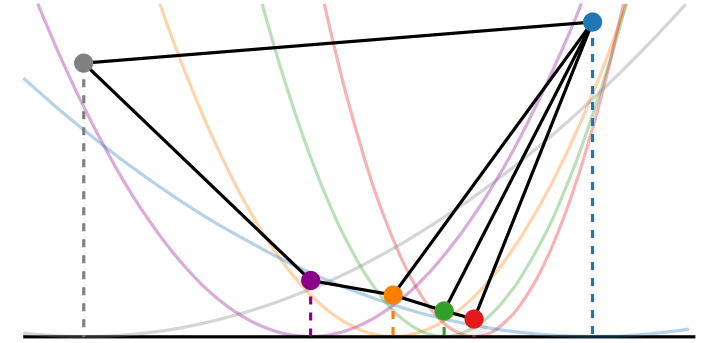
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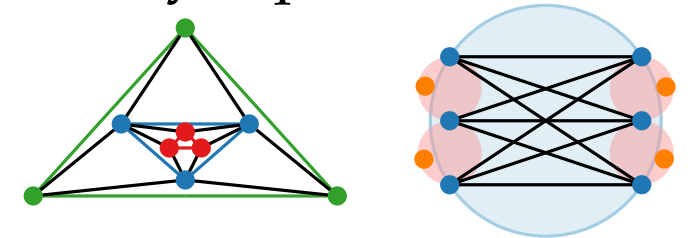
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